The construction of four-dimensional quantum Yang–Mills theory

Under construction. Version of November 11, 2026: this manuscript is incomplete. Sections are added and revised as they are written and checked against the formal proof; see the status table at https://github.com/scottnarmstrong/yang-mills-informalization/tree/main/manuscript .

Abstract

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Part I
The theorem and the architecture of its proof

1 Introduction

2 Wilson theory

The starting measure assigns a group element to each lattice link and weights the curvature around each plaquette. We first fix the matrix normalization: it determines both the Wilson energy and the coefficient of the logarithmic coupling flow.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, coordinate indices run from 1 to 4, and e_a is the ath coordinate vector. Time is the fourth coordinate. For the later relativistic statements write

$$\mathsf{m}(x) = (x^4)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{\mathbf{p} : |\vec{\mathbf{p}}| \le \mathbf{p}^4\}\,.$$

Inner products are antilinear in the first variable, and spacetime integrals use Lebesgue measure. Arity zero is included; empty sums are zero and empty products are one. The group H_4 consists of the signed permutation matrices; $\mathsf{s}_a^{\mathrm{refl}}$ denotes reflection in coordinate a.

Definition 2.1 (Gauge normalization)**.** Fix $n_G \ge 1$ and a closed subgroup $G \subset U(n_G)$ with simple nonabelian real Lie algebra

$$\mathfrak{g} = \{X \in M_{n_G}(\mathbb{C}) : \exp(tX) \in G \text{ for every } t \in \mathbb{R}\}, \qquad \langle X, Y\rangle_{\mathfrak{g}} = -n_G^{-1}\operatorname{Re}\operatorname{tr}(XY)\,.$$

For an orthonormal basis $(e_\alpha^{\mathfrak{g}})$, put

$$d_{\mathfrak{g}} = \dim_{\mathbb{R}} \mathfrak{g}, \qquad C_{\mathfrak{g}} = -d_{\mathfrak{g}}^{-1}\sum_\alpha \operatorname{tr}_{\mathfrak{g}}\big(\operatorname{ad}(e_\alpha^{\mathfrak{g}})^2\big),$$

$$b_0 = \frac{11 C_{\mathfrak{g}}}{48\pi^2}, \qquad \kappa_G = 2b_0\,.$$

This fixes the normalization of the running coupling.

Here $\operatorname{ad}(X)Y = [X, Y]$ and $\operatorname{tr}_{\mathfrak{g}}$ is the real operator trace on $\mathfrak{g}$. Simplicity means $[\mathfrak{g}, \mathfrak{g}] \ne 0$ and no proper nonzero real ideal. Connectedness of G is not assumed. The adjoint maps are skew-adjoint for this metric; nonabelianity therefore gives $C_{\mathfrak{g}} > 0$ and $\kappa_G > 0$. For $u \in G$, set

$$\operatorname{pr}(u) = 1 - n_G^{-1}\operatorname{Re}\operatorname{tr} u, \qquad \operatorname{ch}(u) = n_G^{-1/2}\|u - 1\|_{\mathrm{F}}, \qquad \operatorname{ch}(u)^2 = 2\operatorname{pr}(u)\,. \tag{2.1}$$

The first quantity is the plaquette energy; the second measures its distance from the identity. Both are invariant under conjugation and inversion.

Definition 2.2 (Wilson measure)**.** For an integer $s \ge 1$ and $\beta \in \mathbb{R}$, put $\Lambda_s = (\mathbb{Z}/s\mathbb{Z})^4$. A configuration assigns $U(y, a) \in G$ to every positive link, with inverse on the reversed link. Define

$$U_{y;ab} = U(y, a)U(y + e_a, b)U(y + e_b, a)^{-1}U(y, b)^{-1},$$

$$E_s^{\mathrm{W}}(U) = \sum_{y\in\Lambda_s}\sum_{a<b}\operatorname{pr}(U_{y;ab}),$$

$$d\mu_{s,\beta}(U) = Z_{s,\beta}^{-1}e^{-\beta E_s^{\mathrm{W}}(U)}\prod_{y,a} dh(U(y, a))\,.$$

Here dh is probability Haar measure and $Z_{s,\beta}$ normalizes the measure. This is the lattice theory to be scaled.

The normalization is the positive finite integral

$$Z_{s,\beta} = \int e^{-\beta E_s^{\mathrm{W}}(U)} \prod_{y,a} dh(U(y,a)) \,.$$

Indeed, the integrand is continuous and strictly positive on the compact configuration space. A site gauge transformation $\gamma_{\mathrm{g}} : \Lambda_s \to G$ acts by

$$U^{\gamma_{\mathrm{g}}}(y,a) = \gamma_{\mathrm{g}}(y) U(y,a) \gamma_{\mathrm{g}}(y+e_a)^{-1} \,.$$

It changes each plaquette by conjugation at its base point, so the energy is gauge invariant. Invariance of probability Haar measure under left and right multiplication then gives gauge invariance of the Wilson measure. Affine lattice isometries $x \mapsto \rho_{\mathrm{lat}} x + b$, with $\rho_{\mathrm{lat}} \in H_4$ and $b \in \Lambda_s$, transport links and invert reversed ones. The measure is invariant under these transformations as well; the finite-lattice identities are proved in Section 10.

A general Wilson family, indexed by $N, p \in \mathbb{N}$, consists of strictly increasing positive integer sequences (L_p) and (D_N), real couplings $\beta_{N,p}$, a label set $\mathcal{T}$, and complex insertions $I_{N,p}(v, y; U)$. The index N controls the spacing, and p controls the physical volume. Set

$$a_N = D_N^{-1}, \qquad \mu_{N,p} = \mu_{L_p D_N, \beta_{N,p}}, \qquad \hat{y} = a_N \big(r_{L_p D_N}(y^a)\big)_{a=1}^4 \,, \tag{2.2}$$

where $r_s(j)$ is the integer representative of $j \in \mathbb{Z}/s\mathbb{Z}$ in $(-s/2, s/2]$. Thus the physical torus has side L_p.

Definition 2.3 (Lattice moments)**.** For a label list $v \in \mathcal{T}^n$, sites $y \in \Lambda_{L_p D_N}^n$, and a test $f : (\mathbb{R}^4)^n \to \mathbb{C}$, define

$$M_{N,p}^{\mathrm{lat}}(v; y) = \int \prod_{i=1}^n I_{N,p}(v_i, y_i; U)\, d\mu_{N,p}(U),$$
$$C_{N,p}^{\mathrm{lat}}(v; f) = a_N^{4n} \sum_{y \in \Lambda_{L_p D_N}^n} f(\hat{y}) M_{N,p}^{\mathrm{lat}}(v; y) \,.$$

These are the point correlations and their physical Riemann sums.

The construction below chooses the spacings, couplings and local insertions together. Its bare inverse coupling has logarithmic slope κ_G:

$$\frac{\beta_{N,p}}{\log(1/a_N)} \longrightarrow \kappa_G \qquad (N \to \infty), \quad \text{uniformly in } p \,. \tag{2.3}$$

This running law is proved in Section 3. At each fixed physical volume the ultraviolet limit will use the full sequence $N \to \infty$; only the subsequent infinite-volume limit requires a subsequence.

3 Renormalization

Blocking integrates one scale against a probability kernel. A curvature action, a vacuum scalar and two activity families describe the new density. The inductive step controls these coordinates and the coupling; the multiscale theorem chooses the bare coupling to reach one prescribed endpoint.

Fix $\mathsf{b} > 0$. Throughout we use the common choices of Sections 10–14, made once for this group realization and penalty in the order of Section 9. Their existence is proved with Theorems 3.5 and 3.6. Constants may depend on these choices; further dependence is stated where it occurs.

3.1 Scales and the block map

Levels decrease under blocking. With odd ratios $\ell_m \geq 3$, set

$$D_0 = 1, \qquad D_{m+1} = \ell_m D_m, \qquad a_m = D_m^{-1}, \qquad s_m = \log D_m \,. \tag{3.1}$$

The level-m torus has side $L_p D_m$ in lattice units and side L_p in physical units. Starting at cutoff N, blocking successively reaches levels $N-1, \dots, 0$.

The stage J_m records the available derivative order and bounds. The stages satisfy $4 \leq J_m \leq J_{m+1} \leq J_m + 1$ and $J_m \to \infty$. Choose the reference endpoint $\bar{\beta}$ sufficiently large, as in row 7 of Section 9; in particular $\bar{\beta} > 1$. Define

$$\widehat{\beta}_m = \bar{\beta} + \kappa_G s_m, \qquad M_m^{\mathrm{cut}} = \exp\bigl((\log\log \widehat{\beta}_m)^2\bigr), \qquad t_m = \widehat{\beta}_m^{-1/2} M_m^{\mathrm{cut}}, \tag{3.2}$$

$$B_m = \lceil (\log \widehat{\beta}_m)^2 \rceil, \qquad \epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}} \,. \tag{3.3}$$

The reference coupling $\widehat{\beta}_m$ tracks the leading running slope. The factor M_m^{cut} enlarges the Gaussian field scale; t_m is the threshold separating small and large plaquette fields. The radius B_m sets the scale of the support weights. The quantities $\epsilon_m^{\mathrm{reg}}$ and $\epsilon_m^{\mathrm{for}}$ bound the regular and forced activity sums, respectively. These thresholds depend on the reference coupling, so they remain fixed as the actual coupling varies.

The allowed coupling interval is

$$\mathfrak{B}_m = [\widehat{\beta}_m/2, 2\widehat{\beta}_m] \,.$$

At stage j, write $\mathfrak{q}_j$ for the derivative order, ϖ_j for the spatial power, σ_j for the support exponent and $\omega_j^{\mathrm{RG}} \geq 0$ for the coupling drift bound. Their choices and growth conditions are in Section 9.

At a step $m+1 \to m$, the fine links U and coarse links V lie on $\Lambda_{L_p D_{m+1}}$ and $\Lambda_{L_p D_m}$. Put $j = J_{m+1}$ and $\ell = \ell_m$. Write $\mathbb{B}_R = \{-R, \dots, R\}^4$. In a patch of radius $(j+3)\ell$ about the fine site ℓy, let $h_{y,z}$ be the chosen frame at offset z. For a coarse edge $e = (y, a)$ the framed straight connector is

$$Y_{e,z} = h_{y,z} \left(\prod_{r=0}^{\ell-1} U(\ell y + z + r e_a, a) \right) h_{y+e_a,z}^{-1} \,. \tag{3.4}$$

This compares the fine transport between the two coarse endpoints in their chosen frames. The averaging weights and the predictor on its minimizing branch are

$$w_z^{\mathrm{av}} = \bigl(\mathrm{Unif}(\mathbb{B}_{(\ell-1)/2})\bigr)^{*j}(z), \qquad Q_e^{\mathrm{pred}} = \operatorname*{argmin}_{u \in G} \sum_z w_z^{\mathrm{av}} \,\mathrm{pr}(Y_{e,z} u^{-1}) \,. \tag{3.5}$$

Here Unif is uniform probability mass and $*j$ denotes j-fold convolution. This branch is used when the displayed minimum is below the fixed predictor cutoff; otherwise a tag selects $Y_{e,z}$ with mass w_z^{av}. Symmetric minimizing frames and the alternative axial frames are specified in Section 11, Definitions 11.1–11.3. The finite tag list α_{tag} specifies all frame and predictor choices. Their product mass satisfies

$$\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) \geq 0, \qquad \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) = 1 \,. \tag{3.6}$$

Definition 3.1 (Exact block map)**.** For fine links U and coarse links V, define

$$k_\beta^{(m)}(U, V) = \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) \prod_{e \text{ coarse}} \frac{e^{-\mathsf{b}\beta\,\mathrm{pr}(V_e Q_e^{\mathrm{pred}}(U, \alpha_{\mathrm{tag}})^{-1})}}{\int_G e^{-\mathsf{b}\beta\,\mathrm{pr}(u)}\, dh(u)},$$

$$(\mathcal{B}_\beta^{(m)} f)(V) = \int k_\beta^{(m)}(U, V) f(U)\, dU \,.$$

Both product measures are probability Haar measures, and $\int k_\beta^{(m)}(U,V)\,dV = 1$. The map integrates one scale without changing the partition integral.

For fixed U and tags, right invariance of Haar measure makes every coarse-link factor integrate to one. Summing the tag masses proves the normalization. Consequently, for integrable f,

$$\int (\mathcal{B}_\beta^{(m)} f)(V)\,dV = \int f(U)\,dU\,. \tag{3.7}$$

The branch choices, measurability and symmetry properties of this kernel are proved in Section 11.

3.2 The density and its norms

At level m, write $\Lambda = \Lambda_{L_p D_m}$. Let $\mathrm{Flag}_m(U)$ be the sites incident to a plaquette whose chord exceeds t_m. A support is healthy if it avoids this set. A regular activity E_i^{reg} has a site support X_i and an anchor $o_i \in X_i$. A forced activity K_i^{for} has a support Y_i and a nonempty inventory $I_i \subset Y_i$ of flagged sites. Its eligibility event is

$$\mathcal{E}_i^{\mathrm{elig}} = \{U : Y_i \cap \mathrm{Flag}_m(U) = I_i\} \cap \mathcal{E}_i^{\mathrm{extra}}, \qquad K_i^{\mathrm{for}}(U) = 0 \quad (U \notin \mathcal{E}_i^{\mathrm{elig}})\,. \tag{3.8}$$

The first test specifies exactly which sites of Y_i are flagged. The extra event $\mathcal{E}_i^{\mathrm{extra}}$ is a fixed Borel gauge-invariant test supplied with that activity; it reads only links with both endpoints in Y_i and the flags in Y_i. It restricts which local configurations can contribute that forced term. The activity families are countable.

The path action separates the kinetic coefficient from the quadratic corrections generated by integration. For the fixed level data, write $c_{\mathrm{dir}} > 16 c_{\mathrm{lin}} > 0$ for the clipping constants, chosen with $c_{\mathrm{dir}} \le \min(\gamma_{J_m}, 1)$ as in row 4 of Section 9, and $R_{J_m}^{\mathrm{rel}} > 0$ for the relaxation length. Distances use the periodic maximum-coordinate metric in lattice units: on a torus of integer side n, $\mathrm{dist}(x,y) = \min_{z \in n\mathbb{Z}^4} |\tilde{x} - \tilde{y} + z|_\infty$ for any lifts $\tilde{x}, \tilde{y} \in \mathbb{Z}^4$. Set

$$\begin{aligned} \mathrm{clip}_t(r) &= \begin{cases} c_{\mathrm{dir}} r^2/2, & r \le t, \\ c_{\mathrm{lin}} t r, & r > t, \end{cases} \\ \chi_{m,x}^{\mathrm{hlth}}(U) &= \mathbf{1}\Big\{ \mathrm{ch}(U_{y;ab}) \le t_m e^{\mathrm{dist}(x,y)/R_{J_m}^{\mathrm{rel}}} \text{ for every } y, a < b \Big\}\,. \end{aligned} \tag{3.9}$$

The clip keeps a positive linear penalty beyond the small-field threshold, while $\chi_{m,x}^{\mathrm{hlth}}$ switches off a quadratic row when a plaquette violates the threshold prescribed by its distance from x. For each plaquette orientation $\mathfrak{p} = (a,b)$, let $\mathcal{F}_{\mathfrak{p}i}(U,x)$ be a curvature $\mathrm{proj}_{\mathfrak{g}}(U_\square - U_\square^{-1})/2$, transported along a specified finite path to x, and put

$$\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}(U,x) = \sum_i r_{\mathfrak{p}i} \mathcal{F}_{\mathfrak{p}i}(U,x)\,.$$

The projection is orthogonal for the trace metric of Definition 2.1. Let $\mathfrak{a}_i^{(q)}(U,x)$, $q = 2,3,4$, denote the fixed real invariant q-linear expressions in transported curvatures, multiplied by the health indicator of their combined site supports. Then

$$\begin{aligned} (P^{\mathrm{path}})_{N,m}^{\mathrm{raw}}(\beta,U) &= \beta \sum_{x,\mathfrak{p}} \Big[\mathrm{clip}_{t_m}(\mathrm{ch}(U_{x;\mathfrak{p}})) + \tfrac{1}{2} \chi_{m,x}^{\mathrm{hlth}}(U) |\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}(U,x)|^2 \Big] \\ &\quad + \beta \sum_{x,i} (\mathfrak{a}_i^{(3)} + \mathfrak{a}_i^{(4)})(U,x) + \sum_{x,i} \mathfrak{a}_i^{(2)}(U,x), \\ P_{N,m}^{\mathrm{path}}(\beta,U) &= |H_4|^{-1} \sum_{h \in H_4} (P^{\mathrm{path}})_{N,m}^{\mathrm{raw}}(\beta, h^* U)\,. \end{aligned} \tag{3.10}$$

Here $x \in \Lambda$ and $i \in \mathbb{N}$; h^*U is the link transport by the lattice isometry h. The path coefficients and multilinear expressions are independent of p. Their absolutely convergent series and the clipping restrictions are given in Section 12, Definitions 12.1–12.2 and Lemmas 12.3–12.4. In particular the action is affine in β; its last quadratic sum is independent of β.

For these fixed path actions and thresholds, a *representation* at level m is the datum

$$\rho = \big(\beta, v_{\mathrm{vac}}, (X_i, o_i, E_i^{\mathrm{reg}})_i, (Y_i, I_i, \mathcal{E}_i^{\mathrm{extra}}, K_i^{\mathrm{for}})_i\big)\,. \tag{3.11}$$

The scalar v_{vac} is its vacuum term, and $\beta(\rho) = \beta$ its running kinetic coefficient. These coordinates describe the integrated density below.

Definition 3.2 (Effective density)**.** For regular activities $(X_i, o_i, E_i^{\mathrm{reg}})$ and forced activities $(Y_i, I_i, K_i^{\mathrm{for}})$, define

$$\operatorname{dens}\rho(U) = e^{v_{\mathrm{vac}}|\Lambda| - P^{\mathrm{path}}_{N,m}(\beta,U) - \sum_i \mathbf{1}_{X_i \text{ healthy}} E_i^{\mathrm{reg}}(U)} \cdot \sum_{\substack{\mathcal{A}_{\mathrm{for}} \subset \mathbb{N} \text{ finite} \\ Y_i \ (i\in\mathcal{A}_{\mathrm{for}}) \text{ pairwise disjoint} \\ \bigcup_{i\in\mathcal{A}_{\mathrm{for}}} I_i = \mathrm{Flag}_m(U)}} \prod_{i\in\mathcal{A}_{\mathrm{for}}} K_i^{\mathrm{for}}(U)\,.$$

The sum covers every flagged site; each inventory is a nonempty subset of its support. This represents the density after integration.

If there are no flagged sites, only the empty family occurs and the forced sum equals one. When flags are present, their exact coverage and the disjointness of the supports assign each flagged site to precisely one selected forced term. Regular activities enter only on healthy supports. Their local regularity and cancellations, and the restrictions on forced activities, are specified in Section 13, Definitions 13.1–13.2.

For a link e, let e^- be its initial site. The left derivative $D^L_{\dot{\mathcal{A}}}$ differentiates the increments $U_e \mapsto e^{t\dot{\mathcal{A}}_e} U_e$, with $\dot{\mathcal{A}}_e \in \mathfrak{g}$ and norm from Definition 2.1.

Definition 3.3 (Local derivative seminorm)**.** For a regular activity anchored at o_i, set

$$\|E_i^{\mathrm{reg}}\|_m = \sup_{0\le q\le \mathfrak{q}_{J_m}} \sup_{U,\dot{\mathcal{A}}_1,\dots,\dot{\mathcal{A}}_q} \big| D^L_{\dot{\mathcal{A}}_1} \cdots D^L_{\dot{\mathcal{A}}_q} E_i^{\mathrm{reg}}(U)\big|\,.$$

The second supremum is over configurations healthy on X_i and directions satisfying

$$|(\dot{\mathcal{A}}_h)_e| \le t_m\big(1 + \operatorname{dist}(e^-, o_i)\big)^{\varpi_{J_m}}\,.$$

This measures ordered local derivatives in units of the permitted field size.

For a support X, let $\mathrm{cov}_{B_m}(X)$ be the minimum number of radius-B_m lattice balls with connected union covering X. The exponential weight below penalizes supports that require many such balls.

Definition 3.4 (Weighted activity sums)**.** For nonnegative majorants $e_i^{\mathrm{maj}} \ge \|E_i^{\mathrm{reg}}\|_m$ and $k_i^{\mathrm{maj}} \ge \sup_U |K_i^{\mathrm{for}}(U)|$, define

$$\mathcal{E}_m = \sup_x \sum_{o_i = x} e^{\sigma_{J_m}\,\mathrm{cov}_{B_m}(X_i)} e_i^{\mathrm{maj}},$$

$$\mathcal{K}_m = \sup_x \sum_{x\in Y_i} e^{\sigma_{J_m}\,\mathrm{cov}_{B_m}(Y_i)} k_i^{\mathrm{maj}}\,.$$

These sums measure regular and forced terms separately, with a penalty for large supports.

The chosen admissible classes $\mathcal{A}_{N,m,p}$, initial classes $\mathcal{I}_{N,p} \subset \mathcal{A}_{N,N,p}$, and adjacent-level relations

$$\mathcal{R}_{N,m,p} \subset \mathcal{A}_{N,m,p} \times \mathcal{A}_{N,m+1,p}$$

are constructed in Section 13, 13.4. They retain the Wilson initial representations, the coarse representations produced by the inductive assembly, and the estimates needed to continue that assembly and perform endpoint tuning. For every member, 13.3 supplies majorants with convergent weighted sums satisfying

$$\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}}\,. \tag{3.12}$$

These inequalities are properties of the chosen classes; the classes also retain the restrictions imposed by the inductive assembly. The relation $\mathcal{R}_{N,m,p}$ records its permitted adjacent pairs. The density identity and drift bound below hold for the pairs produced by RG and for the trajectories chosen in MT.

Along a continuous curve, supports, anchors, inventories and extra tests are fixed, and the coupling and vacuum scalar are continuous. Differences of regular derivatives on healthy configurations and of forced values admit majorants whose weighted sums over all addresses tend to zero as the parameter approaches any point. The weights are those of Definition 3.4 at the level of the representation, including t_m in the derivative seminorm. Thus the input and output curves below use levels $m+1$ and m, respectively. This is the continuity of Definition 13.5 in Section 13: regular derivatives are measured on the healthy domain, and forced terms are compared in supremum norm on all configurations. No derivative of a flag indicator is taken.

3.3 The inductive step

Exact integration removes one scale. Extracting its marginal curvature-square term into the coupling leaves a bounded drift error and returns a density in the next admissible class.

Theorem 3.5 (RG. One step)**.** *For every N, p, $m < N$, reals $u_0 \le u_1$ and continuous $\rho'_u \in \mathcal{A}_{N,m+1,p}$ with $\beta'_u = \beta(\rho'_u) \in \mathfrak{B}_{m+1}$ for all $u \in [u_0, u_1]$, there is a continuous $\rho_u \in \mathcal{A}_{N,m,p}$ with $(\rho_u, \rho'_u) \in \mathcal{R}_{N,m,p}$ and, for every $u \in [u_0, u_1]$ and coarse configuration V,*

$$\begin{aligned} \operatorname{dens}\rho_u(V) &= (\mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens}\rho'_u)(V), \\ \beta(\rho_u) &= \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le \omega^{\mathrm{RG}}_{J_{m+1}}\,. \end{aligned}$$

The proof occupies Sections 11–13, ending with 13.6. Symmetric gauge fixing and the stationary background give the small-field integration; the large-field lower bound controls the forced terms. The Gaussian determinant and the Haar-coordinate Jacobian contribute to the one-loop action. Its quadratic curvature coefficient changes the inverse coupling by $-\kappa_G \log \ell_m$, up to the displayed error. Fourth-order Wilson matching fixes the remaining local terms, and the scalar coupling reset absorbs the residual kinetic contribution in Section 12. Section 13 assembles the activities with the bounds (3.12).

For the fixed construction, ω^{RG}_j depends only on the stage; it is independent of N, p, the interval and the curve. Since $\widehat{\beta}_m = \widehat{\beta}_{m+1} - \kappa_G \log \ell_m$, a step changes $\beta - \widehat{\beta}$ by e_u only. RG imposes a band condition only on the input coupling. The trajectories of MT below also have their output couplings in the appropriate bands. Section 13, 13.6 supplies the continuous initial curve

$$\beta \longmapsto \rho^{\mathrm{W}}_\beta \in \mathcal{I}_{N,p}, \qquad \beta(\rho^{\mathrm{W}}_\beta) = \beta, \qquad \operatorname{dens}\rho^{\mathrm{W}}_\beta = e^{-\beta E^{\mathrm{W}}_{L_p D_N}} \quad (\beta \in \mathfrak{B}_N)\,, \tag{3.13}$$

whose vacuum scalar is zero. This is the base case for the induction.

3.4 Multiscale trajectories

A trajectory is a tuple $\rho_{N,p,m} \in \mathcal{A}_{N,m,p}$, $0 \le m \le N$, satisfying

$$\rho_{N,p,N} \in \mathcal{I}_{N,p}, \qquad (\rho_{N,p,m}, \rho_{N,p,m+1}) \in \mathcal{R}_{N,m,p} \quad (m < N)\,.$$

Write $\beta_{N,p,m} = \beta(\rho_{N,p,m})$; here m is the final index, whereas it is the middle index of $\mathcal{A}_{N,m,p}$ and $\mathcal{R}_{N,m,p}$. To select the bare coupling, start with the continuous Wilson curve across $\mathfrak{B}_N$ and apply RG. At the step $m+1 \to m$, the drift bound makes the endpoint output couplings bracket $[\bar\beta + (\kappa_G/2)s_m, \bar\beta + 2\kappa_G s_m] \subset \mathfrak{B}_m$. The intermediate value theorem gives a closed parameter subinterval whose endpoint couplings are the ends of this interval and whose couplings remain within it. Repeating this restriction reaches $\bar\beta$ at level zero. This is the shooting argument. It chooses the couplings for all N, p before an increasing sequence of volume indices $\mathsf{p} : \mathbb{N} \to \mathbb{N}$ is prescribed.

Theorem 3.6 (MT. Multiscale trajectory)**.** *There are bare couplings $\beta_{N,p}$ and trajectories $(\rho_{N,p,m})_{m=0}^{N}$, for all $N, p \in \mathbb{N}$, with $\rho_{N,p,N} = \rho^{\mathrm{W}}_{\beta_{N,p}}$ and $\beta_{N,p,N} = \beta_{N,p}$, satisfying*

$$\beta_{N,p,0} = \bar\beta, \qquad \beta_{N,p,m} \in \mathfrak{B}_m \quad (0 \le m \le N),$$
$$\operatorname{dens} \rho_{N,p,m} = \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} \operatorname{dens} \rho_{N,p,m+1} \quad (m < N)\,.$$

For every $\varepsilon > 0$, some $m_0(\varepsilon)$ satisfies

$$|\beta_{N,p,m} - \kappa_G s_m| \le \varepsilon s_m \qquad (N \ge m \ge m_0,\ p \in \mathbb{N})\,.$$

The proof is the shooting argument of Section 14, 14.1, with 14.1–14.2. Fix this family of trajectories for all subsequent constructions. The exact drift bound of RG holds on every adjacent pair of these trajectories, and all their integrals equal $Z_{L_p D_N, \beta_{N,p}}$ by (3.7) and the Wilson start. The path arrays remain independent of p.

The running law follows by summing the drift from the fixed endpoint:

$$|\beta_{N,p,m} - \widehat{\beta}_m| \le \sum_{i=0}^{m-1} \omega^{\mathrm{RG}}_{J_{i+1}}\,. \tag{3.14}$$

Row 5 of Section 9 bounds the sum by $(\kappa_G/4)s_m$, which also places the trajectory couplings in $\mathfrak{B}_m$. Its stronger bound $\omega^{\mathrm{RG}}_{J_{i+1}} / \log \ell_i \le J_{i+1}^{-2}$ makes the sum $o(s_m)$: split at a fixed level, then bound the tail by $\sup_{i \ge i_0} J_{i+1}^{-2}$ times s_m. Since $J_i \to \infty$ and $s_m \to \infty$, the finite initial sum and $\bar\beta$ disappear after division by s_m, uniformly in $N \ge m$ and p.

The same level choices give $\log \ell_m \to \infty$ and $\log \ell_m / s_m \to 0$. The physical periods L_p are strictly increasing multiples of four, and every positive integer divides L_p for infinitely many p.

Endpoint tuning varies the prescribed coupling at level zero. Section 14, 14.3 proves that there is a width $w_{\mathrm{tun}} > 0$, independent of N, p, such that the unique bare coupling $\beta_{N,p}(s)$ within the specified recursion that reaches $\bar\beta + w_{\mathrm{tun}} s$ satisfies $\beta_{N,p}(0) = \beta_{N,p}$ and extends to a function $b_{N,p}$, holomorphic on $|z| < 1$, with

$$b_{N,p}(s) = \beta_{N,p}(s) \qquad (-1 < s < 1)\,. \tag{3.15}$$

The real endpoint parameter ranges over $s \in [-1, 1]$. This analyticity is used in the full ultraviolet limit; it does not assert a common analyticity width for all correlation functions.

3.5 Local fields

Local fields are normalized along the same trajectories. Their polynomial labels specify the continuum Taylor coefficients, while transfer to a fixed reference level determines the insertion at the fine cutoff.

The labels use radial, or Fock–Schwinger, gauge: the condition in the next definition sets the radial component of the connection to zero.

Definition 3.7 (Polynomial labels)**.** Let $\mathcal{J}$ consist of formal $\mathfrak{g}$-valued arrays $\mathcal{A}_a(x) = \sum_{k\ge 0} \mathcal{A}_{k,a}(x)$, with $\mathcal{A}_{k,a}$ homogeneous of degree $k+1$ and

$$\sum_a x^a \mathcal{A}_{k,a}(x) = 0, \qquad V_d = \mathbb{C}[\mathcal{J}]^{\mathrm{Ad}(G)}_{\mathrm{weight}\le d}\,.$$

Each coordinate of $\mathcal{A}_k$ has bidegree $(k+2,1)$: engineering weight and number of polynomial factors. Fix bihomogeneous bases, the supremum coefficient norm, conjugation and the induced H_4 action. These labels describe invariant curvature polynomials; curvature has weight two and a covariant derivative adds one.

Write $\mathbf{1}$ for the constant polynomial one. Section 10, 10.2 constructs a local section $P_d^{\mathrm{loc}}[\mathbf{c}](y;U)$, linear in $\mathbf{c}\in V_d$ and polynomial in the real and imaginary parts of link matrix entries on a fixed finite stencil $\Sigma_d \subset \mathbb{Z}^4$. It is gauge invariant, commutes with complex conjugation, is H_4 covariant, and sends the polynomial unit to one.

Its Taylor normalization compares lattice holonomies with a smooth connection jet. For $\mathsf{j} = (\mathcal{A}_k)_k \in \mathcal{J}$, put $\mathcal{A}^{\le d} = \sum_{k+2\le d} \mathcal{A}_k$ and let $U(h_{\mathrm{lat}},\mathsf{j})$ be its test holonomies at spacing h_{lat}. On a positive link (y,a), with $y\in\mathbb{Z}^4$, these are defined by

$$U(h_{\mathrm{lat}},\mathsf{j})(y,a) = u(1), \qquad u'(t) = u(t) h_{\mathrm{lat}} \mathcal{A}_a^{\le d}(h_{\mathrm{lat}}(y+te_a)), \qquad u(0) = 1_G\,. \tag{3.16}$$

The jet dilation is $\mathrm{sc}_{h_{\mathrm{lat}}}\mathsf{j} = (h_{\mathrm{lat}}^{k+2}\mathcal{A}_k)_k$. For every $\mathbf{c}\in V_d$ and $0\le q\le d$, 10.2 proves

$$\frac{d^q}{dh_{\mathrm{lat}}^q} P_d^{\mathrm{loc}}[\mathbf{c}](0;U(h_{\mathrm{lat}},\mathsf{j}))\Big|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q}\mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}}\mathsf{j})\Big|_{h_{\mathrm{lat}}=0}\,. \tag{3.17}$$

Thus a homogeneous label of engineering weight w specifies the coefficient at order h_{lat}^w in this expansion.

Choose the reference level $m_{\mathrm{ref}}(d)$ with $J_{m_{\mathrm{ref}}(d)} \ge d+3$ and $\mathfrak{q}_{J_m} \ge d+1$, $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$, as in Section 14, 14.5. For a basis vector $b_{d,i}$ of bidegree (w_i,k_i) define the factor-count scaling by

$$\mathsf{D}_t^{\mathrm{fac}} b_{d,i} = t^{k_i} b_{d,i}\,.$$

This rescales each polynomial factor in units of the small-field threshold. For $N \ge m_{\mathrm{ref}}(d)$ and terminal input $\mathbf{v}\in V_d$, the marked density at the origin is initialized and evolved by

$$H^{\mathrm{ins}}_{N,p,N;d}(\mathbf{v};U) = \mathrm{dens}\,\rho_{N,p,N}(U)\, P_d^{\mathrm{loc}}[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathbf{v}](0;U), \tag{3.18}$$

$$H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v}) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H^{\mathrm{ins}}_{N,p,m+1;d}(\mathbf{v}) \quad (m_{\mathrm{ref}}(d)\le m < N)\,. \tag{3.19}$$

For fixed N,p,d, the transfer construction chooses coefficient vectors $\mathbf{x}_m(\mathbf{v})$ and regular and forced remainders $R^{\mathrm{reg}}_{m,i}(\mathbf{v})$ and $Q^{\mathrm{for}}_{m,i}(\mathbf{v})$. All are linear in $\mathbf{v}$. Write $\mathbf{x}_m = x_{m,0}\mathbf{1} + \mathbf{x}_m^\circ$, and abbreviate

$\rho_{N,p,m}$ by ρ_m. The regular remainders have supports $Z_{m,i}$ containing the origin. Define their sum and the forced remainder by

$$\begin{aligned}\mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) &= \sum_i \mathbf{1}_{Z_{m,i}\ \mathrm{healthy}} R^{\mathrm{reg}}_{m,i}(\mathbf{v};U),\\ \mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U) &= e^{v_{\mathrm{vac}}|\Lambda| - P^{\mathrm{path}}_{N,m}(\beta(\rho_m),U) - \sum_i \mathbf{1}_{X_i\ \mathrm{healthy}} E^{\mathrm{reg}}_i(U)}\\ &\quad\cdot \sum_{\mathcal{A}_{\mathrm{for}}\ \mathrm{eligible}} \sum_{i\in\mathcal{A}_{\mathrm{for}}} Q^{\mathrm{for}}_{m,i}(\mathbf{v};U) \prod_{j\in\mathcal{A}_{\mathrm{for}}\setminus\{i\}} K^{\mathrm{for}}_j(U)\,. \end{aligned} \tag{3.20}$$

All unmarked quantities in this formula are taken from ρ_m, and eligibility is the disjointness and exact flag coverage in Definition 3.2. The marked density has the exact decomposition

$$\begin{aligned}H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v};U) = \mathrm{dens}\,\rho_m(U)\left[x_{m,0}(\mathbf{v}) + \mathbf{1}_{\Sigma_d\ \mathrm{healthy}} P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_m^{-1}}\mathbf{x}^{\circ}_m(\mathbf{v})](0;U)\right]\\ + \mathrm{dens}\,\rho_m(U)\mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) + \mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U)\,.\end{aligned} \tag{3.21}$$

Here the stencil Σ_d is placed at the origin on the level-m torus. If $10\,\mathrm{dist}(x,0) < L_pD_m$ for every $x \in Z_{m,i}$, the regular remainder $R^{\mathrm{reg}}_{m,i}$ has zero radial-jet Taylor coefficients through order d; the forced remainders vanish outside their unmarked eligibility events. Their format and bounds are proved in Section 14, 14.7–14.8. We use the fixed decomposition selected there. It supplies the invertible terminal map of 14.5:

$$\mathbf{x}_N(\mathbf{v}) = \mathbf{v}, \qquad \mathbf{x}_{m_{\mathrm{ref}}(d)}(\mathbf{v}) = \mathsf{Z}_{N,p,d}\mathbf{v}\,.$$

For $N < m_{\mathrm{ref}}(d)$ choose invertible maps to define the insertions; transfer normalization is required only at and above the reference level.

Definition 3.8 (Renormalized insertions)**.** Put $\mathcal{T} = \bigsqcup_{d\ge 0} V_d$ and define, at every cutoff,

$$I_{N,p}((d,\mathbf{c}),y;U) = O^{(d)}_{N,p}(\mathbf{c},y;U) = a_N^{-d} P^{\mathrm{loc}}_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathsf{Z}^{-1}_{N,p,d}\mathbf{c}](y;U)\,.$$

This fixes the insertion along the trajectory. Its distinguished unit coefficient is $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, whose insertion equals one for $N \ge m_{\mathrm{ref}}(d)$. The scaling $\mathsf{D}^{\mathrm{fac}}_t$ counts factors, not engineering weight.

The inverse terminal map makes the transferred coefficient equal to the prescribed $\mathbf{c}$ at the reference level. Thus $\mathbf{c}$ is specified at the fixed physical scale $a_{m_{\mathrm{ref}}(d)}$, in that level's lattice units. The factor $\ell^d_m = (a_m/a_{m+1})^d$ uses the degree label of the whole space V_d; the terminal map allows mixing between its engineering weights. The factor a_N^{-d} converts the normalized lattice insertion to physical units.

The unit identity follows from $\mathbf{x}_m(a^d_N\mathbf{1}) = a^d_m\mathbf{1}$, proved with the transfer symmetries in Section 14, 14.6. This explains why the coefficient giving the constant insertion is $a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, even though V_d also contains polynomials of lower engineering weight.

The bare couplings of MT and these insertions now specify one Wilson family. Each insertion is continuous, gauge invariant and local on its fixed stencil. MT at $m = N$ gives the volume-uniform asymptotic slope $\kappa_G > 0$. These are the lattice inputs to the continuum limits.

4 The continuum limit

We first remove the lattice spacing at each fixed volume, then let the volume increase along one refinement. Throughout, the Wilson family and insertions are those of Section 3; all couplings $\beta_{N,p}$ and insertion maps are chosen before the increasing sequence of volume indices is prescribed. Constants may depend on the fixed group, its realization, the penalty and the construction choices of Section 9. Additional dependence is stated below.

For positions in $(\mathbb{R}^4)^n$, set

$$\mathrm{Conf}_n = \{x : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{Gen}_n = \{x : x_i^a \neq x_j^a \text{ for } i \neq j,\ a = 1, \dots, 4\},$$
$$\mathrm{NoHit}_{L,n} = \{x : x_i - x_j \notin L\mathbb{Z}^4 \text{ for } i \neq j\}.$$

The last domain excludes collisions on a torus of physical side L. A separated test is a smooth function on $(\mathbb{R}^4)^n$ with compact closed support in Conf_n. It is p-admissible if its support lies in $((-L_p/2, L_p/2)^4)^n$. Write $\mathrm{sep}_\infty(x) = \min_{i\neq j} |x_i - x_j|_\infty$, with value $+\infty$ for $n \leq 1$.

For a finite list $\boldsymbol{l}$ of pairs $(i, a) \in \{1, \dots, n\} \times \{1, \dots, 4\}$, define

$$\Delta_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} a_N^{-1}(\tau_{i,a} - 1), \qquad \partial_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} \partial_{x_i^a}.$$

Here $\tau_{i,a}$ shifts slot i forward by one lattice link. These differences measure derivatives in physical units. Torus separation is the minimum, over distinct slots, of $\min_{z\in L_p\mathbb{Z}^4} |\hat{y}_i - \hat{y}_j + z|_\infty$, with value $+\infty$ for $n \leq 1$. An element $v_i = (d_i, c_i) \in \mathcal{T}$ retains both the polynomial label and its degree; we call it a token.

Marked transfers along the trajectories of Theorem 3.6 give bounds independent of the volume. They control compactness for each fixed list of fields and derivatives.

Proposition 4.1 (UB. Uniform difference bounds)**.** *For every $v \in \mathcal{T}^n$, finite difference list $\boldsymbol{l}$ and $r > 0$, there are $B(v, \boldsymbol{l}, r) \geq 0$ and $N_0(v, \boldsymbol{l}, r)$ such that, for every $N \geq N_0$, every p and every lattice tuple of torus separation at least r,*

$$|\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v; y)| \leq B.$$

The proof is in Section 14, 14.7–14.11. Marked transfers control insertion coefficients and remainders. The chessboard estimate reflects an insertion into neighbouring blocks and bounds a mixed correlation by the product of the resulting reflected moments. Choosing an intermediate level whose physical blocks fit below the prescribed separation permits its use for the differentiated insertions. The reflected-moment constant in 14.10 is chosen before that level, although its cutoff threshold may depend on it. No common bound or cutoff threshold is asserted over unbounded labels or difference orders.

The common endpoint can be varied analytically. More precisely, 14.3 supplies a width $w_{\mathrm{tun}} > 0$, independent of N, p, such that for each $s \in [-1, 1]$ exactly one bare coupling $\beta_{N,p}(s)$ within the specified recursion reaches $\bar{\beta} + w_{\mathrm{tun}} s$ at level zero. It equals $\beta_{N,p}$ at $s = 0$ and extends holomorphically to $|s| < 1$.

This tuning gives convergence along the full ultraviolet sequence. The limiting kernel is periodic in each position and smooth away from periodic collisions, including configurations where positions share some coordinates.

Theorem 4.2 (UV. Fixed-volume limit)**.** *For every p, n and $v \in \mathcal{T}^n$, there is $M^{\mathrm{UV}}_{p,v} \in C^\infty(\mathrm{NoHit}_{L_p,n})$ such that, for every finite $\boldsymbol{l}$ and every p-admissible separated test f,*

$$\lim_{N\to\infty} a_N^{4n} \sum_{y\in\Lambda^n_{L_pD_N}} f(\hat{y})\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v;y) = \int f(x)\partial_{\boldsymbol{l}} M^{\mathrm{UV}}_{p,v}(x)\,dx\,.$$

For $\boldsymbol{l} = \varnothing$, the left side is $\lim_N C^{\mathrm{lat}}_{N,p}(v;f)$. Thus the inner limits in Definition 4.3 below exist at every admissible volume. Admissibility places the test support inside $\mathrm{NoHit}_{L_p,n}$, so the displayed integrand is compactly supported and integrable.

The proof is in Section 14, 14.19–14.20. At a fixed intermediate level, the vacuum scalar cancels between the correlation numerator and its partition function. Analytic tuning and the fixed-level transfer estimates make the normalized integrals of the transferred insertions Cauchy as N increases. For insertion sites obtained by repeated blocking from common finer sites, comparison with the original correlations has error bounded by a constant times $e^{-B_m/64}$. First choose the intermediate level m, then take both cutoffs large; the triangle inequality gives the Cauchy property for the original correlations. UB extends convergence from the dense set of these sites and identifies every derivative. 14.20 also treats multilinear interpolants evaluated at any sequence $x_N \to x \in \mathrm{NoHit}_{L_p,n}$, including when the points cross a boundary of the chosen fundamental cell. Thresholds may depend on p.

The following definition keeps the order of limits explicit. Write $C^{\mathrm{UV}}_p(v;f)$ for a fixed-volume smeared limit.

Definition 4.3 (Ordered convergence)**.** A Wilson family converges to K along $P : \mathbb{N} \to \mathbb{N}$ if, for every n, v and separated test f, the test is eventually $P(k)$-admissible and there are numbers $C^{\mathrm{UV}}_p(v;f)$ with

$$\begin{aligned} C^{\mathrm{lat}}_{N,p}(v;f) &\longrightarrow C^{\mathrm{UV}}_p(v;f) \quad (N\to\infty) && \text{at every admissible } p,\\ C^{\mathrm{UV}}_{P(k)}(v;f) &\longrightarrow \int f(x)K_n(v;x)\,dx\,. \end{aligned}$$

Values at inadmissible volumes are unrestricted.

A table assigns to each degree list $(d_i)_{i=1}^n$ and $x \in \mathrm{Conf}_n$ a complex multilinear function $S_n(c;x)$ of $c_i \in V_{d_i}$. For $v_i = (d_i, c_i)$ write also $S_n(v;x)$, and for the word $w = (v_i(x_i))_{i=1}^n$ write $S(w)$. Conjugation and the signed permutation action are those of the polynomial labels; a unit slot carries the distinguished coefficient $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$ of Definition 3.8. The following conditions specify the Euclidean properties of the table to be constructed.

Definition 4.4 (E1–E3. Euclidean symmetries)**.** E1 is simultaneous permutation symmetry, $S_0 = 1$ and deletion of a slot carrying $\mathbf{u}_d$. E2 is

$$S_n(hc;hx) = S_n(c;x) \quad (h\in H_4), \qquad S_n(\bar{c};x) = \overline{S_n(c;x)}\,.$$

E3 is translation invariance.

A word is positive-side for axis a when all its positions have positive ath coordinate. The product of words concatenates their lists, so reflection positivity tests a correlation with one reflected group of insertions.

Definition 4.5 (E4–E5. Reflection positivity)**.** E4 requires, for every axis a, positive-side separated words w_i and complex coefficients z_i,

$$\sum_{i,j} \bar{z}_i z_j S(\mathsf{s}_a^{\mathrm{refl}} w_i^* w_j) \geq 0\,.$$

The star reverses the word and applies conjugation to its labels; $\mathsf{s}_a^{\mathrm{refl}}$ reflects positions and acts on labels through H_4. E5 requires the same positivity for finite sums of words smeared against joint smooth compactly supported tests on positive-side separated configurations, and absolute integrability of every mixed pairing.

For a fixed label list, D_x^k denotes the kth Fréchet derivative in the positions, with the product Euclidean norm. The next bound controls approach to collision sets uniformly over translations and over configurations with large diameter.

Definition 4.6 (E6–E7. Euclidean regularity)**.** E6 is smoothness on Conf_n. E7 requires, for each n, d_i, c_i, k, constants $C(n,d,c,k) \geq 0$ and an integer $A(n,d,c,k) \geq 0$ such that

$$\|D_x^k S_n(c;x)\| \leq C r^{-A} \qquad (0 < r \leq 1,\ \mathrm{sep}_\infty(x) \geq r)\,.$$

One volume refinement will serve every field and all subsequent constructions.

Theorem 4.7 (IV. Infinite-volume limit)**.** *For every strictly increasing* $\mathsf{p} : \mathbb{N} \to \mathbb{N}$*, there are a strictly increasing* $\mathsf{q} : \mathbb{N} \to \mathbb{N}$ *and a multilinear table* S *such that the Wilson family converges to* S *along* $P = \mathsf{p} \circ \mathsf{q}$ *in the sense of Definition 4.3. The table satisfies E1–E7.*

The proof is in Section 15, Lemmas 15.1–15.5, using the ultraviolet estimates of Lemmas 14.13–14.18. The extraction gives local smooth volume limits on Conf_n from Proposition 4.1 and Lemma 14.20, as recorded in (15.1) (E6). On the periodic generic domain the fixed-volume kernels are the restrictions of $M_{p,v}^{\mathrm{UV}}$, retaining UV's full inner limit at every admissible volume. The chessboard root bounds give E7 at order zero; the corrected derivative maps of Lemma 15.4 give it at positive orders. Lattice reflection positivity for finite sums of smeared words passes to the limit, giving E5; E4 follows from E5 together with E1 and E6. A further volume refinement in Lemma 15.4 fixes the corrected coefficient maps, which adjust labels of different degrees so their correlations agree or differ by a positional derivative; ordered convergence persists. No uniqueness of the infinite-volume limit is asserted; fix p, its refinement q and the resulting table S from now on.

Write $v(x) = (v_1(x_1), \dots, v_n(x_n))$ for a spectator word. Two polynomial labels represent the same observable when no separated correlation distinguishes them.

Definition 4.8 (Observable fields)**.** Set

$$\begin{aligned}
\mathcal{N}_d(S) &= \big\{ c \in V_d : S_{n+1}(c(x_0), v(x)) = 0 \\
&\qquad\qquad \text{for every } n, v \text{ and } x \in \mathrm{Conf}_{n+1} \big\}, \\
F_d &= V_d/\mathcal{N}_d(S), \qquad \overline{[c]} = [\bar{c}], \qquad h[c] = [hc] \quad (h \in H_4), \\
\iota_d[c](v;x) &= S_{n+1}(c(x_0), v(x)), \qquad \mathcal{L} = \mathrm{span} \bigcup_d \iota_d F_d\,.
\end{aligned}$$

The maps ι_d give the responses to all spectators. Choose conjugation-invariant quotient norms.

E2 makes $\mathcal{N}_d(S)$ invariant under conjugation and H_4, so both displayed actions on F_d are well defined. Permutation symmetry makes correlations independent of representatives in every slot: for $\boldsymbol{A} = ([c_1], \dots, [c_n])$, put $S_n(\boldsymbol{A}; x) = S_n(c; x)$. The unit field is $\mathbf{1}_d = [\mathbf{u}_d]$, with the normalization of Definition 3.8.

To state the dimension of F_d, introduce the free comparison space.

Definition 4.9 (Free Maxwell comparison)**.** For $d_{\mathfrak{g}}$ independent colours, take

$$\mathbb{E}A_a^\alpha(x)A_b^\beta(y) = \delta_{\alpha\beta}\delta_{ab}\Gamma(x-y), \qquad \Gamma(x) = \frac{1}{4\pi^2|x|^2}, \qquad F_{ab} = \partial_a A_b - \partial_b A_a\,.$$

Let M_d^{Max} consist of complex $\mathrm{Ad}(G)$-invariant Wick polynomials in curvature jets of weight at most d, modulo antisymmetry, the linear Bianchi identities and $\sum_a \partial_a F_{ab} = 0$. Their separated Wick correlations are S^{Max}.

The constant one is the Maxwell unit, coefficient conjugation gives its real structure, and inclusions retain the same polynomial at a larger weight bound. Free field differentiation is $\partial_a^{\mathrm{Max}}(:P:) =: \partial_a P :$ and the natural rotation action is $R_d^{\mathrm{Max}}(h)(:P:) =: hP :$, where the jet action is determined by

$$hF_{ab}^\alpha = \sum_{i,j} h_{ia}h_{jb}F_{ij}^\alpha, \qquad h\partial_a = \sum_i h_{ia}\partial_i h \quad (h \in O(4))\,.$$

These operations preserve the defining relations and give the structures used in the comparison below.

The quotient fields admit inclusions that preserve correlations and derivatives that act as positional derivatives. Their dimensions equal those of the Maxwell spaces.

Proposition 4.10 (FC. Field calculus)**.** *For the table S of Theorem 4.7, there are complex linear maps $\jmath_{d,e} : F_d \to F_e$ for $d \le e$ and $\partial_a : F_d \to F_{d+1}$ for every d, a such that, for every $A \in F_d$,*

$$\iota_e \circ \jmath_{d,e} = \iota_d, \qquad \iota_{d+1}(\partial_a A) = \partial_{x_0^a}\iota_d(A),$$
$$\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}\,.$$

The proof is in Section 15, 15.4, using the corrected coefficient maps chosen with S. Each ι_d is injective by Definition 4.8, so the inclusions are unique and injective, compose as $\jmath_{e,f}\jmath_{d,e} = \jmath_{d,f}$ and satisfy $\jmath_{d,d} = \mathrm{id}$. By E1 they carry $\mathbf{1}_d$ to $\mathbf{1}_e$. Their images are nested, hence $\mathcal{L} = \bigcup_d \iota_d F_d$. These inclusions are characterized by correlations; they are not defined by the polynomial embedding $V_d \subset V_e$, since (d, c) and (e, c) have different insertion normalizations.

Permutation symmetry gives, for every arity n, degree list (d_i), $\boldsymbol{A} \in \prod_{i=1}^n F_{d_i}$, $x \in \mathrm{Conf}_n$, slot i and axis a,

$$\partial_{x_i^a} S_n(\boldsymbol{A}; x) = S_n(A_1, \dots, \partial_a A_i, \dots, A_n; x)\,.$$

E2 and the injectivity of the response maps imply that inclusions and derivatives commute with conjugation, that inclusions commute with H_4, and that

$$h\partial_a = \sum_b h_{ba}\partial_b h \qquad (h \in H_4)\,.$$

5 Stress and rotations

We compare the fields of Section 4 with Maxwell fields at small distances, then normalize a conserved tensor by its shell fluxes. The volume refinement and the table S remain fixed throughout. Subsequences taken below concern only comparison radii.

The comparison permits the basis of local fields to vary with the radius. This accommodates mixing among fields in a fixed finite-dimensional space.

Definition 5.1 (Scaling frames)**.** A scaling sequence consists of $r_k > 0$, $r_k \to 0$, and complex linear isomorphisms

$$\mathsf{U}^{\mathrm{sc}}_{k,d} : M_d^{\mathrm{Max}} \longrightarrow F_d \,.$$

It is stable if, for every $D \in \mathbb{N}$, these maps eventually preserve conjugation, H_4, units and inclusions exactly in all degrees $d \le e \le D$. Scaling dimensions are included in the maps: an ordinary comparison insertion is $\mathsf{U}^{\mathrm{sc}}_{k,d} u$ at $r_k x$, with no further factor r_k^d.

The frames compare fields without identifying their null ideals.

After rescaling positions and choosing these frames, the interacting correlations approach their Maxwell counterparts. The first comparison measures the error by its integral on separated compact sets; one sequence works for every fixed list of fields.

Proposition 5.2 (MC. Maxwell comparison)**.** *There is one stable scaling sequence such that, for every arity n, degree list (d_i) and $u_i \in M_{d_i}^{\mathrm{Max}}$,*

$$S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i} u_i; r_k \cdot\,) \longrightarrow S_n^{\mathrm{Max}}(u; \cdot) \quad \textit{in } L^1_{\mathrm{loc}}(\mathrm{Conf}_n) \,.$$

The proof is in Section 16, 16.2. The lattice comparison stops near the requested physical scale and compares the resulting correlations with the free Gaussian ones. Finitely many separating correlations make the comparison maps injective; the upper bound $\dim_{\mathbb{C}} F_d \le \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$ from Proposition 4.10 makes them isomorphisms. Finite-dimensional corrections impose the eventual equalities of Definition 5.1.

The integral norm retains uniformity over tests. Indeed, write $g_k(x)$ for the difference of the two smooth kernels in Proposition 5.2. For compact $\mathcal{K} \subset \operatorname{int} \mathcal{K}' \subset \mathcal{K}' \subset \mathrm{Conf}_n$, smooth approximation to $\overline{g_k}/|g_k|$ with a cutoff equal to one on $\mathcal{K}$ gives

$$\int_{\mathcal{K}} |g_k(x)|\, dx \le \sup_{\substack{f \in C_c^\infty,\ \operatorname{supp} f \subset \mathcal{K}' \\ \|f\|_\infty \le 1}} \left| \int f(x) g_k(x)\, dx \right| .$$

Conversely, the tested error on $\mathcal{K}$ is bounded by $\|f\|_\infty \int_{\mathcal{K}} |g_k|$. Thus local integral convergence is equivalent to convergence uniform over the sup-norm unit ball of tests on each fixed compact set. This comparison is used to construct and normalize the conserved current before the stress tensor is available.

To read translation and angular charge, integrate a conserved symmetric tensor through a shell surrounding just one insertion. Let $A \in F_d$, let $\boldsymbol{B}$ be a list of n spectator fields at x, and take $(x_0, x) \in \mathrm{Conf}_{n+1}$.

Definition 5.3 (Shell fluxes)**.** For $J_{ab} = J_{ba} \in F_4$ with $\sum_a \partial_a J_{ab} = 0$ and $K_{\mathrm{rot}} \in \mathfrak{so}(4)$, choose a real shell cutoff $\chi \in C_c^\infty(\mathbb{R}^4)$ equal to one near x_0 and supported away from spectators. Define

$$\mathcal{P}_b^J = -\sum_a \int \partial_a \chi(y) S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy,$$
$$\mathcal{Y}_{K_{\mathrm{rot}}}^J = -\sum_{a,b} \int \partial_a \chi(y)(K_{\mathrm{rot}}(y - x_0))_b S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy\,.$$

These integrals measure the flux without meeting an insertion point.

The fluxes are independent of the shell cutoff. For two choices, their difference vanishes near every insertion; integration by parts and Proposition 4.10 remove the divergence term. In the angular flux the additional term is $\sum_{a,b}(K_{\mathrm{rot}})_{ba} J_{ab} = 0$, since K_{rot} is skew-symmetric and J is symmetric.

A field is real when it is fixed by conjugation. A family $J_{ab} \in F_4$ is H_4-tensorial when

$$hJ_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij} \qquad (h \in H_4)\,.$$

It is conserved when $\sum_a \partial_a J_{ab} = 0$ in F_5 for every b, and centered when every component has zero one-point function. Section 16, 16.3 proves that the centered real symmetric H_4-tensorial conserved tensors of degree four form a real line. Along a refinement of the sequence in Proposition 5.2, positive multiples of the rescaled nonzero current approach the free stress line in the degree-four frames. This identifies the current needed for the shell laws.

For a nonzero member J of this line, 16.4 constructs complex linear charge maps $\mathsf{p}_b : F_d \to F_{d+1}$ and $\mathsf{Y}_{K_{\mathrm{rot}}} : F_d \to F_d$ satisfying, for every degree, field, spectator list, separated configuration and shell cutoff χ,

$$\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^J = S((\mathsf{Y}_{K_{\mathrm{rot}}} A)(x_0), \boldsymbol{B}(x))\,.$$

The angular map is real linear in K_{rot}.

Section 16, 16.5 identifies these maps with differentiation and rotations. It constructs continuous complex linear representations R_d of $O(4)$, extending H_4, fixing units and commuting with conjugation and inclusions. For a real $c_\star \neq 0$ common to every degree, the generators and charge maps satisfy

$$\mathsf{L}_d(K_{\mathrm{rot}}) = \frac{d}{dt} R_d(e^{tK_{\mathrm{rot}}})\Big|_{t=0}, \qquad \mathsf{p}_b = c_\star \partial_b, \qquad \mathsf{Y}_{K_{\mathrm{rot}}} = c_\star \mathsf{L}_d(K_{\mathrm{rot}})\,.$$

The proof controls exterior angular flux before extending the rotation identity to all separated configurations. The resulting table has full covariance: for every arity n, degree list (d_i), $\boldsymbol{A} \in \prod_i F_{d_i}$, $x \in \mathrm{Conf}_n$ and $h \in O(4)$, 16.6 gives

$$S_n(R_{d_i}(h) A_i; hx) = S_n(\boldsymbol{A}; x)\,.$$

For these representations, $O(4)$-tensorial means $R_4(h) J_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij}$ for every $h \in O(4)$, with the same index convention as the discrete action above.

The common constant $c_\star$ fixes the normalization of the stress tensor by its action on fields.

Theorem 5.4 (ST. Stress tensor). *There is a real symmetric $O(4)$-tensorial family $T_{ab} \in F_4$ with $\sum_a \partial_a T_{ab} = 0$ such that, for every degree, field, spectator list, separated configuration and shell cutoff of Definition 5.3, every b and every $K_{\mathrm{rot}} \in \mathfrak{so}(4)$,*

$$\mathcal{P}_b^T = \partial_{x_0^b} S(A(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^T = S((\mathsf{L}_d(K_{\mathrm{rot}})A)(x_0), \boldsymbol{B}(x)) .$$

The proof is in Section 16, 16.5–16.6. Divide the current J by $c_\star$ and average its tensor transform over $O(4)$ with probability Haar measure. Covariance and the transformation laws for derivatives and generators give the same shell actions for each transformed current; integrating these identities preserves both actions. Symmetry, reality, centering and conservation also pass through this finite-dimensional integral. Fix this averaged normalized tensor for all subsequent uses; it is centered, since J is centered and R_4 preserves zero one-point values.

Write $\mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}}) = \frac{d}{dt} R_d^{\mathrm{Max}}(e^{tK_{\mathrm{rot}}})\big|_{t=0}$ for the free rotation generators. The free stress T^{Max} is normalized by the same shell laws, with $\partial_a^{\mathrm{Max}}$ and $\mathsf{L}_d^{\mathrm{Max}}$, in 16.1. Their sign gives

$$T_{ab}^{\mathrm{Max}} = -\sum_{c,\alpha} : F_{ac}^\alpha F_{bc}^\alpha : + \frac{\delta_{ab}}{4} \sum_{i,j,\alpha} : F_{ij}^\alpha F_{ij}^\alpha : .$$

The final comparison includes any chosen stress insertions. For a free list $u_1, \dots, u_n$ with $u_i \in M_{d_i}^{\mathrm{Max}}$, a stress-slot choice is a subset $I \subset \{1, \dots, n\}$ with component pairs (a_i, b_i) for $i \in I$. Put

$$A_i^0 = \begin{cases} T_{a_i b_i}^{\mathrm{Max}}, & i \in I, \\ u_i, & i \notin I. \end{cases}$$

The shell laws fix the stress normalization, so its comparison insertion needs only the engineering factor r_k^4. In the next limit D_x^j differentiates the whole function $x \mapsto S_n(\boldsymbol{A}_k; r_k x)$, including the rescaled positions.

Proposition 5.5 (SC. Differentiated short-distance comparison). *There is one stable scaling sequence such that, for every fixed free list and stress-slot choice, every derivative order j and compact $\mathcal{K} \subset \mathrm{Conf}_n$, with $A_{k,i} = r_k^4 T_{a_i b_i}$ on I for the fixed tensor of Theorem 5.4 and $A_{k,i} = \mathsf{U}_{k,d_i}^{\mathrm{sc}} u_i$ elsewhere,*

$$D_x^j S_n(\boldsymbol{A}_k; r_k x) \longrightarrow D_x^j S_n^{\mathrm{Max}}(\boldsymbol{A}^0; x) \quad \text{uniformly on } \mathcal{K} .$$

The proof is in Section 16, 16.7. With $I = \varnothing$ and $j = 0$, this gives Proposition 5.2 in uniform form along the same sequence: an integral against f is bounded by $\|f\|_\infty$ times the volume of its compact support times the uniform error. The earlier comparison is needed first to construct the tensor. Along the sequence of Proposition 5.5, for every fixed d, a, K_{rot}, the transported infinitesimal actions satisfy the operator-norm limits

$$r_k (\mathsf{U}_{k,d+1}^{\mathrm{sc}})^{-1} \partial_a \mathsf{U}_{k,d}^{\mathrm{sc}} \longrightarrow \partial_a^{\mathrm{Max}},$$
$$(\mathsf{U}_{k,d}^{\mathrm{sc}})^{-1} \mathsf{L}_d(K_{\mathrm{rot}}) \mathsf{U}_{k,d}^{\mathrm{sc}} \longrightarrow \mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}}) .$$

6 Phases and reconstruction

Keep the table and quotient fields of Sections 4–5. Compression to zero energy separates the mixed correlations into phases. The slab estimate of 17.1 makes word insertion bounded after a positive time translation. The coordinate reflection $\mathsf{s}_a^{\mathrm{refl}}$ acts on labels through H_4.

Definition 6.1 (OS space and compression)**.** For words at distinct points in $x^a > 0$, let $\mathcal{H}_a^{\mathrm{OS}}$ be the completion of their complex span modulo the null space of

$$\langle w, v\rangle_a = S(\mathsf{s}_a^{\mathrm{refl}} w^* v), \qquad e^{-tH_a^{\mathrm{OS}}}[v] = [\tau_{te_a} v] \quad (t \geq 0)\,.$$

Let Ω_a^{OS} be the empty-word vector and P_a^0 the projection onto $\ker H_a^{\mathrm{OS}}$. For $h_{\mathrm{sl}} > 0$ and w with times in $(0, h_{\mathrm{sl}})$, define, on positive-time words v,

$$A_w v = w\, \tau_{h_{\mathrm{sl}} e_4} v, \qquad M^0(w) = P_4^0 \mathsf{B}_{h_{\mathrm{sl}}}(w)|_{\ker H_4^{\mathrm{OS}}}\,,$$

where $\mathsf{B}_{h_{\mathrm{sl}}}(w)$ extends A_w boundedly on OS classes, by 17.1.

Translate any separated word into a slab to define its compression; 17.2 proves independence of width and placement. The compressed operators generate the compression algebra

$$\mathfrak{A}_0 = C^*(M^0(w) : w \text{ separated})\,.$$

Section 17, 17.2 proves that this algebra is separable, commutative and unital, and that Ω_4^{OS} is cyclic and separating. Its vacuum state is

$$a \longmapsto \langle \Omega_4^{\mathrm{OS}}, a\Omega_4^{\mathrm{OS}}\rangle, \qquad S(w) = \langle \Omega_4^{\mathrm{OS}}, M^0(w)\Omega_4^{\mathrm{OS}}\rangle\,.$$

The second identity uses translation invariance. The measure below represents this state on the character space; the separating property gives full support. A one-dimensional mixed ground space would make $M^0(w) = S(w)\mathrm{Id}$ and leave a single character.

Characters separate the mixed correlations without changing which fields are detectable. Call a character ξ a phase and S^ξ its correlation table.

Theorem 6.2 (PH. Phase decomposition)**.** *The algebra $\mathfrak{A}_0$ has a compact metric character space X with a full-support probability measure ν such that, for every separated word w,*

$$S^\xi(w) = \xi(M^0(w)), \qquad S(w) = \int_{\mathsf{X}} S^\xi(w)\, d\nu(\xi)\,.$$

Each phase satisfies E1–E7 and has exactly the null spaces $\mathcal{N}_d(S^\xi) = \mathcal{N}_d(S)$. The identities of FC, $O(4)$ covariance and ST hold with S^ξ in place of S, for the same maps and tensor T. For fixed w, $\xi \mapsto S^\xi(w)$ is continuous.

The proof is in Section 17, 17.1–17.3. Two positive coordinate directions give opposite orders to the middle two of four widely separated words. Rotation covariance identifies the limits, and the positive translation semigroups converge to the ground projections; the resulting identity makes the compressed operators commute.

For each phase apply Definition 6.1 to S^ξ, obtaining $\mathcal{H}_{4,\xi}^{\mathrm{OS}}$, $H_{4,\xi}^{\mathrm{OS}}$ and Ω_ξ^{OS}. Fix ξ_* in the full-measure set with one-dimensional time-axis OS ground space supplied by 17.3. This choice uses no trace condition. Theorem 6.15 will extend ground-space uniqueness to every phase.

Write $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{4n}; \mathbb{C})$ and $\mathcal{S} = \mathcal{S}_1$. The following definitions specify the relativistic theory.

Definition 6.3 (Relativistic fields)**.** Take a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$ and $\Omega \in \mathcal{D}$. Fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ are linear operators, complex bilinear in $(\ell, f) \in \mathcal{L} \times \mathcal{S}$. Include an antilinear map $\ell \mapsto \ell^*$, strongly continuous unitary translations $\mathsf{T}(z)$, $z \in \mathbb{R}^4$, and a Borel projection-valued measure E, with finite diagonal measures $\mu_\psi(B) = \langle \psi, \mathsf{E}(B)\psi\rangle$ for Borel $B \subset \mathbb{R}^4$.

Definition 6.4 (Poincaré action)**.** Take a topological group $\mathcal{P}$ with topological group isomorphism $\kappa_{\mathrm{P}} : \mathcal{P} \to SO^{+}(1,3) \ltimes \mathbb{R}^4$, writing $\kappa_{\mathrm{P}}(\gamma) = (\Lambda_\gamma, z_\gamma)$. Here $SO^{+}(1,3)$ preserves m, has determinant one and preserves $\overline{V}_{+}$. Include a strongly continuous unitary representation U with $\mathsf{U}(\kappa_{\mathrm{P}}^{-1}(1,z)) = \mathsf{T}(z)$ and linear bijections $\rho_{\mathrm{lab}}(\gamma)$ of $\mathcal{L}$. Set $(\gamma \cdot f)(x) = f(\Lambda_\gamma^{-1}(x - z_\gamma))$.

In group arguments we identify $\mathcal{P}$ with $SO^{+}(1,3) \ltimes \mathbb{R}^4$ through κ_{P}. The antilinear label map need not be involutive. The label action has no additional continuity, dimension or translation restriction. The spectral relation is imposed next.

Definition 6.5 (W0. Spectrum and vacuum)**.** For every $\psi \in \mathcal{H}$ and $z \in \mathbb{R}^4$, require

$$\langle \psi, \mathsf{T}(z)\psi \rangle = \int e^{i\mathbf{p}\cdot z}\, d\mu_\psi(\mathbf{p}), \qquad \mathsf{E}(\mathbb{R}^4 \setminus \overline{V}_{+}) = 0\,,$$

with Euclidean pairing. Every vector fixed by all translations belongs to $\mathbb{C}\Omega$.

Definition 6.6 (W1. Word domain and adjoints)**.** Require $\|\Omega\| = 1$, density of $\mathcal{D}$, and

$$\mathcal{D} = \operatorname{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1)\cdots\Phi(\ell_k, f_k)\Omega : k \geq 0\}\,.$$

For each $\ell \in \mathcal{L}$, $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ to $\mathcal{H}$ in norm. For $\psi, \chi \in \mathcal{D}$, require $\langle \Phi(\ell, f)\psi, \chi\rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi\rangle$.

Definition 6.7 (W2. Covariance)**.** Require $\rho_{\mathrm{lab}}(1) = 1$ and $\rho_{\mathrm{lab}}(\gamma\gamma') = \rho_{\mathrm{lab}}(\gamma)\rho_{\mathrm{lab}}(\gamma')$. For every $\gamma \in \mathcal{P}$, $\ell \in \mathcal{L}$, $f \in \mathcal{S}$ and $\psi \in \mathcal{D}$, require $\mathsf{U}(\gamma)\mathcal{D} \subset \mathcal{D}$, $\mathsf{U}(\gamma)\Omega = \Omega$, and

$$\mathsf{U}(\gamma)\Phi(\ell, f)\psi = \Phi(\rho_{\mathrm{lab}}(\gamma)\ell, \gamma \cdot f)\mathsf{U}(\gamma)\psi\,.$$

Definition 6.8 (W3. Locality)**.** For $f, h \in \mathcal{S}$ with $\mathsf{m}(x - y) < 0$ for every $x \in \operatorname{supp} f$, $y \in \operatorname{supp} h$, require $[\Phi(\ell, f), \Phi(\ell', h)]\psi = 0$ for all $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$.

Definition 6.9 (Q2. Wightman axioms)**.** A relativistic datum of Definitions 6.3–6.4 is a Wightman theory if it satisfies W0–W3.

Definition 6.10 (Faithful labels)**.** The labels are faithful if, for every nonzero $\ell \in \mathcal{L}$, some $f \in \mathcal{S}$ has $\Phi(\ell, f) \neq 0$.

Definition 6.11 (Bounded weak commutant)**.** A bounded operator B belongs to the weak commutant if, for every smeared field Φ and all $u, v \in \mathcal{D}$, one has $u \in \operatorname{Dom}\Phi^*$ and $\langle \Phi^* u, Bv\rangle = \langle u, B\Phi v\rangle$. Here Φ^* is the unbounded Hilbert-space adjoint.

Definition 6.12 (Hermitian scalar labels)**.** A label is Hermitian if $\ell^* = \ell$, and Lorentz scalar if $\rho_{\mathrm{lab}}(\Lambda, 0)\ell = \ell$ for every $\Lambda \in SO^{+}(1,3)$. These conditions give the adjoint and scalar covariance identities through W1 and W2.

For reconstruction use the response space $\mathcal{L}$ of Definition 4.8. Put $\mathbb{F} = \bigsqcup_d F_d$ and $\iota(d, A) = \iota_d A$.

Definition 6.13 (Boundary condition)**.** For a phase ξ, a relativistic datum $(\mathcal{H}_\xi, \mathcal{L}, \mathcal{D}_\xi, \Omega_\xi, \Phi_\xi, \ldots)$ of Definitions 6.3–6.4, linear maps $\iota_d : F_d \to \mathcal{L}$, and functions F_n^ξ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$, require, for every $n \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{F}^n$, real $h_1 < \cdots < h_n$ and $f_i \in \mathcal{S}$,

$$\lim_{r \downarrow 0} \int \mathsf{F}_n^\xi(\boldsymbol{A}; (x_i^4 + irh_i)_i, (\vec{x}_i)_i) \prod_i f_i(x_i)\, dx \\ = \langle \Omega_\xi, \Phi_\xi(\iota(A_1), f_1)\cdots\Phi_\xi(\iota(A_n), f_n)\Omega_\xi\rangle\,.$$

For this formula alone a nonintegrable function has integral zero.

This boundary condition does not assert positive-height integrability, joint-test convergence or complex spatial analyticity.

Definition 6.14 (Ordered-time continuation)**.** For a phase ξ, put $\mathcal{Z}_n = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n\}$. A continuation F_n^ξ is holomorphic there at each fixed real spatial configuration and satisfies

$$\mathsf{F}_n^\xi(\boldsymbol{A}; (it_i)_i, \vec{x}) = S_n^\xi(\boldsymbol{A}; (\vec{x}_i, t_i)_i) \qquad (t_1 < \cdots < t_n)\,.$$

Together with Definition 6.13, it connects Euclidean and real-time correlations.

For a theory at phase ξ in the next theorem, write $\mathcal{H}_\xi$, $\mathcal{D}_\xi$, Ω_ξ, Φ_ξ and T_ξ for its Hilbert space, word domain, vacuum, fields and translations. Define its energy generator by $\mathsf{T}_\xi(te_4) = e^{itH_\xi}$. Its OS comparison map means the bounded linear map

$$\mathsf{J}_\xi^{\mathrm{OS}} : \mathcal{H}_\xi \longrightarrow \mathcal{H}_{4,\xi}^{\mathrm{OS}}, \qquad \mathsf{J}_\xi^{\mathrm{OS}}\Omega_\xi = \Omega_\xi^{\mathrm{OS}}\,,$$

sending each smeared field-word vector to the norm limit of its heat-regularized positive-time OS vectors as the ordered heights tend to zero. Section 17, 17.5 constructs this map and proves its uniqueness.

Each character yields relativistic fields on the common response labels. The continuation and boundary condition specify their relation to the phase correlations.

Theorem 6.15 (RC. Reconstruction)**.** *For every $\xi \in \mathsf{X}$, there are a Wightman theory (Definition 6.9) on $\mathcal{L}$, faithful in the sense of Definition 6.10, and functions F_n^ξ satisfying Definitions 6.14 and 6.13 with the response maps ι_d of Definition 4.8. Its OS comparison map $\mathsf{J}_\xi^{\mathrm{OS}}$ is onto, and*

$$\ker H_\xi = \mathbb{C}\Omega_\xi, \qquad \ker H_{4,\xi}^{\mathrm{OS}} = \mathbb{C}\Omega_\xi^{\mathrm{OS}}\,.$$

Every bounded operator in the weak commutant of its fields (Definition 6.11) is scalar.

The proof is in Section 17, 17.4–17.7. Slab bounds control the analytic boundary values of these continuations. A low-energy bound uniform in ξ, continuity of the vacuum-subtracted heat pairings and full support of ν extend almost-everywhere ground-space uniqueness to every phase. Thus the earlier choice of ξ_* remains valid. 17.4–17.7 also give joint Schwartz distributions, phase averages and stress continuation.

The reconstructed label maps are common to all phases. In 17.4 the adjoint is induced by time reflection followed by conjugation:

$$(\iota_d A)^* = \iota_d\big(R_d(\mathsf{s}_4^{\mathrm{refl}})\overline{A}\big)\,.$$

The Lorentz label action is the continuation of R_d: spatial rotations retain their Euclidean action. For $i = 1, 2, 3$, the boost along e_i has generator $-i\mathsf{L}_d(K_{4i})$, where $K_{4i} = e_4 e_i^{\mathsf{T}} - e_i e_4^{\mathsf{T}}$ generates rotation in the (e_4, e_i) plane (17.4). In particular, a real $O(4)$-invariant field gives a Hermitian Lorentz scalar label. Retain the theory at ξ_* and suppress its phase subscript in $\mathcal{H}, \mathcal{D}, \Omega, \Phi$ and the other relativistic data.

7 Trace and products

The trace is centered in the phase already selected in Section 6. Its short-distance logarithm will distinguish the resulting theory from finite Wick comparisons.

Definition 7.1 (Centered trace)**.** For the tensor T fixed after Theorem 5.4 and $r > 0$, put

$$\tau = \sum_{a=1}^{4} T_{aa}, \qquad e_{\mathrm{tr}} = S^{\xi_*}(\tau), \qquad \Theta = \tau - e_{\mathrm{tr}}\mathbf{1}_4,$$
$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

The one-point value is position independent. Centering removes the vacuum contribution in the selected phase.

Adding a real multiple of $\delta_{ab}\mathbf{1}_4$ to T preserves its shell laws, the limit in Proposition 5.5 and its centered trace Θ; no uniqueness is needed. The fixed stress tensor is centered in the mixture, but a single-phase one-point value need not vanish. The next limit detects a nonzero centered trace and determines its power and logarithm.

Theorem 7.2 (TR. Trace law)**.** *The selected field Θ is nonzero, real and $O(4)$ invariant; its label $\iota_4\Theta$ is Hermitian and Lorentz scalar (Definition 6.12). For every $f \in \mathcal{S}$,*

$$\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0\,.$$

As $r \downarrow 0$ through all positive radii,

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0\,.$$

The proof is in Section 18, 18.1–18.3. To identify the amplitude, use the free scalar curvature square

$$\mathcal{O}_{F^2} = \frac{1}{4}\sum_{a,b,\alpha} :F^\alpha_{ab}F^\alpha_{ab}:, \qquad S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}, \mathcal{O}_{F^2}; -e_4/2, e_4/2) = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

Here $\alpha = 1, \dots, d_{\mathfrak{g}}$ indexes an orthonormal colour basis. At running coupling β, the scalar trace coefficient is $4d_2/\beta$, with $d_2 = -b_0/2$. Since $\kappa_G = 2b_0$, its leading coefficient at radius r is $4d_2/(\kappa_G \log(1/r)) = -1/\log(1/r)$. Squaring it and using the free correlation gives A_G. The comparison uses adjustable stopping levels and fourth-order information beyond Proposition 5.2; its estimates are uniform in the phase, although TR is stated only at the already selected ξ_*. For every prescribed sequence of positive radii tending to zero, the comparison holds along a subsequence. The positive spectral representation below makes G_Θ real and nonincreasing; a sandwich argument gives the limit on that subsequence. The subsequence criterion then gives the full-radius limit.

The same proof identifies centering on the word domain:

$$\Phi(\iota_4\Theta, f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}\left(\int f\right)\mathrm{Id}_{\mathcal{H}}\,.$$

It also constructs the unique positive tempered-growth spectral measure μ_Θ, supported in $\overline{V}_+$, for which

$$d\mu_{\Phi(\iota_4\Theta,f)\Omega} = |\widehat{f}_+|^2\, d\mu_\Theta, \qquad G_\Theta(r) = \int e^{-r\mathbf{p}^4}\, d\mu_\Theta(\mathbf{p}) \quad (r > 0)\,.$$

Here $\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\, dx$. The measure's Lorentz invariance and its high-energy limits against compactly supported continuous tests and Schwartz tests are stated in 18.2–18.3.

To define the comparison class, fix $m_{\mathrm{sh}} \geq 0$, integers $j_h, k_E \geq 0$ and continuous real h_{mom} on $\mathbb{R}^3$ with $h_{\mathrm{mom}}(t\mathbf{k}) = t^{j_h} h_{\mathrm{mom}}(\mathbf{k})$ for $t > 0$.

Definition 7.3 (Finite Wick profiles)**.** Put $\omega_{m_{\mathrm{sh}}}(\mathbf{k}) = \sqrt{m_{\mathrm{sh}}^2 + |\mathbf{k}|^2}$ and, for $r > 0$,

$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \int_{\mathbb{R}^3} \frac{\omega_{m_{\mathrm{sh}}}(\mathbf{k})^{k_E} h_{\mathrm{mom}}(\mathbf{k})}{2\omega_{m_{\mathrm{sh}}}(\mathbf{k})} e^{-r\omega_{m_{\mathrm{sh}}}(\mathbf{k})}\, d^3\mathbf{k}\,.$$

A finite Wick profile is a function

$$W_{\mathrm{Wick}}(r) = \sum_{i=1}^{N} c_i \prod_{j=1}^{n_i} K_{m_{ij},k_{ij},h_{ij}}(r) \quad (r > 0), \qquad c_i \in \mathbb{C}\,,$$

with finitely many terms and factors of the preceding kind; empty products are one. A zero denominator at zero momentum gives integrand zero.

The momentum integrals converge absolutely for $r > 0$. For a free scalar field of mass m_{sh}, the time-axis two-point function is $(2\pi)^{-3} K_{m_{\mathrm{sh}},0,1}(r)$. Derivatives give momentum factors, and Wick contractions give finite sums of products at the same radius.

Such profiles cannot reproduce the factor $r^{-8}\log^{-2}(1/r)$ in Theorem 7.2.

Proposition 7.4 (NF. Finite-profile exclusion)**.** *For every finite Wick profile W_{Wick} and every $\delta > 0$, there is $0 < r < \delta$ such that*

$$G_\Theta(r) \neq W_{\mathrm{Wick}}(r)\,.$$

The proof is in Section 18, 18.4. The mass-shell kernels have expansions to every finite order in integer powers of r and nonnegative powers of $\log r$. Finite products and sums therefore give, to bounded error, a finite sum of monomials $r^a(\log r)^k$ with $a \in \mathbb{Z}$ and $k \in \mathbb{N}$. Multiplication by $r^8 \log^2(1/r)$ makes the terms with $a > -8$ vanish. The remaining nonzero leading term diverges, so any finite limit is zero. The positive limit in Theorem 7.2 therefore excludes equality on every interval $(0, \delta)$. Only that limit and its positivity enter this exclusion. 18.6 identifies polynomial mass-shell comparisons with their Laplace profiles and applies 18.4; 18.9–18.10 give the Fock countertests. Their comparison classes are defined there.

The product expansion is stated for the Euclidean tables on their common quotient fields. Write $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$ for a state index (a phase or the mixture), with $S^{\mathrm{mix}} = S$, and retain degree tags by $\mathbb{F} = \bigsqcup_d F_d$, $\deg(d, A) = d$. The underlying prescription consists of the original fields and their correlation functionals. From it, Section 19, 19.4 constructs finite channel sets Ch_j, real fields $O_{j,\lambda} \in F_j$ and, for every finite I, functions

$$c_{j,\lambda} : \mathbb{F}^I \times (\mathbb{R}^4)^I \times \mathbb{R}^4 \longrightarrow \mathbb{C}\,.$$

Fix these choices once, before spectators, state indices and truncations. The fields form complex bases of conjugation-invariant complements of F_{j-1} in F_j, with $F_{-1} = 0$. The degree-zero vector spans the unit line; it need not equal the unit. For finite cluster and spectator index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$, take positions $x \in (\mathbb{R}^4)^{I_{\mathrm{cl}}}$, $y \in (\mathbb{R}^4)^{I_{\mathrm{sp}}}$ and expansion center $o \in \mathbb{R}^4$.

Definition 7.5 (Product remainder)**.** For finite index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$ and $D \in \mathbb{N}$, cluster and spectator lists $\boldsymbol{A} \in \mathbb{F}^{I_{\mathrm{cl}}}$, $\boldsymbol{B} \in \mathbb{F}^{I_{\mathrm{sp}}}$, put $|\boldsymbol{A}|_{\deg} = \sum_i \deg A_i$ and

$$\mathcal{R}_D^\zeta = S^\zeta(\boldsymbol{A}, x; \boldsymbol{B}, y) - \sum_{j=0}^{D} \sum_{\lambda \in \mathrm{Ch}_j} c_{j,\lambda}(\boldsymbol{A}; x, o) S^\zeta(O_{j,\lambda}(o), \boldsymbol{B}(y))\,.$$

Use a fixed enumeration of $I_{\mathrm{cl}} \sqcup I_{\mathrm{sp}}$ for the concatenated correlations. A pair (x, o) is an $(r, M_{\mathrm{geo}}, \eta)$-cluster if $|x_i - o| \le M_{\mathrm{geo}} r$ and $|x_i - x_{i'}| \ge \eta r$ for distinct slots. Spectators are δ_{sp}-clear if $|o - y_j| \ge \delta_{\mathrm{sp}}$ and $|y_j - y_{j'}| \ge \delta_{\mathrm{sp}}$ for distinct slots. The remainder measures the error through degree D.

If $M_{\mathrm{geo}} r < \delta_{\mathrm{sp}}$, the concatenated positions are separated: $|x_i - y_j| \ge \delta_{\mathrm{sp}} - M_{\mathrm{geo}} r > 0$. Thus the remainder is defined in the small-radius range used below.

One field expansion approximates a cluster in every phase and in the mixture. Its coefficients are independent of all spectators.

Theorem 7.6 (OP. Operator product expansion)**.** *For every finite list $\boldsymbol{A}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\lambda \in \mathrm{Ch}_j$, some $r_0 > 0$ gives, for $0 < r < r_0$ and every $(r, M_{\mathrm{geo}}, \eta)$-cluster,*

$$|c_{j,\lambda}(\boldsymbol{A}; x, o)| \le r^{j - |\boldsymbol{A}|_{\deg} - \varepsilon} .$$

For every pair of finite lists $\boldsymbol{A}, \boldsymbol{B}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$ and $D \in \mathbb{N}$, some $r_1 > 0$ gives, for $0 < r < r_1$, such clusters, δ_{sp}-clear spectators and all state indices ζ,

$$|\mathcal{R}_D^{\zeta}| \le r^{D+1-|\boldsymbol{A}|_{\deg} - \varepsilon} .$$

The proof is in Section 19, 19.4–19.5. A fixed left inverse recovers $A \in F_j$ from finitely many values $S(A(0), B)$ against chosen probe words B. Apply it to cluster correlations, average under rotations and conjugation, and recursively subtract lower degrees. The lattice product estimate bounds the discarded part in the compression algebra. Since $|\xi(a)| \le \|a\|$ for $a \in \mathfrak{A}_0$, that bound holds in every phase, and integration gives the mixture, with one radius threshold. The exponents compare output and input degrees. Since $C r^{\varepsilon/2} \log^m(1/r) \to 0$ for fixed C, m, the loss ε absorbs logarithms and constants, giving coefficient one for smaller radii. The thresholds have separate dependencies.

19.6–19.11 give integrated and nested estimates; the nested error is a sum of gains from proper clusters.

Contact subtractions use the original fields and correlations underlying 19.4, with separate projections extracting fields through finite correlation probes. Their clearance is a radius keeping the collapsed cluster separated from these probes, independent of spectator clearance δ_{sp}.

A collision forest is a family of subsets of slots, any two disjoint or nested. The contact construction subtracts normal Taylor jets along their collision diagonals. The next result concerns this fixed subtraction; its local order bounds are stated in 19.17.

Theorem 7.7 (CT. Contact extension)**.** *For every state index ζ, arity n and list $\boldsymbol{A} \in \mathbb{F}^n$, the contact construction of Section 19, 19.2, fixed on the underlying prescription of 19.4, gives a distribution $\mathfrak{D}_{\boldsymbol{A}}^{\zeta}$ on $\mathbb{R}^{4n}$ such that, for every separated test f,*

$$\mathfrak{D}_{\boldsymbol{A}}^{\zeta}(f) = \int f(x) S_n^{\zeta}(\boldsymbol{A}; x)\, dx .$$

The proof is in Section 19, 19.15–19.17. For separated f, all its jets on collision diagonals vanish, so the subtractions leave the correlation integral unchanged. For general f, the contact projections retain compatible field components and the lattice shell bounds make the subtracted sums converge to a continuous functional. 19.2 specifies the real stratum bases, scales and local order bounds, independently of OP's estimates. CT enters Theorem B; reconstruction and Theorem A use separated correlations.

8 Main theorems

The existence theorem concerns one Wilson family $\mathcal{W}$, chosen before any prescribed volume sequence. Its tokens $v \in \mathcal{T}$ label the lattice insertions. We first specify its continuum conclusion; the field calculus and stress tensor established above are additional properties of the construction.

Definition 8.1 (Q0. Hierarchy data)**.** Hierarchy data over $\mathcal{W}$ consist of:

- complex Hilbert and vector spaces $\mathcal{H}, \mathcal{L}$, and the relativistic data of Definitions 6.3–6.4;
- finite-dimensional complex normed spaces F_d, linear maps $\iota_d : F_d \to \mathcal{L}$, and $\pi_{\mathrm{quo}} : \mathcal{T} \to \mathbb{F}$, where $\mathbb{F} = \bigsqcup_d F_d$;
- a compact metric space X, a Borel measure ν and $\xi_* \in \mathsf{X}$;
- functions $S_n^\zeta : \mathbb{F}^n \times (\mathbb{R}^4)^n \to \mathbb{C}$ for $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, and $\mathsf{F}_n : \mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n \to \mathbb{C}$;
- finite sets Ch_j, fields $O_{j,\lambda} \in F_j$, and coefficients $c_{j,\lambda} : \mathbb{F}^{I_{\mathrm{cl}}} \times (\mathbb{R}^4)^{I_{\mathrm{cl}}} \times \mathbb{R}^4 \to \mathbb{C}$ for every finite index set I_{cl};
- an element $\Theta \in F_4$.

No further condition is part of Q0. Put $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$. Here the spaces and maps are arbitrary: no response realization, stress interpretation, multilinearity or channel basis property is implicit. Correlations are defined at all positions; the axioms constrain their separated values.

In Q2–Q6, all cited definitions and formulas are read with these hierarchy data. In definitions indexed by a phase, read Ω_{ξ_*}, Φ_{ξ_*} and $\mathsf{F}_n^{\xi_*}$ as the Ω, Φ and F_n of these data.

Definition 8.2 (Q1. Regular Wilson family)**.** A Wilson family is regular if its insertions are continuous and gauge invariant, each token v has a finite stencil $\Sigma_v \subset \mathbb{Z}^4$ valid for every N, p, y, and some $b_* > 0$ satisfies

$$\forall \varepsilon > 0 \; \exists N_0 \; \forall N \geq N_0 \; \forall p : \quad \left| \frac{\beta_{N,p}}{\log(1/a_N)} - b_* \right| < \varepsilon \, .$$

Only links with both ends in $y + \Sigma_v$ modulo the torus are read.

The relativistic axioms Q2 are Definition 6.9. The following conditions specify the labels, phase average and selected continuation.

Definition 8.3 (Q3. Labels and phases)**.** Every nonzero $\ell \in \mathcal{L}$ has $\Phi(\ell, f) \neq 0$ for some $f \in \mathcal{S}$. Each ι_d is injective, their images cover $\mathcal{L}$, π_{quo} is onto, and $\Phi(\iota(A), f) \neq 0$ for some $A \in \mathbb{F}$, $f \in \mathcal{S}$. The measure ν is a probability measure of full support. Each S_n^ζ is simultaneously permutation symmetric and smooth on Conf_n. For every n, $\boldsymbol{A} \in \mathbb{F}^n$ and $x \in \mathrm{Conf}_n$,

$$\xi \longmapsto S_n^\xi(\boldsymbol{A}; x) \text{ is continuous}, \qquad S_n^{\mathrm{mix}}(\boldsymbol{A}; x) = \int_{\mathsf{X}} S_n^\xi(\boldsymbol{A}; x) \, d\nu(\xi) \, .$$

Definition 8.4 (Q4. Selected continuation)**.** The functions F_n satisfy the ordered-time continuation condition of Definition 6.14 and the boundary condition of Definition 6.13 at ξ_*, using the relativistic datum and the maps ι_d of the hierarchy data.

Definition 8.5 (Q5. Products)**.** For the channels, fields, coefficients and tables S^ζ of Q0, form $\mathcal{R}_D^\zeta$ by Definition 7.5. These coefficients and remainders satisfy the two inequalities of Theorem 7.6: the coefficient threshold depends on $I_{\mathrm{cl}}, \boldsymbol{A}, M_{\mathrm{geo}}, \eta, \varepsilon, j, \lambda$; the remainder threshold depends on $I_{\mathrm{cl}}, I_{\mathrm{sp}}, \boldsymbol{A}, \boldsymbol{B}, M_{\mathrm{geo}}, \eta, \delta_{\mathrm{sp}}, \varepsilon, D$. Both precede positions and states. Coefficients have no spectator, state or truncation argument.

Definition 8.6 (Q6. Trace test)**.** For the chosen $\Theta \in F_4$, put

$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

Require

$$\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0 \qquad (f \in \mathcal{S})\,,$$

and

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0 \quad \text{in } \mathbb{C}\,,$$

and require the finite-profile exclusion of Proposition 7.4.

For the lattice limit, $S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ denotes the kernels

$$K_n(v; x) = S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1), \dots, \pi_{\mathrm{quo}}(v_n); x)\,.$$

Q2–Q4 describe a relativistic theory continued from one phase; Q5 controls products, and Q6's inverse logarithm excludes every finite Wick profile. In Theorem A, Q6 places a condition on a degree-four field without requiring a stress-tensor interpretation.

Theorem A (Existence)**.** *For every gauge group of Definition 2.1, there is a regular* (Q1) *Wilson family $\mathcal{W}$ such that, for every strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$, there are a strictly increasing q and hierarchy data over $\mathcal{W}$* (Q0) *satisfying* Q2–Q6*, with*

$$\mathcal{W} \longrightarrow S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}} \quad \textit{along } \mathsf{p} \circ \mathsf{q} \quad \textit{in the sense of Definition 4.3}\,.$$

Step 10 below deduces Theorem A from Theorem B.

In the construction theorem, the constructed hierarchy means the following Q0 data. Its spaces $F_d = V_d/\mathcal{N}_d(S)$, response labels $\mathcal{L}$ and maps ι_d are those of Definition 4.8, and

$$\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])\,.$$

Its mixture and phase tables are the quotient correlations of S and Theorem 6.2 on Conf_n, extended arbitrarily on collision diagonals. Its phase space, measure and selected point are those of that phase decomposition, with ξ_* chosen as in 17.3. Its relativistic data and continuation F_n come from Theorem 6.15 at ξ_*. Its Θ is the centered trace of the tensor in Theorem 5.4, as in Definition 7.1; its channels, fields and coefficients are those constructed in Section 19, 19.4.

The conclusions B1–B6 retain the following properties of these objects. The tags name the preceding results; the letter-and-number references in Theorem B specify the additional conclusions in Part II.

B1. The kernel, spacings, divisible periods, Taylor and reference normalizations and specified recursion are those fixed above. MT (Theorem 3.6) gives the trajectories and uniform running slope. Endpoint tuning in 14.3 gives a width $w_{\mathrm{tun}} > 0$, independent of N, p: the bare coupling reaching $\bar\beta + w_{\mathrm{tun}} s$ is unique within the specified recursion for $-1 \le s \le 1$ and extends holomorphically to $|s| < 1$.

B2. UV (Theorem 4.2) and 14.19–14.20 give the full fixed-volume limit, including derivatives, interpolation at moving positions and smearing.

B3. IV (Theorem 4.7) and FC (Proposition 4.10) give the quotient calculus and Maxwell dimensions. MC (Proposition 5.2) and SC (Proposition 5.5) give the ordinary and differentiated comparisons. ST (Theorem 5.4) with 16.5–16.6, and PH (Theorem 6.2), give the mixed and phase Euclidean, rotation and stress laws, with Θ identified as above.

B4. RC (Theorem 6.15) reconstructs every phase, with the OS comparison maps, ground-space and weak-commutant conclusions. 17.4–17.7 also give low-energy bounds, continuation of the stress tensor and phase averages of joint distributions.

B5. TR (Theorem 7.2) and NF (Proposition 7.4) give the full-radius law and finite-profile exclusion. Sections 18.1 and 18.2 also give the centering identity, unique spectral measure, high-energy limits and polynomial mass-shell and Fock countertests.

B6. OP (Theorem 7.6) and 19.4–19.11 give one product prescription with separated, integrated and nested estimates. CT (Theorem 7.7) and 19.15–19.17 give contact extensions, real bases and subtraction scales. Both use the same original fields and correlation functionals, with separate contact projections, separating functionals and clearance.

The next theorem realizes Theorem A's hierarchy by these quotient fields, with Θ the centered trace of a normalized stress tensor and with every phase reconstructed.

Theorem B (Construction)**.** *For every gauge group of Definition 2.1 and* $\mathsf{b} > 0$*, there is one regular Wilson family* $\mathcal{W}$*, given by Definitions 3.1 and 3.8, satisfying* MT, 14.1–14.3, 10.2, 14.5–14.6 (B1)*, and* UV, 14.19–14.20 (B2)*. For every strictly increasing* p*, there are a strictly increasing* q*, a table* S *as in* IV *and a constructed hierarchy over* $\mathcal{W}$ *for this* S*, satisfying* Q2–Q6*, with* $\mathcal{W} \to S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ *along* $\mathsf{p} \circ \mathsf{q}$ (Definition 4.3)*. Its table, fields and tensor obey* IV, FC, 15.4–15.5, MC, ST, SC, PH, 16.2–16.4, 16.5–16.7, 17.3 (B3)*; its phase theories obey* RC, 17.4–17.7 (B4)*; its* Θ *obeys* TR, NF, 18.1–18.3, 18.4–18.10 (B5)*; its product and contact constructions obey* OP, 19.4–19.11, CT, 19.15–19.17 (B6)*, with the same underlying prescription.*

Steps 1–9 below prove Theorem B. Additional position-space normalization results for degree-four fields are given with 18.3, and estimates along general collision subspaces with 19.17; these are separate from the conclusions above. Restricting the full ultraviolet limit also gives convergence along every spacing subsequence.

Deduction of Theorems A and B.

1. Fix G and $\mathsf{b} > 0$. Section 9 and Sections 10–14 supply the common choices, admissible classes, recursion and later lattice estimates. Wilson initialization 13.6 starts the induction; Theorem 3.5 advances it; Theorem 3.6 and 14.3 tune trajectories to the common endpoint. These choices precede any prescribed volume sequence.

2. Fix the local sections, reference levels and terminal maps on those trajectories. Definition 3.8 defines every insertion, including below its reference level. The gauge-invariant local sections are polynomial in the link entries on their stencils; the terminal coefficient maps preserve continuity, gauge invariance and tokenwise locality. Taking $m = N$ in Theorem 3.6 gives Q1 with $b_* = \kappa_G > 0$. This fixes the regular Wilson family.

3. Proposition 4.1 and fixed-volume comparison with analytic tuning give Theorem 4.2 and 14.19–14.20, for every volume and fixed field/difference request. Thus each admissible test has a full inner limit before volume extraction.

4. Given any strictly increasing p, Theorem 4.7 chooses q, S with the convergence of Definition 4.3 along $\mathsf{p} \circ \mathsf{q}$ and E1–E7. The same extraction fixes the corrected coefficient maps.

5. Define $F_d = V_d/\mathcal{N}_d(S)$ and $\mathcal{L}$ by Definition 4.8. Proposition 4.10 and 15.4 give inclusions, derivatives, dimensions and the response realization. Proposition 5.2 and 16.3–16.4 give the conserved line and charge maps; 16.5–16.6 give the representations R_d and full covariance, and Theorem 5.4 gives the normalized tensor; Proposition 5.5 gives the differentiated comparison. The volume refinement is unchanged.

6. Theorem 6.2 and 17.1–17.3 of Section 17 construct X, ν, S^{ξ} on the same quotient spaces. Choose ξ_* from the full-measure set with one-dimensional time-axis OS ground space. This selection precedes reconstruction and uses only that ground-space property. Theorem 6.15 reconstructs every character simultaneously; 17.4–17.7 retain this selected theory and give B4.

7. Set $\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])$. Representatives make this map onto. Every response lies in one $\iota_d F_d$, and each ι_d is injective. The unit response is nonzero since its one-point function is one; faithfulness gives a nonzero acting field. Extend the quotient mixture and phase correlations arbitrarily off Conf_n. Separated tests vanish on collision diagonals, so these extensions do not affect Definition 4.3. On Conf_n, $S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}} v; x) = S_n(v; x)$ by the representative independence after Definition 4.8, so Theorem 4.7 gives the convergence in Theorem A. Faithfulness in Theorem 6.15 and the preceding facts give the label clauses of Q3; E1, E6 and Theorem 6.2 give its phase clauses; reconstruction gives Q2 and Q4. These identifications use the all-spectator quotient.

8. Center the trace of this tensor in this selected phase. Theorem 7.2 gives Q6's vacuum-centering, amplitude and full-radius clauses; 18.1–18.3 of Section 18 give the operator identity and the unique spectral measure. Proposition 7.4 gives Q6's finite-profile exclusion; 18.6–18.10 give the additional countertests in B5, with ξ_* unchanged.

9. Choose the prescription of 19.4 once. Theorem 7.6 gives Q5; 19.6–19.11 give the integrated and nested bounds in B6. Theorem 7.7 and 19.15–19.17 of Section 19 use the same underlying prescription, with separate contact projections, functionals and clearance, to give the extensions, real bases and subtraction scales. This uses its own lattice estimates.

10. Steps 1–9 give B1–B6 on the same family, table, quotient fields, tensor, phases and selected theory. Retain Q1–Q6 and set $\mathsf{b} = 1$. Their quantifier order, lattice approximation, reconstruction, products and trace test give Theorem A.

□

9 Constants and order of choice

Table 1 gives the choices for the construction. At each fixed stage the requirements in Sections 11–14, 16.2, 18.1 and 19 impose finitely many lower bounds on a ratio or an entry level, depending only on earlier data. Costs determining a ratio precede that ratio; increasing an entry level preserves earlier requirements. The displayed inequalities are properties of these choices. They do not define a class of constructions for which the conclusions are asserted.

Table 1: Choices in dependence order. Stage choices are made at each fixed $j \geq 4$ before entry levels.

	Choice	Dependence, restrictions and purpose
1	Gauge normalization; $\mathsf{b} > 0$	Fix G, its realization, probability Haar measure and metric, then $d_{\mathfrak{g}}, C_{\mathfrak{g}}, b_0, \kappa_G$ as in Definition 2.1; $\kappa_G > 0$. All later choices use these data.
2	Frame and predictor cutoffs; axial factor	Depend on G. The frame and predictor cutoffs select the unique-minimizer branches of the block kernel. On a radius-R patch, the total link chord in an eligible axial frame is at most $a_{\mathrm{ax}}(2R+1)$ times the total plaquette chord, with $a_{\mathrm{ax}} \geq 16$. Choose the cutoffs small enough for branch validity on every patch (Lemma 11.4).
3	Homogeneous bases, norms, sections, stencils	Depend on G, d. Impose Taylor matching, reality and H_4 covariance (Lemma 10.2 and Definitions 3.7–3.8).
4	Stage parameters	Depend on rows 1–3 and j. Choose derivative orders $\mathfrak{q}_j \geq 6$, spatial powers $\varpi_j \geq 2$ and support exponents $\sigma_j = 100$. In Section 12, 12.3, $w_j^{\mathrm{path}} > 0$ multiplies path length in the exponential weight, and $A_j^{\mathrm{path}} > 0$ bounds the weighted row, cubic and quartic sums. The constant $C_j \geq 1$ bounds marked coefficient transport (14.7). Let $\gamma_j > 0$ be half a uniform positive lower bound for the free effective plaquette forms in Lemma 11.5, chosen for the top form and both possible incoming stages. Choose clipping coefficients with $0 < 16c_{\mathrm{lin},j} < c_{\mathrm{dir},j} \leq \min(\gamma_j, 1)$; the remaining free form then has the positive row factor used in Definition 12.2. Its health cutoff relaxes with distance through $e^{\mathrm{dist}/R_j^{\mathrm{rel}}}$, where $R_j^{\mathrm{rel}} > 0$ and $R_j^{\mathrm{rel}} w_j^{\mathrm{path}} \geq 12$. Let $d_j \geq 0$ be the block-ratio-independent one-loop bound of Lemma 12.9. Fix the reset half-width $r_{\mathrm{res}} = 1$, then choose $\omega_j^{\mathrm{RG}} \geq d_j + 1$ for Lemma 12.10. Orders and powers tend to infinity and, along levels, are nondecreasing, as are support exponents. All ratio costs are now fixed.
5	Odd ratios ℓ_j^{st}	Depend on rows 1–4; choose $\ell_j^{\mathrm{st}} \geq 3$ nondecreasing and above every ratio threshold. At $j = J_{m+1}$ and $\ell_m = \ell_j^{\mathrm{st}}$ require $$\frac{\omega_j^{\mathrm{RG}}}{\log \ell_m} \leq \min(\kappa_G/4, j^{-2}),$$ $$\frac{\log(1 + A_j^{\mathrm{path}} + (w_j^{\mathrm{path}})^{-1} + C_j + R_j^{\mathrm{rel}} + \sigma_j)}{\log \ell_m} \leq j^{-2}.$$ These bounds control the drift and logarithmic stage costs relative to $\log \ell_m$.

Table 1 (continued)

	Choice	Dependence, restrictions and purpose
6	Exponents and coupling entry bounds	Depend on rows 1–5. The positive antitone exponents θ_j control the quadratic loop bound in row 9. The bounds β_j^{ent} are lower bounds on the reference coupling when stage j is used. Choose them to meet the finite maximum of the remaining stage requirements, including those of Sections 16.2, 18.1 and 19. Entry bounds may be raised.
7	$\bar\beta$, entry levels m_j^{ent}, minimum periods $n_j^{\min}$	Depend on rows 1–6. The levels m_j^{ent} delay the first use of stage j; $n_j^{\min}$ are its minimum lattice periods. Choose $\bar\beta > 1$ large enough for the initial coupling requirements and for $t_m \le 1$ at every level. Choose entry levels and periods to satisfy the remaining level and geometric requirements. Delaying entry preserves earlier requirements.
8	$J_m, \ell_m, D_m, a_m, L_p$, $m_{\mathrm{ref}}(d)$	Depend on rows 1–7. Set $D_0 = 1$, $D_{m+1} = \ell_m D_m$, $a_m = D_m^{-1}$, $s_m = \log D_m$, with $\ell_m = \ell^{\mathrm{st}}_{J_{m+1}}$ and $4 \le J_m \le J_{m+1} \le J_m + 1$, $J_m \to \infty$. Require $0 < \theta_{J_m} \le J_m^{-1}$, $\log \ell_m \to \infty$, $\log \ell_m / s_m \to 0$. Choose strictly increasing L_p with $4 \mid L_p$ and every positive integer dividing infinitely many L_p. Require $L_p D_m \ge 100(B_m + 1)$ and $L_p D_{m+1} \ge 100((J_{m+1}+3)\ell_m + 1)$. Choose $J_{m_{\mathrm{ref}}(d)} \ge d + 3$, with $\mathfrak{q}_{J_m} \ge d + 1$, $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$.
9	Derived level scales	No new choices. Set $$\widehat\beta_m = \bar\beta + \kappa_G s_m > 1, \qquad M_m^{\mathrm{cut}} = \exp((\log\log\widehat\beta_m)^2),$$ $$t_m = \widehat\beta_m^{-1/2} M_m^{\mathrm{cut}}, \qquad B_m = \lceil (\log\widehat\beta_m)^2 \rceil,$$ $$\epsilon_m^{\mathrm{reg}} = \widehat\beta_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}},$$ $$\Lambda_m^{\mathrm{loop}} = 2(M_m^{\mathrm{cut}})^{\theta_{J_m}}, \qquad \mathfrak{B}_m = [\widehat\beta_m/2, 2\widehat\beta_m].$$ These give the field thresholds, activity bounds and coupling bands. The quantity $\Lambda_m^{\mathrm{loop}}$ bounds the weighted quadratic loop sum in Section 12, 12.3.
10	Actions, classes, recursion, trajectories, couplings; width w_{tun}	Depend on rows 1–9. Choose the path actions and admissible subsets, the specified recursion and its trajectories for every finite N, p, all with the common level-zero endpoint (Theorem 3.6). In Section 14, 14.3, $w_{\mathrm{tun}} > 0$ is the half-width of the endpoint interval $\bar\beta + [-w_{\mathrm{tun}}, w_{\mathrm{tun}}]$, independent of N, p. The tuned bare coupling for endpoint $\bar\beta + w_{\mathrm{tun}} s$ extends holomorphically in s to $\lvert s\rvert < 1$.
11	Marked transfers, terminal maps, insertions	Depend on rows 1–10; impose reference normalization (Definition 3.8). Fix them before any prescribed volume sequence.
12	p, then $\mathfrak{q}, S, F_d, \partial_a, \mathcal{L}$	Prescribe arbitrary increasing p first. Theorem 4.7 and Proposition 4.10 give the refinement and continuum fields. Theorem 4.2 retains the full inner limit.
13	Comparison radii, frames, J, charges, $c_\star, R, T$	Depend on row 12 and the fixed lattice family. Use Proposition 5.2, then Theorem 5.4 and Proposition 5.5. Refine only comparison radii.

Table 1 (continued)

	Choice	Dependence, restrictions and purpose
14	Slab operators, X, ν, S^{ξ}, then ξ_*	Depend on earlier data. Theorem 6.2 gives the phase decomposition. Choose ξ_* from the full-measure set with one-dimensional time-axis OS ground space (17.3). This choice precedes reconstruction and uses only that ground-space property.
15	Every-phase theories and continuations; Θ, μ_Θ	Depend on earlier data. Theorem 6.15 reconstructs the theories. Definition 7.1 centers the trace in the already selected phase; Theorem 7.2 gives its short-distance law; 18.2 supplies its spectral measure.
16	OPE prescription; contact projections, distributions, bases, scales	Depend on rows 12–14 and the fixed family. Theorems 7.6 and 7.7 use the same original fields and correlation functionals. The contact projections, separating functionals and clearance are chosen separately from those used for the product coefficients.
17	Estimate constants and thresholds	Depend only on the fixed data preceding them in each statement. They have no later position, volume or state dependence wherever uniformity is asserted.

Part II
Proofs

10 Local polynomial fields

The polynomial labels of Definition 3.7 specify continuum Taylor coefficients, whereas the Wilson measure averages functions of group-valued links. We relate them by local sections: polynomials in the real and imaginary parts of finitely many link entries that reproduce each label on radial test holonomies through its engineering degree. They initialize the renormalized insertions in Definition 3.8, hence the Wilson family of Theorem B, independently of the lattice volume and later block ratios.

We must recover the connection coefficients visible to a label from finitely many measurements. Transported plaquette curvatures have linearizations given by ordinary square circulations. Vanishing of all these circulations forces the highest homogeneous curvature to vanish; radial gauge then forces the corresponding potential to vanish. Descending in degree proves injectivity. Finite dimensionality reduces this to finitely many plaquettes. A local inverse of their joint map, followed by Taylor expansion, gives the required polynomials. Averaging over signed permutations imposes covariance without changing the Taylor coefficients.

Each curvature factor also contributes one small-field threshold factor to the bounds in Lemma 10.3. For the equation projections in Section 14, Lemma 10.4 constructs auxiliary sections with exact exterior zero moments. Their labels undergo a fixed invertible triangular change. We use Definitions 2.1, 2.2, 3.3 and 3.7, the chord (2.1), and the order of choices in Section 9; no earlier Part II result is used.

Labels and test holonomies

Write $\mathcal{J}_j$ for the real space of homogeneous radial potentials of degree $j+1$ in Definition 3.7. Every label in V_d is a polynomial on the finite-dimensional space

$$\mathcal{J}_{\le d} = \bigoplus_{j+2\le d} \mathcal{J}_j \,.$$

A coordinate of $\mathcal{A}_j$ has weight $j+2$, so only finitely many monomials have weight at most d. Adjoint action and conjugation preserve both gradings. Real and imaginary parts span each invariant component over $\mathbb{C}$, giving bihomogeneous bases as in Definition 3.7. Conjugation in this real basis is coefficientwise and agrees with conjugation of polynomial values on real jets. Write k_i for the number of jet-coordinate factors in $b_{d,i}$. We identify a coefficient vector with its polynomial:

$$\mathbf{c}(\mathrm{j}) = \sum_i c_i b_{d,i}(\mathrm{j}), \qquad b_{d,0} = \mathbf{1} \,.$$

We use the test holonomies (3.16), on the lattice $\mathbb{Z}^4$ covering the torus. With $\mathcal{A}^{\le d} = \sum_{j+2\le d} \mathcal{A}_j$, their integral equation is

$$U(h_{\mathrm{lat}}, \mathrm{j})(x,a) = u(1), \qquad u(\tau) = 1 + \int_0^\tau u(\eta) h_{\mathrm{lat}} \mathcal{A}_a^{\le d}(h_{\mathrm{lat}}(x+\eta e_a))\, d\eta \,. \tag{10.1}$$

The defining linear matrix equation has a unique global solution for continuous coefficients [5, Theorem 3.9]. Piecewise constant Lie-algebra coefficients give products of exponentials in G; uniform convergence on compact intervals, by Gronwall's inequality [5, Lemma 2.7], and closedness of G put the solution in G. Homogeneity and uniqueness give

$$U(h_{\mathrm{lat}}, \mathrm{j}) = U(1, \mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j}), \qquad \mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j} = (h_{\mathrm{lat}}^{j+2} \mathcal{A}_j)_j, \qquad U(0,\mathrm{j}) = 1 \,. \tag{10.2}$$

For a cover plaquette $\square$ based at z and a path γ from 0 to z, write U_γ for the ordered product of successive oriented links, read from the initial point to the endpoint. Define its transported curvature by

$$\sigma_{\gamma,\square}(U) = \mathrm{Ad}(U_\gamma) F_\square(U), \qquad F_\square(U) = \mathrm{proj}_{\mathfrak{g}} \frac{U_\square - U_\square^{-1}}{2} \,.$$

The projection is orthogonal for the normalized Frobenius metric of Definition 2.1. These measurements transform by the adjoint action at 0. They are real polynomial functions of the real and imaginary parts of link entries, since $U_e^{-1} = U_e^*$.

Definition 10.1 (Local sections)**.** For a finite site stencil $\Sigma_d \subset \mathbb{Z}^4$, a local section has the form

$$P_d^{\mathrm{loc}}[\mathbf{c}](U) = \sum_{\nu,\mu} p_{\nu\mu}^{(d)}(\mathbf{c}) \prod_{e,\alpha,\delta} U_e(\alpha,\delta)^{\nu_{e\alpha\delta}} \overline{U_e(\alpha,\delta)}^{\mu_{e\alpha\delta}} \,.$$

The product uses positive links with both endpoints in Σ_d and matrix indices $1 \le \alpha, \delta \le n_G$. The sum is finite over nonnegative multi-indices; each coefficient is complex linear in $\mathbf{c}$. The construction below chooses such a map to represent each continuum label.

For a torus of side n_{tor} (the side s of Definition 2.2), write $[x]$ for reduction of $x \in \mathbb{Z}^4$ modulo n_{tor}. Placement at a torus site y means evaluation on the periodic extension $U^{[y]}(x,a) = U(y+[x],a)$;

denote it by $P_d^{\mathrm{loc}}[\mathbf{c}](y;U)$. On the cover, $P_d^{\mathrm{loc}}[\mathbf{c}](0;U) = P_d^{\mathrm{loc}}[\mathbf{c}](U)$. For $\rho \in H_4$, define the configuration and jet actions by

$$(\rho^* U)(x,a) = \begin{cases} U(\rho x, b), & \rho e_a = e_b, \\ U(\rho x - e_b, b)^{-1}, & \rho e_a = -e_b, \end{cases}$$
$$(\rho \mathrm{j})_{j,a}(z) = \sum_b \rho_{ab} \mathcal{A}_{j,b}(\rho^{-1} z), \qquad (\rho \mathbf{c})(\mathrm{j}) = \mathbf{c}(\rho^{-1}\mathrm{j})\,.$$

This action preserves homogeneity and radial gauge. Reality and covariance mean

$$P_d^{\mathrm{loc}}[\overline{\mathbf{c}}](U) = \overline{P_d^{\mathrm{loc}}[\mathbf{c}](U)}, \qquad P_d^{\mathrm{loc}}[\mathbf{c}](\rho^* U) = P_d^{\mathrm{loc}}[\rho \mathbf{c}](U) \quad (\rho \in H_4),$$

with conjugation as fixed above.

The following result supplies the sections specified in Section 3. The unit normalization is exact at every configuration; the Taylor assertion concerns the test configurations near zero spacing.

Lemma 10.2 (Taylor-normalized sections)**.** *For every d, a box stencil $\Sigma_d = [-\varrho_d, \varrho_d]^4 \cap \mathbb{Z}^4$ and sections can be chosen gauge invariant, real and H_4 covariant, with $P_d^{\mathrm{loc}}[\mathbf{1}] = 1$, so that, for every $\mathbf{c} \in V_d$, radial jet j and $0 \le q \le d$,*

$$\frac{d^q}{dh_{\mathrm{lat}}^q} P_d^{\mathrm{loc}}[\mathbf{c}](U(h_{\mathrm{lat}}, \mathrm{j}))\Big|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q} \mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j})\Big|_{h_{\mathrm{lat}}=0}\,. \tag{10.3}$$

Fix the sections constructed in the following proof.

Proof. We use the metric and lattice transports of Definitions 2.1–2.2 and the labels of Definition 3.7. If $d < 2$, the labels are constant and their constant sections suffice. Fix $d \ge 2$.

Step 1: Curvatures determine the truncated jet.

Use the $\mathrm{Ad}(G)$-invariant coefficient norm on $\mathcal{J}_{\le d}$ obtained from the metric of Definition 2.1. The Volterra term of order l is bounded by $(C_{x,a,d}|\mathrm{j}|)^l/l!$, giving locally uniform convergence with all parameter derivatives and smooth measurements. The first variation of a link solution is its edge integral; that of a plaquette measurement is their circulation. Transport along γ contributes nothing at first order, because $F_\square(1) = 0$.

Suppose all these linearized measurements vanish on $\mathcal{A} = \sum_{j=0}^m \mathcal{A}_j$. For $a < b$ set

$$N_{j;ab}(x,\tau) = \int_0^1 \int_0^1 (\partial_a \mathcal{A}_{j,b} - \partial_b \mathcal{A}_{j,a})(x + \tau(\xi e_a + \eta e_b))\, d\xi\, d\eta\,.$$

The fundamental theorem of calculus on each edge identifies the unit-square circulation with $\sum_{j \le m} N_{j;ab}(x,1)$. It is zero for every $x \in \mathbb{Z}^4$. Homogeneity gives $N_{j;ab}(\tau x, \tau) = \tau^j N_{j;ab}(x,1)$ and hence

$$\sum_{j=0}^m \tau^{m-j} N_{j;ab}(\tau x, \tau) = 0 \qquad (x \in \mathbb{Z}^4,\ \tau > 0)\,.$$

For $z \in \mathbb{R}^4$, set $x_\tau = (\lfloor z^a/\tau \rfloor)_a$, so $|\tau x_\tau - z|_\infty \le \tau$. Continuity as $\tau \downarrow 0$ leaves only the term $j = m$, giving $(\partial_a \mathcal{A}_{m,b} - \partial_b \mathcal{A}_{m,a})(z) = 0$. Antisymmetry covers the other ordered pairs. Differentiating the radial condition and using Euler's identity gives

$$(m+2)\mathcal{A}_{m,b}(z) = \sum_a z^a (\partial_a \mathcal{A}_{m,b} - \partial_b \mathcal{A}_{m,a})(z) = 0\,.$$

Remove this component and descend in m. The common kernel is zero. Adding a measurement whenever the kernel is nonzero decreases its dimension, so finitely many measurements $\sigma_1, \dots, \sigma_{M_{\mathrm{meas}}}$ give a joint map

$$\mathsf{S} : \mathcal{J}_{\le d} \longrightarrow \mathfrak{g}^{M_{\mathrm{meas}}}, \qquad \mathsf{S}(\mathsf{j}) = (\sigma_l(U(1,\mathsf{j})))_{l=1}^{M_{\mathrm{meas}}}$$

with injective derivative $D\mathsf{S}(0)$.

Step 2: A polynomial inverse to the required order. Uniqueness of the link equation shows that a constant gauge transformation takes the test links of j to those of $\mathrm{Ad}(v)\mathsf{j}$. Thus S and its derivative at zero are adjoint equivariant. The range of $D\mathsf{S}(0)$ and its orthogonal complement in $\mathfrak{g}^{M_{\mathrm{meas}}}$ are adjoint invariant. Compose its inverse on the range with orthogonal projection to obtain $C : \mathfrak{g}^{M_{\mathrm{meas}}} \to \mathcal{J}_{\le d}$. Then

$$C D\mathsf{S}(0) = I, \qquad C\,\mathrm{Ad}(v) = \mathrm{Ad}(v) C \quad (v \in G)\,.$$

Apply the inverse function theorem [3, Theorem 9.24] to $C\mathsf{S}$ on this finite-dimensional real space: it is smooth near zero, sends zero to zero and has derivative I. Its local inverse φ is smooth, by differentiating $D\varphi = (D(C\mathsf{S}) \circ \varphi)^{-1}$. Choose an invariant source ball on which $C\mathsf{S}$ is injective, and then an invariant target ball whose inverse image lies in that source ball. Both exist because the adjoint actions are isometric. Put $f = \varphi \circ C$. Equivariance of $C\mathsf{S}$ and injectivity on the source ball give, near zero,

$$f\mathsf{S}(\mathsf{j}) = \mathsf{j}, \qquad f(0) = 0, \qquad f(\mathrm{Ad}(v)z) = \mathrm{Ad}(v) f(z) \quad (v \in G)\,.$$

Consequently the Taylor polynomial

$$T_d f(z) = \sum_{k=1}^{d} \frac{1}{k!} D^k f(0)[z, \dots, z]$$

is adjoint equivariant and has no constant term. Define

$$Q_i(z) = b_{d,i}(T_d f(z)), \qquad \Psi_{\mathbf{c}}(U) = \sum_i c_i Q_i(\sigma_1(U), \dots, \sigma_{M_{\mathrm{meas}}}(U))\,.$$

The polynomials Q_i are invariant and real on real arguments, and $Q_0 = 1$. If $b_{d,i}$ has factor count k_i, every monomial of Q_i has at least k_i factors.

Taylor's formula gives $T_d f(\mathsf{S}(X)) = X + O(|X|^{d+1})$ near 0. Substitute $X = \mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j}$ truncated to $\mathcal{J}_{\le d}$, and use (10.2). Both sides are smooth in h_{lat}, and

$$\Psi_{\mathbf{c}}(U(h_{\mathrm{lat}}, \mathsf{j})) = \mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j}) + O_{\mathbf{c},\mathsf{j}}(|h_{\mathrm{lat}}|^{d+1})\,.$$

Their derivatives through order d at zero agree. The polynomial Q_i itself need not have degree at most d in the curvature measurements.

Step 3: Covariance and a common stencil. Average the section by

$$P_d^{\mathrm{loc}}[\mathbf{c}](U) = \frac{1}{|H_4|} \sum_{\rho \in H_4} \Psi_{\rho^{-1}\mathbf{c}}(\rho^* U)\,. \tag{10.4}$$

Changing the summation variable to $\omega\rho$ proves $P_d^{\mathrm{loc}}[\mathbf{c}](\omega^* U) = P_d^{\mathrm{loc}}[\omega\mathbf{c}](U)$. Simultaneous adjoint covariance of the measurements gives gauge invariance. The real basis and signed-permutation action give reality; the constant column remains one. The action preserves factor count, so averaging retains the lower bound on the number of curvature factors.

For a positive image direction, the pulled-back link equation has coefficient

$$h_{\mathrm{lat}} \sum_{b} \rho_{ba} \mathcal{A}_{b}^{\leq d}(h_{\mathrm{lat}} \rho(x + \tau e_a)),$$

the coefficient for $\rho^{-1}\mathsf{j}$. Reversing the parameter and inverting the transport covers a negative image direction. Uniqueness gives $\rho^{*} U(h_{\mathrm{lat}}, \mathsf{j}) = U(h_{\mathrm{lat}}, \rho^{-1}\mathsf{j})$. Each term of (10.4) therefore has Taylor polynomial $\mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}}\, \mathsf{j})$ through order d. A signed image of a transported curvature is again a transported curvature, up to sign: reversal changes $F_{\square}$ to $-F_{\square}$, and a change of base point gives conjugation by the intervening boundary transport. Enclose the finitely many resulting paths and plaquettes in a box $\Sigma_d = [-\varrho_d, \varrho_d]^4 \cap \mathbb{Z}^4$. This stencil contains zero and is signed-permutation invariant. Expanding the finite polynomial in matrix entries gives Definition 10.1. This proves (3.17). □

Bounds and lattice symmetries

We bound normalized initial insertions using the ordered left derivatives of Definition 3.3. At a placement y, set $Y = y + [\Sigma_d]$; being healthy on Y means that every plaquette meeting Y has chord at most t. A direction is permitted when

$$|\dot{\mathcal{A}}_e| \leq t(1 + \mathrm{dist}(e^{-}, y))^{P}\,.$$

Here $[\Sigma_d]$ is the torus image of the stencil, and $\mathrm{dist}(x, y) = \min_{z \in n_{\mathrm{tor}}\mathbb{Z}^4} |\tilde{x} - \tilde{y} + z|_{\infty}$ is the periodic maximum-coordinate distance, for any lifts $\tilde{x}, \tilde{y}$.

Lemma 10.3 (Section bounds)**.** *Each basis column is a polynomial in transported curvatures with at least k_i factors per monomial. For all $d, k, P \in \mathbb{N}$ there are finite constants $C_{d,k,P}$ and C_d such that, on every torus, for every placement y, $0 < t \leq 1$, configuration healthy on Y, and permitted direction list of length $q \leq k$,*

$$\left| D^{L}_{\dot{\mathcal{A}}_1} \cdots D^{L}_{\dot{\mathcal{A}}_q} P_d^{\mathrm{loc}}[b_{d,i}](y; U) \right| \leq C_{d,k,P} t^{k_i}\,. \tag{10.5}$$

For arbitrary configurations,

$$\sup_{i,y,U} |P_d^{\mathrm{loc}}[b_{d,i}](y; U)| \leq C_d\,. \tag{10.6}$$

The proof uses the curvature polynomials constructed in Lemma 10.2.

Proof. First note that $|F_{\square}(U)| \leq \mathrm{ch}(U_{\square})$: orthogonal projection is contractive and $\|U_{\square} - U_{\square}^{*}\|_{\mathrm{F}} \leq 2\|U_{\square} - 1\|_{\mathrm{F}}$. Adjoint transport is isometric, so an undifferentiated curvature measurement is bounded by t. An order-q left derivative of a fixed path product is a finite sum of products of unitary matrices and exactly q direction matrices. On the fixed stencil their norms are at most a constant times t, since periodic distance is bounded by cover distance. Thus each differentiated measurement is $O(t^q)$, and hence $O(t)$ when $q \geq 1$. This argument also applies when cover links coincide after placement: one first differentiates the cover expression and then assigns the same direction to coincident links.

Apply the product rule to the curvature polynomials of (10.4). Every differentiated monomial still has at least k_i factors, each bounded by a constant times t. There are finitely many terms; since $t \leq 1$, their sum gives (10.5), including order zero. Each entry of a unitary matrix has absolute value at most one, so the finite matrix-entry polynomial gives (10.6). The formula also gives smoothness, measurability and locality. The constants depend on the sections, G, d, k, P as applicable, and finite maxima make them nondecreasing in the integer parameters. They are fixed before the stage block ratios in Section 9. □

By linearity, Lemma 10.3 bounds permitted derivatives of $P_d^{\mathrm{loc}}[\mathsf{D}_{t^{-1}}^{\mathrm{fac}}\mathbf{c}]$ by $C_{d,k,P}\sum_i|c_i|$ on the healthy set: the factor t^{-k_i} cancels t^{k_i}. Taking $t=t_N$, $k=\mathfrak{q}_{J_N}$ and $P=\varpi_{J_N}$ bounds the initial marked density of (3.18) on the healthy set, in the sense of Definition 3.3. On unhealthy configurations the global bound gives $C_d\sum_i|c_i|t^{-k_i}$; Lemmas 14.7–14.8 place this contribution in the forced remainders of (3.21).

Finite-lattice invariance. For the identities deferred from Definition 2.2, telescoping the path products shows that gauge transformations act on plaquette holonomies by conjugation. An affine isometry $h_{\mathrm{iso}}(x)=\rho x+b$, with $\rho\in H_4$, changes an image plaquette holonomy by conjugation, possibly composed with inversion. Both operations preserve pr, hence the Wilson energy. Fixed left and right multiplication, inversion and permutation preserve product probability Haar measure [1, Corollary 2.28 and Proposition 2.31]. Consequently, for every integer side $n_{\mathrm{tor}}\geq 1$, real β, site gauge transformation γ_{g} and bounded measurable F,

$$\int F(U^{\gamma_{\mathrm{g}}})\,d\mu_{n_{\mathrm{tor}},\beta}(U)=\int F(h_{\mathrm{iso}}^{*}U)\,d\mu_{n_{\mathrm{tor}},\beta}(U)=\int F(U)\,d\mu_{n_{\mathrm{tor}},\beta}(U)\,. \tag{10.7}$$

All integrals exist: $0\leq\mathrm{pr}\leq 2$ gives $0\leq E_{n_{\mathrm{tor}}}^{\mathrm{W}}\leq 12n_{\mathrm{tor}}^4$ and $0<Z_{n_{\mathrm{tor}},\beta}<\infty$. The terminal transfer normalization in Definition 3.8 is proved in Lemmas 14.5–14.6.

Auxiliary equation sections

Exact lattice equation identities require different sections, still polynomial and local with the same small-field bounds. On $\mathcal{J}_j$ define the Maxwell subspace and its polynomial ideal by

$$K_j=\left\{\mathcal{A}_j:\ \sum_a\partial_a(\partial_a\mathcal{A}_{j,b}-\partial_b\mathcal{A}_{j,a})=0\text{ for every }b\right\},\qquad \mathcal{I}_d=\{p\in V_d:p|_{\prod_j K_j}=0\}\,.$$

Here restriction uses only $j+2\leq d$. Substitution into $\prod_j K_j$ preserves bidegree, so $\mathcal{I}_d$ is the direct sum of its bihomogeneous parts. Order bidegrees (w,k) by the lexicographic order of (k,w). A regrading is a complex-linear map whose difference from the identity sends each bihomogeneous polynomial into strictly later bidegrees. Such a map preserves the filtration by minimum factor count. For a torus site y, put

$$\Omega_r(y)=\{x:\mathrm{dist}(x,y)>r\}\,.$$

An exterior spectator is a bounded measurable function of the positive links having both endpoints in $\Omega_r(y)$; it need not be gauge invariant.

The following identity supplies the auxiliary equation observables used in Section 14. Its proof constructs them from derivatives of the Wilson energy with multipliers independent of the varied link.

Lemma 10.4 (Auxiliary equation sections)**.** *For every d there are a local section V_d^{aux} satisfying Lemmas 10.2 and 10.3 with its own stencil and constants, an invertible regrading R_d fixing $\mathbf{1}$ and commuting with conjugation and H_4, and integers r_d,M_d such that every $n_{\mathrm{tor}}\geq M_d$, $\beta\neq 0$, placement y, $p\in\mathcal{I}_d$ and exterior spectator S on $\Omega_{r_d}(y)$ satisfy*

$$\int V_d^{\mathrm{aux}}[\mathsf{R}_d p](y;U)S(U)\,d\mu_{n_{\mathrm{tor}},\beta}(U)=0\,. \tag{10.8}$$

The exact-equation subspace for these sections is $\mathsf{R}_d(\mathcal{I}_d)$. Its comparison with the selected primary family is deferred to Section 13. All choices in the following proof depend only on G, d and the label basis.

Proof. We use Definitions 2.1, 2.2, 3.7 and Lemmas 10.2–10.3.

Step 1: One-link equations. Extend the curvature $F_\square$ defined above to $F_{ab}(U;x)$ for ordered axes by antisymmetry, with $F_{aa}=0$. For a positive link $e=(x,a)$ set

$$\mathcal{E}_e(U)=\sum_{b\neq a}\big(F_{ab}(U;x)-\mathrm{Ad}(U(x-e_b,b)^{-1})F_{ab}(U;x-e_b)\big)\ .$$

On a torus of side at least two, direct differentiation gives

$$D^L_{e,A}E^{\mathrm{W}}(U)=\langle A,\mathcal{E}_e(U)\rangle_{\mathfrak{g}}\,. \tag{10.9}$$

Indeed, the forward plaquette contributes $-n_G^{-1}\,\mathrm{Re}\,\mathrm{tr}(AU_\square)=\langle A,F_\square\rangle_{\mathfrak{g}}$. The backward plaquette, rebased at x, contributes the negative of the transported curvature in the display. These are the two incidences for each transverse axis; antisymmetry gives the signs for the opposite ordering.

If a continuous multiplier g is independent of U_e, and a bounded measurable S is also independent of U_e, left invariance of Haar measure gives

$$0=\frac{d}{d\epsilon}\int g(U)S(U)e^{-\beta E^{\mathrm{W}}(U^{e,A,\epsilon})}\,dU\Big|_{\epsilon=0}=-\beta Z_{n_{\mathrm{tor}},\beta}\int\langle A,\mathcal{E}_e\rangle_{\mathfrak{g}}gS\,d\mu_{n_{\mathrm{tor}},\beta}\,.$$

Here $U_e^{e,A,\epsilon}=e^{\epsilon A}U_e$ and all other links are fixed. Compactness bounds the derivative of the smooth density uniformly, justifying differentiation even for measurable S.

Choose a path γ_e from 0 to e^- avoiding e. If g also avoids e, then

$$\Theta(U)=\langle\xi,\mathrm{Ad}(U_{\gamma_e})\mathcal{E}_e(U)\rangle_{\mathfrak{g}}g(U)$$

has zero moment for every $\xi\in\mathfrak{g}$ against every exterior spectator. To see this, expand $\mathrm{Ad}(U_{\gamma_e}^{-1})\xi$ in a fixed orthonormal basis of $\mathfrak{g}$ and apply the preceding identity to each coefficient times gS. These multipliers are independent of U_e. There is one radius and one minimum torus side for any finite collection of such expressions: enlarge the radius to contain both endpoints of every varied link, and the side until distinct links in their finite cover supports remain distinct after wrapping. The same choices work at every placement.

Step 2: Enough measurements with avoidance. Any path can avoid one specified link: replace a traversal of that link by three sides of a square in a transverse direction. For each plaquette orientation only finitely many base points give a plaquette containing that link. Removing these points does not affect the shrinking-square argument in Lemma 10.2: for $z\neq 0$, the integer points $x_\tau=\lfloor z/\tau\rfloor$ eventually avoid any finite set, and continuity gives the conclusion at $z=0$. Thus curvature measurements whose paths and plaquettes avoid e still have injective joint linearization. Finite-dimensional duality therefore realizes every linear functional on $\mathcal{J}_{\le d}$ by finitely many components.

For every linear functional ℓ on $\mathcal{J}_j$ vanishing on K_j, a finite combination of force linearizations equals $\ell\circ\mathrm{pr}_j$ on $\mathcal{J}_{\le j+2}=\bigoplus_{i\le j}\mathcal{J}_i$. Write $N_{j;ab}$ for the square averages in the proof of Lemma 10.2, extended antisymmetrically. The force linearization at $e=(x,a)$ is

$$L_a(\mathsf{j};x)=\sum_{i,b}\big(N_{i;ab}(x,1)-N_{i;ab}(x-e_b,1)\big)\,.$$

For a homogeneous component define

$$H_{j;ab}(z,\tau) = \int_{[0,1]^3} \partial_b(\partial_a \mathcal{A}_{j,b} - \partial_b \mathcal{A}_{j,a})\big(z + \tau(\xi e_a + (\eta + \zeta - 1)e_b)\big)\, d\xi\, d\eta\, d\zeta\,.$$

The fundamental theorem of calculus in the backward direction gives

$$N_{j;ab}(z,\tau) - N_{j;ab}(z - \tau e_b, \tau) = \tau H_{j;ab}(z,\tau)\,.$$

If all $L_a(\mathrm{j}; x)$ vanish for jets through degree $j = m$, homogeneity gives $\sum_{j \le m,b} \tau^{m-j} H_{j;ab}(\tau x, \tau) = 0$ for $\tau > 0$. Taking $x = x_\tau$ and $\tau \downarrow 0$ shows $\sum_b \partial_b(\partial_a \mathcal{A}_{m,b} - \partial_b \mathcal{A}_{m,a}) = 0$, hence $\mathcal{A}_m \in K_m$. A functional of this top component annihilating K_m therefore annihilates the common kernel of the force measurements. Finite-dimensional duality puts it in their linear span. Transport paths do not enter the linearization and can be chosen to avoid their varied links. On $\mathcal{J}_{\le d}$ the same combination can have additional linear terms only at engineering weights above $m + 2$.

Step 3: Polynomial lifts and the triangular map. Choose linear projections $\pi_j : \mathcal{J}_j \to K_j$ and put $\pi = \prod_j \pi_j$. A polynomial belongs to $\mathcal{I}_d$ precisely when $p \circ \pi = 0$. For a product of linear coordinates,

$$\prod_{i=1}^{k_{\mathrm{tot}}} \ell_i - \prod_{i=1}^{k_{\mathrm{tot}}} (\ell_i \circ \pi) = \sum_{i=1}^{k_{\mathrm{tot}}} \left[\prod_{l<i} (\ell_l \circ \pi)\right] (\ell_i - \ell_i \circ \pi) \prod_{l>i} \ell_l\,.$$

Consequently every bihomogeneous equation label is a finite sum of products with one linear factor vanishing on some K_j, with the same engineering weight and factor count. The individual factors need not be invariant. For an observable O, its Taylor polynomial through engineering weight d means the degree-d Taylor polynomial at zero of $\mathrm{j} \mapsto O(U(1,\mathrm{j}))$ on $\mathcal{J}_{\le d}$, with monomials of weight above d discarded. Every coordinate has weight at least two. By (10.2), matching this polynomial is equivalent to matching the spacing derivatives through order d in (10.3).

Lift that distinguished factor by a finite combination of transported forces from Step 2. For each summand, lift the other factors by curvature measurements avoiding its varied link, matching their full linear parts through engineering weight d. The resulting expression has the zero-moment property of Step 1 and at least k_{tot} curvature factors. Indeed, every force is a difference of two transported curvatures per transverse axis. Its Taylor polynomial through engineering weight d equals the prescribed product plus terms later in (k, w): the additional linear terms of the force have greater engineering weight, and every nonlinear correction increases factor count. The square-measurement lifts have their prescribed full linear parts, so introduce only the latter kind of correction. Taylor polynomials exist by the smooth Volterra dependence used in Lemma 10.2.

Average over constant gauge transformations. All factors depend on a gauge transformation only at the origin, where they transform by adjoint matrices. Their orbit therefore lies in the finite-dimensional span of finitely many fixed-degree colour-coordinate monomials on the same measurements. Each orbit function retains zero moments, with colour vectors rotated by $\mathrm{Ad}(v^{-1})$. The Haar average is a finite linear combination of orbit functions, so preserves zero moments, minimum factor count and triangular Taylor identity, and is gauge invariant. The test-holonomy identity makes its Taylor polynomial invariant too.

In each bidegree extend a basis of $\mathcal{I}_d$ to a basis of the invariant label space in that bidegree, lifting the complementary vectors by Lemma 10.2. Lift $\mathbf{1}$ to 1, since $\mathcal{I}_d$ contains no nonzero

constant. Summing gives a linear lift $p \mapsto \mathcal{Y}_p$. Replace this lift by the two successive averages

$$\mathcal{Y}'_p(U) = \frac{1}{|H_4|} \sum_{\rho \in H_4} \mathcal{Y}_{\rho^{-1}p}(\rho^* U),$$
$$\mathcal{Y}''_p = \frac{1}{2}\left(\mathcal{Y}'_p + \overline{\mathcal{Y}'_{\bar{p}}}\right).$$

The Taylor map commutes with both operations; write $\mathcal{Y}_p$ for the final lift. The real Maxwell equations and the chain rule make K_j, and hence $\mathcal{I}_d$, stable under signed permutations, adjoint transformations and conjugation. The zero-moment property survives the signed average by (10.7), changing the spectator by the inverse isometry; the exterior set is preserved. Conjugation uses $\bar{S}$. A reversed varied link uses a right derivative of the Wilson energy, whose Haar identity is identical. Thus the averaged equation representatives remain finite sums of one-link action derivatives with link-independent multipliers and avoiding transport paths.

Let $\mathsf{R}_d p$ be the Taylor polynomial of the resulting $\mathcal{Y}_p$ through engineering weight d. Then $\mathsf{R}_d - I$ strictly raises (k, w). There are finitely many bidegrees of weight at most d, so $(\mathsf{R}_d - I)^{h_0} = 0$ for some integer h_0, and

$$\mathsf{R}_d^{-1} = \sum_{l=0}^{h_0-1} (-1)^l (\mathsf{R}_d - I)^l, \qquad V_d^{\mathrm{aux}}[c] = \mathcal{Y}_{\mathsf{R}_d^{-1} c}.$$

This gives invertibility and Taylor normalization; the averages give the symmetry and unit identities. The inverse preserves minimum factor count, so Lemma 10.3 applies to these sections with their own constants. A signed-invariant box enclosing all paths and plaquettes gives a common stencil. Take r_d, M_d as maxima for an ideal basis; linearity gives (10.8). □

11 Gauges and block integration

This section is under construction.

For $R = (j+3)\ell$, put $\mathbb{B}_R = \{-R, \dots, R\}^4$. Let $T_U(y \to z; o)$ be the ordered link product along the route changing coordinates in the order $o \in S_4$.

Definition 11.1 (Axial frames)**.** The frame tagged by (z, o), its normalization and its link coordinates are

$$g_y^{z,o} = T_U(z \to y; o), \qquad h_y^{z,o} = (g_0^{z,o})^{-1} g_y^{z,o}, \qquad U_{(y,a)}^{z,o} = g_y^{z,o} U(y,a) (g_{y+e_a}^{z,o})^{-1}.$$

These frames provide the finite alternatives for the symmetric gauge choice.

Definition 11.2 (Symmetric frame choice)**.** When every route transport in every origin-based axial gauge has chord below δ_{fr}, use the minimizer, with $h_0 = 1$, of

$$\frac{1}{24|\mathbb{B}_R|^2} \sum_{y,z \in \mathbb{B}_R} \sum_{o \in S_4} \mathrm{pr}(h_y T_U(y \to z; o) h_z^{-1}).$$

Otherwise use normalized axial frames whose tags satisfy

$$\sum_{e \text{ patch}} \mathrm{ch}(U_e^{z,o}) \le a_{\mathrm{ax}}(2R+1) \sum_{\square \text{ patch}} \mathrm{ch}(U_\square).$$

Tags have equal mass over all tags on the first branch and eligible tags on the second. This makes the choice symmetric.

Definition 11.3 (Coarse predictors)**.** Let $w^{\mathrm{av}} = (\mathrm{Unif}(\mathbb{B}_{(\ell-1)/2}))^{*j}$ and let Y_z be the framed straight connector of (3.4). Define

$$Q^{\mathrm{pred}} = \operatorname*{argmin}_{u\in G} \sum_z w_z^{\mathrm{av}} \,\mathrm{pr}(Y_z u^{-1}) \quad \text{if the minimum is below } \varepsilon_{\mathrm{pred}}\,.$$

Give predictor tags uniform mass on this branch. Otherwise set $Q^{\mathrm{pred}} = Y_z$ with tag mass w_z^{av}. The product of frame and predictor masses is $\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}})$. These choices define every summand of Definition 3.1.

Lemma 11.4 (Branches and normalization)**.** *For the chosen frame and predictor cutoffs and axial factor $a_{\mathrm{ax}} \geq 16$, every stated minimizer is unique on its branch, eligible axial tags are nonempty, and*

$$\sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) = 1, \qquad \int k_\beta^{(m)}(U, V)\, dV = 1\,.$$

The proof of Lemma 11.4 is deferred to this section's branch estimates.

The following free forms fix the quadratic normalization used in Section 12. For a link field a, write $d_1 a$ for its lattice curl and $\mathcal{Q}_m a$ for the linearization of the predictor in (3.5). Set

$$\mathsf{S}_{N,m}^{\mathrm{free}}(a) = \tfrac{1}{2}\langle d_1 a, \mathsf{H}_{N,m} d_1 a\rangle\,.$$

Let W_0 be the free stationary interpolation and put $\mathsf{K}_{N,m} = d_1^* \mathsf{H}_{N,m} d_1$.

Lemma 11.5 (Free quadratic forms)**.** *For the chosen histories, the free forms satisfy*

$$W_0^* \mathsf{K}_{N,m+1} W_0 + \mathsf{b}(\mathcal{Q}_m W_0 - I)^*(\mathcal{Q}_m W_0 - I) = \mathsf{K}_{N,m}, \qquad \widehat{\mathsf{H}}_{N,m}(0) = I,$$
$$\mathsf{H}_{N,m} \geq 2\gamma_{J_m} I, \qquad \mathsf{H}_{N,m} - c_{\mathrm{dir},J_m} I = \rho_{N,m}^* \rho_{N,m}\,.$$

Here $\gamma_k > 0$ is uniform over histories at stage k, including the top and both incoming stages. For $c_{\mathrm{dir},k} \leq \gamma_k$, the row factor is analytic on a fixed strip and has exponentially decaying coefficients, uniformly in N, m, p. At $m = N$ the free form is the identity on curls.

The proof of Lemma 11.5 is deferred to this section's free block integration.

For the analytic stationary problem, v is the coarse chart variable, z the fine axial variable, and $S(v, z)$ the fine leading action plus the block penalty, with the common coupling factor removed. Put $\mathsf{Q} = D_z^2 S(0, 0)$ and $Wv = (v, Dz(0)v)$. For the inverse estimates, use weighted operator norms on a Bloch cell. Put $\mu = \ell^{-1}$ and $k_z(\xi) = (\xi + 2\pi z)/\ell$, with $x, z \in \{0, \dots, \ell - 1\}^4$, and define

$$q_a(k) = e^{ik_a} - 1, \qquad h_z(\xi) = \big(\mu^2 + \sum_a |q_a(\mathrm{Re}\, k_z(\xi))|^2\big)^{1/2},$$
$$\mathcal{U}_\ell(\xi)_{xz} = \ell^{-2} e^{ik_z(\xi)\cdot x}, \qquad T_\ell(\xi) = \mathcal{U}_\ell(\xi)\,\mathrm{diag}(h_z(\xi)^{-1})\mathcal{U}_\ell(0)^{-1}\,.$$

These weights measure one fine derivative. Use T_ℓ on fine link or site fields, the identity on coarse coordinates, direct sums on their products, and $T_\ell(-\xi)^{-T}$ on dual fields. Axial variables inherit the norm of their slice embedding. For an operator kernel set

$$\|K\|_\eta^{\mathrm{bl}} = \sup \|T_{\mathrm{out}}(\xi_{\mathrm{out}})^{-1} K(\xi_{\mathrm{out}}, \xi_{\mathrm{in}}) T_{\mathrm{in}}(\xi_{\mathrm{in}})\|_{\mathrm{op}}\,.$$

The supremum ranges over momenta in $\mathcal{S}_\eta = \{\xi \in \mathbb{C}^4 : |\,\mathrm{Im}\,\xi_a| < \eta\}$, with all shifted momenta also in that strip. For multilinear kernels use the corresponding tensor products. Field insertions are in normalized coarse directions. The entry norms in Lemma 12.6 are taken after these rescalings. The weights themselves need not be analytic.

Lemma 11.6 (Stationary interpolation)**.** *Near zero there is an analytic stationary branch with*

$$D_z S(v, z(v)) = 0, \qquad z(0) = 0, \qquad \mathsf{Q} \geq \nu_j I, \qquad \|\mathsf{Q}^{-1}\|_{\eta_j}^{\mathrm{bl}} + \|W\|_{\eta_j}^{\mathrm{bl}} \leq E_j \,.$$

Here $\nu_j, \eta_j > 0$ and $E_j < \infty$ are uniform in the volume; the coercivity constant ν_j is chosen after the ratio. The free interpolation is $Dz(0)$. The averaging symbol in (3.5) vanishes to second order at every nonzero alias. The zero alias has the curvature normalization of Lemma 11.5.

The proof of Lemma 11.6 is deferred to this section's stationary equation and inverse estimates.

The Gaussian coordinates also determine the chart factor in the one-loop action. Let U_v be the stationary configuration, $M(v)$ its axial Hessian, s the map from axial increments to left increments, and J_{ax} the corresponding Haar-density ratio. Write $\Psi(U, h_{\mathrm{fr}})$ for the rooted minimizing frame, R for infinitesimal gauge insertion and Q for the axial slice. Define

$$X = D_{w=0}[-\log(\Psi(U, h_{\mathrm{fr}})^{-1}\Psi(e^w(U, h_{\mathrm{fr}})))], \qquad \mathcal{L}_{m+1} = \sum_{x,i} \mathfrak{a}_i^{(2)}(\,\cdot\,, x) \,.$$

These coordinates permit comparison with a gauge-completed Hessian.

Lemma 11.7 (Gaussian coordinates)**.** *Near zero, the one-loop action is*

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det M(v) - \log J_{\mathrm{ax}}(v), \qquad J_{\mathrm{ax}} = |\det Ds| > 0 \,.$$

The rooted-frame coordinates satisfy $XR = I$ and $\det(R, Q)^2 = 1$. The Hessian M is positive, and $J_{\mathrm{ax}}(0) = 1$.

The proof of Lemma 11.7 is deferred to this section's Gaussian change of coordinates.

The bounds for the third and fourth vertices separate the Wilson part from the input-dependent correction. Let V_r be the degree-r Taylor kernel of the leading fine action minus the Wilson action, $r = 2, 3, 4$. Let C_3, C_4 be the bounds after insertion of W used in Lemma 12.6.

Lemma 11.8 (Vertex bounds)**.** *The kernels V_2, V_3, V_4 have momentum orders at least $3, 2, 1$. For ℓ above a threshold depending on the input path data,*

$$C_3 \leq a_3 + \ell^{-1} a_3', \qquad C_4 \leq c_4 + \ell^{-1} c_4' \,.$$

The constants a_3, c_4 depend only on the preceding stage data; a_3', c_4' may depend on the input path bound. The mixed insertions in the stationary fourth derivative obey the same block-scaled bounds with the inverse constant E of Lemma 12.6.

The proof of Lemma 11.8 is deferred to this section's block-scaled vertex estimates.

Let Y^{con} be the difference between the quadratic one-loop kernel and the kernel with the predictor contact term removed. This term is the second derivative of the predictor multiplied by the first derivative of the penalty. The following orders explain why it does not alter the coupling coefficient.

Lemma 11.9 (Stationary Ward estimates)**.** *Along the free interpolation, the first penalty derivative vanishes to momentum order two, and the mixed fine/frame trace in the contact correction vanishes to order one. The trace with two frame insertions is analytic. Consequently, near zero,*

$$Y^{\mathrm{con}}(\xi) = O(|\xi|^3) \,.$$

The proof of Lemma 11.9 is deferred to this section's stationary gauge identities.

Finally put $\mu = \ell^{-1}$, $\Delta_U = d_U^* d_U$ and $P_\mu = \mu^2(\Delta_U + \mu^2 I)^{-1}$, with d_U as in Section 12. The inverse and insertions below are evaluated at zero background. Field differentiation inserts the free interpolated coarse field. Write A_μ for either the completed Hessian without its contact term or the folded auxiliary fine Hessian in that section.

Lemma 11.10 (Scaled inverse insertions)**.** *There are $u_j \geq 1$, $\eta_j > 0$ and $C_j^{\mathrm{ins}} < \infty$, depending only on the fixed stage bounds, such that, for every odd $\ell \geq u_j$,*

$$\sup_{\xi \in \mathcal{S}_{\eta_j}} \|P_\mu(\xi)^3\|_1 \leq C_j^{\mathrm{ins}},$$

$$\|D^r P_\mu\|_{\eta_j}^{\mathrm{bl}} + \|\mu^2 D^r (A_\mu^{-1})\|_{\eta_j}^{\mathrm{bl}} \leq C_j^{\mathrm{ins}} \qquad (0 \leq r \leq 2)\,.$$

Here $\|\cdot\|_1$ is trace norm on a Bloch cell. The block-penalty and frame insertion products in the resolvent difference satisfy the same uniform operator bounds in block units.

The proof of Lemma 11.10 is deferred to this section's free inverse and insertion estimates.

12 Path actions and the coupling reset

The path action retains curvature terms through fourth order after each integration. We choose its quadratic part so that β is the coefficient of the squared Wilson curvature at zero momentum at every level. Quadratic terms independent of β have two vanishing displacement moments. The marginal part generated by the determinant is removed from these terms and absorbed into the coupling.

The path bounds enter Theorem 3.6, and the coupling formula enters Theorem 3.5. We evaluate the leading action at its stationary background and express its cubic and quartic terms in transported curvatures. We compare the coupling coefficient of the blocked determinant with that of a regulated fine-lattice determinant, with an error uniform in the block ratio. The vector and ghost bubbles give the logarithmic coefficient; an affine equation bounds the remaining reset. Lemmas 12.5–12.10 carry out these steps. The inputs are the branches, free forms, stationary interpolation, Gaussian coordinates, vertex, Ward and insertion estimates of Lemmas 11.4–11.10. Lemmas 13.3, 13.7–13.9 supply the activity bounds, pre-reset expansion, exact density identity and continuity. Their proofs belong to Sections 11 and 13; the reset here bounds and identifies the correction in that identity.

Throughout, the group and metric are those of Definition 2.1, and the scales are those of Section 3. At a step $m+1 \to m$ put $\ell = \ell_m$ and $j = J_{m+1}$. Arrays depend on N, m and the intervening block ratios, and are chosen on $\mathbb{Z}^4$ before periodic evaluation. They are independent of the volume index p and of the running coupling. Constants declared uniform in ℓ are chosen before ℓ.

Curvature coordinates

A path transports curvature to an anchor, where invariant colour contractions can be taken. Write U_γ for the ordered link product along γ, $\mathfrak{p} = (a, b)$, $a < b$, for an orientation, and $U_{x;\mathfrak{p}}$ for a plaquette holonomy. The footprint of a slot is the set of vertices visited by its path and plaquette boundary. The footprint of an atom is the union of its slot footprints and the origin, translated to its anchor. Health means disjointness from $\mathrm{Flag}_m(U)$: no plaquette with a vertex in that footprint has chord exceeding t_m.

Definition 12.1 (Curvature slots)**.** A slot $s = (\gamma, \mathfrak{p})$, with $\gamma : 0 \to z_s$, has length $|\gamma| + 4$ and value

$$\mathcal{F}_s(U, x) = \mathrm{Ad}(U_{x+\gamma}) \operatorname{proj}_{\mathfrak{g}} \frac{U_{x+z_s;\mathfrak{p}} - U_{x+z_s;\mathfrak{p}}^{-1}}{2} .$$

Transport is from $x + z_s$ to x. Set $\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}} = \sum_i r_{\mathfrak{p}i} \mathcal{F}_{\mathfrak{p}i}$. For $k = 2, 3, 4$, the atoms $\mathfrak{a}_i^{(k)}$ are real invariant k-linear expressions in slots, masked by health on their footprints. Atom length sums slot lengths; $|\mathfrak{a}_i^{(k)}|$ sums absolute colour coefficients. These arrays retain local terms through fourth order.

Here $r_{\mathfrak{p}i} \in \mathbb{R}$ and $\mathcal{F}_{\mathfrak{p}i}$ denotes the slot assigned to row $\mathfrak{p}$ and index $i \in \mathbb{N}$. More explicitly, in the orthonormal colour basis $(e_b^{\mathfrak{g}})_b$ of Definition 2.1, an atom has the form

$$\mathfrak{a}_i^{(k)}(U, x) = \mathbf{1}_{X_i + x \text{ healthy}} \sum_{b_1, \dots, b_k} c_i(b_1, \dots, b_k) \prod_{\nu=1}^{k} \langle e_{b_\nu}^{\mathfrak{g}}, \mathcal{F}_{s_{i,\nu}}(U, x) \rangle_{\mathfrak{g}} .$$

The coefficients are invariant under the simultaneous adjoint action on all slots. The finite set X_i is the footprint just defined.

For each stage k, take $\gamma_k > 0$ from Lemma 11.5, so the free plaquette forms for the top and both incoming stages are at least $2\gamma_k I$. Clipping therefore leaves a strictly positive remainder even at $c_{\mathrm{dir}} = \gamma_k$. Write $c_{\mathrm{dir}} = c_{\mathrm{dir},J_m} > 16 c_{\mathrm{lin}} = 16 c_{\mathrm{lin},J_m} > 0$. The relaxed health test permits a larger chord farther from the anchor.

Definition 12.2 (Path action)**.** Require $c_{\mathrm{dir}} \le \min(\gamma_{J_m}, 1)$, and put

$$\mathrm{clip}_t(u) = \begin{cases} c_{\mathrm{dir}} u^2/2, & u \le t, \\ c_{\mathrm{lin}} t u, & u > t, \end{cases}$$

$$\chi_{m,x}^{\mathrm{hlth}}(U) = \mathbf{1}\{\mathrm{ch}(U_{y;\mathfrak{p}}) \le t_m e^{\mathrm{dist}(x,y)/R_{J_m}^{\mathrm{rel}}} \text{ for every } y, \mathfrak{p}\} .$$

Define

$$\begin{aligned} (P^{\mathrm{path}})_{N,m}^{\mathrm{raw}}(\beta, U) &= \beta \sum_{x,\mathfrak{p}} \Big[\mathrm{clip}_{t_m}(\mathrm{ch}(U_{x;\mathfrak{p}})) + \tfrac{1}{2} \chi_{m,x}^{\mathrm{hlth}} |\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}(U, x)|^2 \Big] \\ &\quad + \beta \sum_{x,i} (\mathfrak{a}_i^{(3)} + \mathfrak{a}_i^{(4)})(U, x) + \sum_{x,i} \mathfrak{a}_i^{(2)}(U, x), \\ P_{N,m}^{\mathrm{path}}(\beta, U) &= \frac{1}{|H_4|} \sum_{\varsigma \in H_4} (P^{\mathrm{path}})_{N,m}^{\mathrm{raw}}(\beta, \varsigma^* U) . \end{aligned}$$

This separates the kinetic coefficient from the loop correction.

The sums over x are over the torus at level m, and $i \in \mathbb{N}$. The bounds below control actual path lengths on every period.

Lemma 12.3 (Path bounds)**.** *For every N and $m \le N$, the construction below chooses arrays independent of p and β, with*

$$\sum_i |r_{\mathfrak{p}i}| e^{w_{J_m}^{\mathrm{path}} \, \mathrm{length}(\mathcal{F}_{\mathfrak{p}i})} \le A_{J_m}^{\mathrm{path}},$$

$$\sum_i |\mathfrak{a}_i^{(k)}| e^{w_{J_m}^{\mathrm{path}} \, \mathrm{length}(\mathfrak{a}_i^{(k)})} \le \begin{cases} \Lambda_m^{\mathrm{loop}}, & k = 2, \\ A_{J_m}^{\mathrm{path}}, & k = 3, 4. \end{cases}$$

All these series converge. The constants $A_k^{\mathrm{path}}, w_k^{\mathrm{path}} > 0$ depend only on the fixed stage data.

The joint proof of Lemmas 12.3 and 12.4 follows Lemma 12.9. To state the cancellations, put

$$s_{\mathfrak{pq}} = \sum_{i:\,\mathrm{orientation}(\mathcal{F}_{\mathfrak{p}i})=\mathfrak{q}} r_{\mathfrak{p}i}\,.$$

For fixed ordered orientations $\mathfrak{q},\mathfrak{r}$ and colours b_1,b_2, let $c_i^{\mathfrak{qr};b_1b_2}$ be the corresponding coefficient of the quadratic atom, zero unless its slots have those orientations. Write $z_{i,1},z_{i,2}$ for their offsets. The cancellations remove the kinetic component from the loop atoms.

Lemma 12.4 (Kinetic and quadratic cancellations)**.** *For each pair of orientations* $\mathfrak{q},\mathfrak{r}$*,*

$$c_{\mathrm{dir},J_m}\delta_{\mathfrak{qr}} + \sum_{\mathfrak{p}} s_{\mathfrak{pq}}s_{\mathfrak{pr}} = \delta_{\mathfrak{qr}}\,.$$

For each ordered orientation-colour pair,

$$\sum_i c_i^{\mathfrak{qr};b_1b_2} = 0, \qquad \sum_i c_i^{\mathfrak{qr};b_1b_2}(z_{i,2}-z_{i,1}) = 0\,.$$

All these moment series converge absolutely.

The joint proof is given after Lemma 12.9.

Fourth-order matching

Near the flat field all health indicators equal one. By $\mathrm{ch}^2 = 2\,\mathrm{pr}$ from Definition 2.1, the leading analytic action is obtained from Definition 12.2 by replacing the clipped term by $c_{\mathrm{dir}}\,\mathrm{pr}$ and omitting the indicators and loop atoms. Denote this action, without its factor β, by $\mathsf{S}_{N,m}$. Its prescribed quadratic part is

$$\frac{1}{2}\langle d_1 a, \mathsf{H}_{N,m} d_1 a\rangle, \qquad \mathsf{H}_{N,m} = c_{\mathrm{dir},J_m} I + \rho_{N,m}^{*}\rho_{N,m}, \qquad \widehat{\mathsf{H}}_{N,m}(0) = I\,.$$

Here d_1 is the lattice curl. Lemma 11.5 gives the exact adjacent free relation and the row factor, with $\mathsf{H}_{N,m} - c_{\mathrm{dir},J_m} I \geq \gamma_{J_m} I$. Placing its coefficients on coordinate paths gives $\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}$. The cell volume contributes ℓ^4 and each curvature contributes ℓ^{-2}; hence $\ell^4(\ell^{-2})^2 = 1$, preserving the coefficient β.

We need a division that expresses a gauge-invariant Taylor difference in curvatures. For a momentum ξ put $q_a(\xi) = e^{i\xi_a} - 1$; its exterior action $d_1(\xi)a = q(\xi)\wedge a$ is the curl symbol. A kernel is transverse in one leg when contraction of that leg with q is zero. Momentum conservation makes one external momentum the negative sum of the others.

Lemma 12.5 (Curl division)**.** *For $k \geq 3$ and $\sigma > 0$, there are $0 < \eta \leq \sigma$ and $K \geq 0$, depending only on k,σ, such that every transverse periodic analytic k-leg Taylor kernel bounded by $A \geq 0$ on the σ-strip factors through curls, with plaquette kernel bounded by KA on the η-strip. Its Fourier coefficients decay exponentially. For two legs, the same holds; momentum order $s+2$ gives plaquette order s for every $s \geq 0$.*

Proof. We give the division for the Taylor differences of Definition 12.2. Scalar division requires vanishing when the divided leg has zero momentum; positive exterior degree requires only closedness.

For one leg let $\delta = \iota_q$ be contraction on exterior tensors. Split a δ-closed tensor as $f = e_a \wedge u + v$, with u, v containing no e_a. Define

$$D_a f(\xi) = \frac{f(\xi) - f(\xi)|_{\xi_a=0}}{q_a(\xi)}, \qquad h_a f = e_a \wedge D_a v, \qquad R_a f = v|_{\xi_a=0}\,.$$

The identity $\delta f = 0$ gives $\delta' v = -q_a u$, where δ' contracts the other axes. Consequently

$$\delta h_a f = q_a D_a v - e_a \wedge D_a \delta' v = v - v|_{\xi_a=0} + e_a \wedge u = f - R_a f\,.$$

The remainder $R_a f$ is again closed and has no component containing e_a. Applying this in the four axes gives the homotopy

$$f = \delta \left(\sum_{a=1}^{4} h_a R_{a-1} \cdots R_1 f \right) + R_4 R_3 R_2 R_1 f\,.$$

The final remainder vanishes in positive exterior degree because it contains none of the four axes. In degree zero it is the restriction to zero momentum of this leg, which must vanish for scalar division. In that case the formula reads $f = \sum_a q_a D_a (R_{a-1} \cdots R_1 f)$.

The quotients are analytic even at the zeros of q_a. On a Fourier monomial the identity

$$\frac{e^{in\xi_a} - 1}{e^{i\xi_a} - 1} = \begin{cases} \sum_{v=0}^{n-1} e^{iv\xi_a}, & n > 0, \\ -\sum_{v=n}^{-1} e^{iv\xi_a}, & n < 0, \\ 0, & n = 0 \end{cases}$$

bounds the quotient on a smaller strip by a constant times $|n|$ times its original exponential weight. The strip loss absorbs $|n|$, so the Fourier series converges absolutely. Each division loses at most one momentum order.

For independent legs, their remaining transverse subspaces do not depend on the divided momentum, so each restriction and division preserves them. For the dependent momentum $\xi_0 = -\sum_{i=1}^{k-1} \xi_i$, first shear the coordinates to replace ξ_1 by ξ_0. The other independent contractions remain unchanged. The one-leg formula gives $\delta_0 X = f$, retaining the contractions in legs $2, \dots, k-1$. Write c_1 for contraction in leg 1. Its defect obeys $\delta_0(c_1 X) = c_1 f = 0$. Solve $\delta_0 Y = c_1 X$ by the same formula. After c_1, leg 1 has exterior degree zero, so restoring that leg will require only scalar division.

On $\xi_1 = 0$, the restriction Y_0 is δ_0-closed. Recursively solve $\delta_0 Z = Y_0$ with one fewer independent momentum and extend Z independently of ξ_1. Then $Y' = Y - \delta_0 Z$ vanishes on $\xi_1 = 0$, and scalar division gives $c_1 D = Y'$. Thus

$$V = X - \delta_0 D, \qquad \delta_0 V = f, \qquad c_1 V = c_1 X - \delta_0 Y' = 0\,.$$

All corrections preserve the other independent contractions. The two successive solves $f \mapsto X$ and $c_1 X \mapsto Y$ each raise the dependent exterior degree by one; contraction c_1 changes only leg 1. Thus the restricted problem for Y_0 has one fewer independent momentum and dependent degree two higher. For three or more legs it terminates above exterior degree four, where the tensor is zero. The independent-leg divisions then finish the factorization.

For two legs, the restriction Y_0 is at zero external momentum. Starting from order $s+2$, the first division loses one order, contraction restores one, and the solve for Y leaves order $s+1 \geq 1$. Hence $Y_0 = 0$ directly. The correction and final division leave order s. This verifies the scalar vanishing hypothesis where it is needed. For the Taylor differences of Definition 12.2, gauge invariance supplies transversality because all lower Taylor differences vanish. □

For $z = (z_1, \dots, z_{k-1})$, contour displacement gives $|K(z)| \le Ce^{-\zeta \sum_\nu |z_\nu|_1}$. Choose the anchor at zero and coordinate paths to z_ν. Their total slot length is at most $4k + \sum_\nu |z_\nu|_1$. Every path width below ζ therefore gives a convergent weighted sum. Reality and simultaneous adjoint invariance pass to the coefficients by conjugation and invariant projection.

At a step, let v be the coarse chart variable and z the fine axial variable. Let $S(v, z)$ be the fine leading action plus $\mathsf{b} \sum_e \mathrm{pr}$ of the predictor discrepancy, without β'. It excludes the inherited loop action. Lemma 11.6 supplies

$$D_z S(v, z(v)) = 0, \qquad z(0) = 0, \qquad \mathsf{Q} = D_z^2 S(0,0) \ge \nu_j I, \qquad \nu_j > 0\,.$$

Put $Wv = (v, Dz(0)v)$ and $\mathsf{B}(v, v)[u] = D^3 S(0)[Wv, Wv, (0, u)]$. Let C_3, C_4 bound the kernels of $D^3 S(0)[(W\cdot)^3]$ and $D^4 S(0)[(W\cdot)^4]$ in the supremum norm of their block-scaled Fourier entries on the fixed smaller strip. The inverse and mixed-insertion bounds of Lemmas 11.6, 11.8 give E. The constants a_3, c_4, E depend only on preceding stage data; a_3', c_4' may also depend on the input path bound.

The next formula constructs the higher arrays while keeping their bound independent of the input amplitude after the block ratio has been enlarged.

Lemma 12.6 (Classical matching)**.** *For ℓ above a threshold depending on the input path data, the stationary action $\mathsf{S}_{\mathrm{cl}}(v) = S(v, z(v))$ satisfies*

$$\begin{aligned} D^3 \mathsf{S}_{\mathrm{cl}}(0)[v^3] &= D^3 S(0)[(Wv)^3], \\ D^4 \mathsf{S}_{\mathrm{cl}}(0)[v^4] &= D^4 S(0)[(Wv)^4] - 3\mathsf{B}(v, v)\mathsf{Q}^{-1}\mathsf{B}(v, v)\,. \end{aligned}$$

Its factorial-normalized kernels have bounds $C_3/6$ and $(C_4 + 3C_3^2 E)/24$, where

$$C_3 \le a_3 + \ell^{-1} a_3', \qquad C_4 \le c_4 + \ell^{-1} c_4'\,.$$

The fine leading vertices of degrees $2, 3, 4$ differ from Wilson by momentum orders $3, 2, 1$.

Proof. Use the block map of Definition 3.1 and Lemmas 11.6, 11.8. Differentiating $D_z S = 0$ once gives $D^2 S(0)[Wv, (0, u)] = 0$; differentiating twice gives $D^2 z(0)[v, v] = -\mathsf{Q}^{-1}\mathsf{B}(v, v)$. In the third derivative of $S(v, z(v))$, the terms containing $D^2 z$ or $D^3 z$ vanish by these two stationary identities. In the fourth derivative the surviving terms involving $w = D^2 z(0)[v, v]$ are

$$6\mathsf{B}(v, v)[w] + 3\langle w, \mathsf{Q}w\rangle = -3\mathsf{B}(v, v)\mathsf{Q}^{-1}\mathsf{B}(v, v)\,.$$

The other term is $D^4 S(0)[(Wv)^4]$. The block-scaled inverse and interpolation bounds give the two displayed kernel estimates, after division by 3! and 4!.

The fine-minus-Wilson vertices have orders $3, 2, 1$ by Lemma 11.8. The quadratic order uses $\widehat{\mathsf{H}}(0) = I$ and two curls. Differentiated gauge invariance then gives the higher orders, including nonlinear curvature transport. Wilson has orders $2, 1, 0$: the extra power becomes ℓ^{-1} under free interpolation and rescaling. The Wilson vertices and penalty give a_3, c_4, independently of the input atom bound; the differences give a_3', c_4'.

Increase ℓ until $\ell^{-1} a_3', \ell^{-1} c_4' \le 1$. Apply Lemma 12.5 to the cubic difference between S_{cl} and the output row action. Subtract the resulting full nonlinear cubic atoms before treating degree four. The remaining quartic difference is transverse, and a second application gives quartic atoms with a fixed bound. This makes the two actions agree through degree four. At $m = N$, the free form acts as the identity on curls, so the same construction matches the Wilson energy through degree four and starts with zero loop atoms. Lemma 13.6 places the remaining top-level difference, including clipping and unhealthy configurations, in the initial activities. □

The common factor β' in the penalty and fine leading action cancels from the stationary equation; hence the arrays are independent of β'. In $\log\det(\beta' M)$, the term $(\dim M)\log\beta'$ is independent of the field and contributes only to the vacuum scalar.

The determinant comparison

The Gaussian change of coordinates in Lemma 11.7 gives the one-loop action

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det M(v) - \log J_{\mathrm{ax}}(v)\,.$$

Here U_v is the stationary fine configuration, $\mathcal{L}_{m+1}$ is the inherited quadratic loop action, M is the axial Hessian without β', and J_{ax} is the Haar chart ratio, normalized to one at zero.

For a quadratic Taylor kernel Y, including the Taylor factor $1/2!$, define

$$[t^2]g(t) = \tfrac{1}{2}g''(0), \qquad \alpha_b(Y) = 2\,\mathrm{Re}[t^2]Y_{(2,b),(2,b)}(te_1), \qquad \alpha_{N,m} = \alpha_b(Y^{\mathrm{blk}}_{N,m})\,.$$

The kernel $Y^{\mathrm{blk}}_{N,m}$ is the infinite-volume quadratic kernel of T. Its periodic evaluation has a small finite-period error retained in the activities. The prescribed kinetic kernel has $\alpha_b = 1$. Simplicity makes $\alpha_{N,m}$ independent of b, as shown in the proof of Lemma 12.4; by construction it is independent of p and β'.

Let J be the left-coordinate Hessian of the joint fine action and block penalty, including the coarse frame. Let R insert infinitesimal gauge transformations and Q the axial slice. For $\mu > 0$, put

$$(d_U\phi)_e = \phi(e^-) - \mathrm{Ad}(U_e)\phi(e^+), \qquad \Delta_U = d_U^* d_U, \qquad \mathfrak{m}_\mu(U) = \mu^2[\mu^2(\Delta_U + \mu^2 I)^{-1}]^9\,.$$

For the rooted frame Ψ of Lemma 11.7, define

$$X = D_{w=0}\big[-\log\big(\Psi(U,h_{\mathrm{fr}})^{-1}\Psi(e^w(U,h_{\mathrm{fr}}))\big)\big]\,.$$

Frame covariance gives $XR = I$. With P_f the projection to fine links, set

$$F_\mu = d_U^* P_f + \mathfrak{m}_\mu X, \qquad \mathsf{C}_\mu = \mathsf{J} + F_\mu^* F_\mu, \qquad \mathsf{G}_\mu = \Delta_U + \mathfrak{m}_\mu\,.$$

The completed Hessian has the following local determinant identity.

Lemma 12.7 (Gauge completion)**.** *For v in a neighbourhood of zero,*

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det\mathsf{C}_\mu(v) - \log\det\mathsf{G}_\mu(v)\,. \tag{12.1}$$

Proof. Use Definitions 2.1, 3.1 and Lemmas 11.6–11.7. Gauge invariance and stationarity give $\mathsf{J}R = 0$. Ordering the axial tree by height makes the gauge block triangular with diagonal entries ± 1, while the slice block is the identity; hence $\det(R,Q)^2 = 1$. Since $F_\mu R = \mathsf{G}_\mu$, in these columns the completed form is

$$\begin{pmatrix} \mathsf{G}_\mu & F_\mu Q \\ 0 & I \end{pmatrix}^{*} \begin{pmatrix} I & 0 \\ 0 & Q^*\mathsf{J}Q \end{pmatrix} \begin{pmatrix} \mathsf{G}_\mu & F_\mu Q \\ 0 & I \end{pmatrix}.$$

Thus $\det\mathsf{C}_\mu = (\det\mathsf{G}_\mu)^2\det(Q^*\mathsf{J}Q)$. If s changes axial coordinates to left increments at the stationary point, then

$$\det M = (\det Ds)^2\det(Q^*\mathsf{J}Q), \qquad J_{\mathrm{ax}} = |\det Ds|\,.$$

The second equality follows by comparing the two Haar densities. Positivity of M, J_{ax} and G_μ near zero permits logarithms and proves (12.1). □

The auxiliary fine determinant used for comparison is

$$\Pi_\mu(U) = \tfrac{1}{2}\log\det(\mathsf{A}(U) + d_U d_U^* + \mathsf{M}_\mu(U)) - \log\det(\Delta_U + \mathfrak{m}_\mu(U))\,.$$

Here A is the left-coordinate Hessian of the fine leading action and M_μ applies $\mathfrak{m}_\mu$ to each link component. Let Y_μ be its infinite-volume quadratic Taylor kernel. The following comparison controls the coupling coefficient. Enlarge the threshold u_j of Lemma 11.10 to include that of Lemma 11.8; rows 5 and 8 of Section 9 choose odd ratios above it.

Lemma 12.8 (Block-to-fine coefficient comparison)**.** *For every odd $\ell \geq u_j$ as in rows 5 and 8 of Section 9, put $\mu = \ell^{-1}$. There is a constant $d_{\mathrm{cmp}} \geq 0$, depending only on the fixed stage bounds, such that*

$$|\alpha_{N,m} - \alpha_b(Y_\mu)| \leq d_{\mathrm{cmp}}\,.$$

The constant is independent of N, p, ℓ.

Proof. Use Lemmas 12.4, 12.7 and Lemmas 11.6, 11.8–11.10. Here Lemma 12.4 is used for the input arrays, in descending induction from $m = N$. The full completed Hessian contains a contact term: the second derivative of the predictor multiplied by the first derivative of the penalty. Along the free interpolation, write $\mathsf{a}(\xi)$ for this penalty derivative and $\mathsf{b}(\xi)$ for the mixed fine/frame trace. Lemma 11.9 gives $\mathsf{a}(\xi) = O(|\xi|^2)$ and $\mathsf{b}(\xi) = O(|\xi|)$. The change in the tested entry on removing the contact term consists of ab, its reflected counterpart, and a term quadratic in a, with bounded analytic factors. It is $O(|\xi|^3) + O(|\xi|^4)$ and has zero α_b.

For the remaining comparison, let D denote external field differentiation. Differentiate the logarithm and use the resolvent identity:

$$D^2 \log\det A_0 = \mathrm{tr}(A_0^{-1}D^2A_0 - A_0^{-1}DA_0\,A_0^{-1}DA_0), \qquad A_0^{-1} - A_1^{-1} = A_0^{-1}(A_1 - A_0)A_1^{-1}\,.$$

The difference consists of the block penalty, coarse-frame terms and regulator terms. The block penalty factors through the coarse links and the frame terms through the coarse frame variables: their dimensions per block are $4d_{\mathfrak{g}}$ and $d_{\mathfrak{g}}$. Lemma 11.10 bounds the resulting products in the block-scaled operator norms. Its trace is therefore bounded by that dimension times the norm, independently of ℓ.

For the regulator put $P_\mu = \mu^2(\Delta_U + \mu^2 I)^{-1}$. At zero background the aliases $k_z = (\xi + 2\pi z)/\ell$, in centered representatives, obey

$$\frac{\mu^2}{|q(k_z)|^2 + \mu^2} \leq \frac{C}{1 + |z|^2}, \qquad \mathrm{tr}\,|P_\mu^3| \leq C\sum_{z\in\mathbb{Z}^4}(1 + |z|^2)^{-3} < \infty\,.$$

The same bounds hold on a fixed smaller complex strip in coarse momentum. In $P_\mu^9 = (P_\mu^3)^3$, two derivatives leave at least one undifferentiated cube. A regulator insertion has the prefactor μ^2, paired with an inverse bounded in block units by $C\mu^{-2}$. Lemma 11.10 supplies these bounds through two field insertions for the inverse and P_μ. Thus the two powers cancel before taking the trace. Put the undifferentiated cube in trace norm and all other factors in operator norm; $|\,\mathrm{tr}\,XY| \leq \mathrm{tr}\,|X|\,\|Y\|$ gives a uniform bound C_* for the analytic difference kernel on a strip of width $\zeta_* > 0$.

To extract the coefficient, restrict its tested entry to $\xi = te_1$. On a circle $|t| = \zeta_*/2$, the analytic bound gives $|[t^2]g| \leq 4C_*/\zeta_*^2$; hence $|\alpha_b(g)| \leq 8C_*/\zeta_*^2$. This gives d_{cmp}. The estimate applies after the zero-coefficient contact term has been removed.

Finally, folding the auxiliary operator into coarse momentum does not alter α_b. The symbol of the weights w^{av} in (3.5) vanishes at every nonzero alias to order at least two in each external leg. These aliases contribute $O(|\xi|^4)$. The zero alias in the tested entry is

$$\ell^4\ell^{-2}Y_\mu(\xi/\ell) + O(|\xi|^3)\,.$$

The factors are the cell volume and ℓ^{-1} per interpolated field. Extracting the quadratic coefficient supplies another ℓ^{-2}, so $\ell^4\ell^{-2}\ell^{-2} = 1$. This proves the comparison. For a period of side n, Fourier coefficients at nonzero multiples of n give an error in each quadratic coefficient tested on Fourier basis modes bounded by $C_\ell|\Lambda|e^{-c_\ell n}$. The Fourier basis modes here have unit amplitude at every site, so the bound includes the volume factor; the preparatory expansion of Lemma 13.7 retains it in the activities. □

The logarithmic coefficient

The vertex comparison in the proof of Lemma 12.6 gives momentum orders $3, 2, 1$ for the quadratic, cubic and quartic differences from Wilson. We use it for the input arrays, in descending induction from $m = N$. The resulting logarithmic term is the part of the one-step drift which remains after the block ratio has been chosen.

Lemma 12.9 (One-loop coefficient)**.** *There is a constant $d_j \geq 0$, depending only on the fixed stage data, such that, for every step $m+1 \to m$ at stage j with odd $\ell_m \geq u_j$ as in Section 9,*

$$|\alpha_{N,m} + \kappa_G \log \ell_m| \leq d_j\,.$$

It is independent of N, p, ℓ_m and the running coupling.

Proof. Use Definition 2.1, Lemmas 12.4, 12.8 and the vertex orders just described. Choose a colour basis index b, external polarization e_2 and external momentum te_1. Signed permutations reduce other transverse axis pairs to this one. For nonzero loop momentum k, write $\varkappa = |k|$. The leading ghost vertex is $-2ik_2\,\mathrm{ad}(e_b^{\mathfrak{g}})$. The vector vertex has the same orbital term on its four components and the spin term

$$2it(\delta_{i2}\delta_{j1} - \delta_{i1}\delta_{j2})\,\mathrm{ad}(e_b^{\mathfrak{g}})\,.$$

Simplicity and the trace metric give $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}(e_b^{\mathfrak{g}})^2) = -C_{\mathfrak{g}}$. The bubble pairs vertices at te_1 and $-te_1$. The orbital factor $-2ik_2$ is unchanged when $k \mapsto k + te_1$ and $te_1 \mapsto -te_1$. Put $S_{\mathrm{sp}} = E_{21} - E_{12}$. Since $\mathrm{tr}\,S_{\mathrm{sp}} = 0$, the orbital–spin cross terms vanish; the spin product is $(2itS_{\mathrm{sp}})(-2itS_{\mathrm{sp}}) = 4t^2S_{\mathrm{sp}}^2$, with trace $-8t^2$. The directional traces of the paired vertices are therefore $4(-2ik_2)^2 - 8t^2$ for the vector and $(-2ik_2)^2$ for the ghost. Divide both by $\varkappa^2|k+te_1|^2$ and differentiate twice at zero. In $\frac{1}{2}\log\det \mathsf{C}_\mu - \log\det \mathsf{G}_\mu$, the bubble terms have weights $-1/4$ for the vector and $+1/2$ for the ghost. The extraction of α_b supplies the factor 2. Thus ghost minus half vector, with the external Taylor factor included, gives

$$h(k) = 4C_{\mathfrak{g}}\left(\frac{k_2^2}{\varkappa^6} - \frac{4k_2^2k_1^2}{\varkappa^8} - \frac{1}{\varkappa^4}\right)\,. \tag{12.2}$$

Expanding the denominator gives $\varkappa^{-4} - 2tk_1\varkappa^{-6} + t^2(4k_1^2\varkappa^{-8} - \varkappa^{-6}) + O(t^3)$; substitution yields (12.2) with its sign.

Let $k_{\mu,b}(k)$ be the integrand of $\alpha_b(Y_\mu)$, obtained with these weights and Taylor factors. Then

$$\alpha_b(Y_\mu) = (2\pi)^{-4}\int_{[-\pi,\pi]^4} \mathrm{Re}\,k_{\mu,b}(k)\,dk\,.$$

Under construction

For constants $A_{\mathrm{int}}, D_{\mathrm{int}} \geq 0$ depending only on the fixed fine-stage bounds, uniformly in μ,

$$|k_{\mu,b}(k)| \leq \frac{A_{\mathrm{int}}}{(\varkappa+\mu)^4}, \qquad |k_{\mu,b}(k) - h(k)| \leq D_{\mathrm{int}} \left(\varkappa^{-3} + \mu^2 \varkappa^{-6}\right) \quad (\varkappa \geq \mu)\,. \tag{12.3}$$

To obtain them, use $q_a(k) = ik_a + O(\varkappa^2)$ and the vertex orders above. For the free inverse $R_\mu(k)$, differentiation of $R_\mu A_\mu = I$ gives

$$\|R_\mu\| \leq C(\varkappa+\mu)^{-2}, \quad \|\partial R_\mu\| \leq C(\varkappa+\mu)^{-3}, \quad \|\partial^2 R_\mu\| \leq C(\varkappa+\mu)^{-4}\,.$$

Each first-order vertex has order $\varkappa$, with bounded first external derivative. In a bubble, two momentum derivatives on one inverse give $\varkappa^2(\varkappa+\mu)^{-2}(\varkappa+\mu)^{-4} \leq (\varkappa+\mu)^{-4}$; moving derivatives to vertices gives the same bound. On $\varkappa \geq \mu$, a lattice-to-leading replacement gains one power of $\varkappa$, giving $C\varkappa^{-3}$. A regulator insertion contributes at most $C\mu^2\varkappa^{-6}$. The terms with a single second-order vertex are bounded by $C(\varkappa^{-2} + \mu^2\varkappa^{-6})$ and so satisfy the same bound on the bounded momentum cube. The product rule gives the same bounds for the ninth-power regulator. Summing proves (12.3).

On the spherical shell $\mu < \varkappa \leq 1$, rotational symmetry gives angular means $k_2^2/\varkappa^2 = 1/4$ and $k_2^2 k_1^2/\varkappa^4 = 1/24$. For the latter, write the fourth moments as $a = \mathbb{E}\theta_1^4$ and $b = \mathbb{E}\theta_1^2\theta_2^2$ on the unit sphere: rotation by $\pi/4$ gives $a = 3b$, and $(\sum_i \theta_i^2)^2 = 1$ gives $4a + 12b = 1$. Since the sphere has area $2\pi^2$,

$$\begin{aligned}(2\pi)^{-4} \int_{\mu<\varkappa\leq 1} h(k)\, dk &= \frac{8\pi^2 C_{\mathfrak{g}}}{(2\pi)^4}\left(\frac{1}{4} - \frac{1}{6} - 1\right)\log \mu^{-1} \\ &= -\frac{11 C_{\mathfrak{g}}}{24\pi^2}\log \mu^{-1} = -\kappa_G \log \mu^{-1}\,.\end{aligned} \tag{12.4}$$

The errors in (12.3) have shell integrals at most $2\pi^2 D_{\mathrm{int}}$ and $\pi^2 D_{\mathrm{int}}$. On $\varkappa \leq \mu$ the bound is $A_{\mathrm{int}}\mu^{-4}$ and the volume is $\pi^2\mu^4/2$; outside $\varkappa \leq 1$ the bound is A_{int}. Hence

$$|\alpha_b(Y_\mu) + \kappa_G \log \mu^{-1}| \leq (2\pi)^{-4}\{A_{\mathrm{int}}\pi^2/2 + A_{\mathrm{int}}(2\pi)^4 + 3\pi^2 D_{\mathrm{int}}\}\,.$$

Combine this bound with Lemma 12.8 and put $\mu = \ell^{-1}$. Their sum gives d_j. The computed slope is $\kappa_G = 2b_0$ of Definition 2.1, with its stated trace metric. □

Proof of Lemmas 12.3 and 12.4. Use Definitions 2.1, 3.1, 12.1–12.2, the free forms of Lemma 11.5, and Lemmas 12.5, 12.6, 12.9. Start at $m = N$ and descend. The condition $c_{\mathrm{dir},k} \leq \gamma_k$ supplies the positive row factor at every level. Its zero-momentum entries give

$$\widehat{\mathsf{H}}(0)_{\mathfrak{q}\mathfrak{r}} = c_{\mathrm{dir},J_m}\delta_{\mathfrak{q}\mathfrak{r}} + \sum_{\mathfrak{p}} s_{\mathfrak{p}\mathfrak{q}} s_{\mathfrak{p}\mathfrak{r}} = \delta_{\mathfrak{q}\mathfrak{r}}\,.$$

This proves the first cancellation and also forces $c_{\mathrm{dir},J_m} \leq 1$ by taking a diagonal entry.

Subtract $\alpha_{N,m}$ times the kinetic kernel from the quadratic kernel of T. The residual is transverse. Its colour bilinear form is scalar: its self-adjoint representing operator commutes with the adjoint action, and its eigenspaces are ideals, so simplicity leaves one eigenvalue. Signed permutations identify the diagonal orientation coefficients. Reflection in an axis belonging to exactly one of two distinct orientations cancels the off-diagonal coefficient. Thus there is one degree-two coefficient, removed by the subtraction. Reflection through the origin removes degree three. The residual link kernel consequently vanishes to order four.

Apply the two-leg part of Lemma 12.5. Removing its two curl symbols leaves a plaquette kernel of order two. If its Fourier coefficients for a fixed orientation-colour pair are $a_{\mathrm{loop}}(z)$, then

$$\sum_z a_{\mathrm{loop}}(z) = 0, \qquad \sum_z z_a a_{\mathrm{loop}}(z) = 0 \quad (a = 1, \dots, 4) .$$

Exponential decay permits termwise differentiation. Put the coefficients on coordinate paths to obtain the quadratic atoms and their absolutely convergent moments. Inherited loop atoms already have these moments, hence a link kernel of order four. Pullback by the free interpolation preserves this order. Only the quadratic Taylor kernel of T is assigned to these atoms; its higher terms enter the preparatory expansion of Lemma 13.7.

For the bounds, contour displacement and the path-length estimate after Lemma 12.5 give a sufficiently small width $w_k^{\mathrm{path}} > 0$. Choose A_k^{path} to bound the top construction and the two possible incoming stages $k, k+1$. Lemma 12.6 permits this choice before the corresponding ratios, by making the input-dependent terms at most one. Choose the widths small enough for these same three constructions.

The new loop contribution has a weighted bound $a_j \ell^4 + b_j \log \ell + c_j$. Here ℓ^4 bounds the trace dimension per coarse cell, and the logarithm comes from the subtracted coefficient. For an inherited order-four link kernel, rescaling gives

$$\ell^4 \ell^{-2} O(|\xi/\ell|^4) = O(\ell^{-2}|\xi|^4) .$$

Its weighted pullback is therefore bounded by $C_j^{\mathrm{pb}} \ell^{-2} \Lambda_{m+1}^{\mathrm{loop}}$. Choose ℓ so that $C_j^{\mathrm{pb}} \ell^{-2} \le 1/4$, then raise the coupling entry threshold until

$$2(a_j \ell^4 + b_j \log \ell + c_j) \le \Lambda_m^{\mathrm{loop}}, \qquad \Lambda_{m+1}^{\mathrm{loop}} \le 2\Lambda_m^{\mathrm{loop}} .$$

The sum is at most $\frac{1}{2}\Lambda_m^{\mathrm{loop}} + \frac{1}{4} \cdot 2\Lambda_m^{\mathrm{loop}} = \Lambda_m^{\mathrm{loop}}$. These choices are in the order of Section 9: widths and amplitudes, ratios, then entry thresholds. The first inequality is possible because $2(M_m^{\mathrm{cut}})^{\theta_{J_m}} \to \infty$ as $\widehat{\beta}_m \to \infty$ with ℓ_m, J_m fixed. For the second, antitonicity of θ_k, $J_m \le J_{m+1} \le J_m + 1$, and $\theta_k \le 1$ give, once $M_m^{\mathrm{cut}} > 1$,

$$\frac{\Lambda_{m+1}^{\mathrm{loop}}}{\Lambda_m^{\mathrm{loop}}} \le \left(\frac{M_{m+1}^{\mathrm{cut}}}{M_m^{\mathrm{cut}}} \right)^{\theta_{J_m}} \le \frac{M_{m+1}^{\mathrm{cut}}}{M_m^{\mathrm{cut}}} \longrightarrow 1$$

as $\widehat{\beta}_m \to \infty$ with ℓ_m fixed. The descent starts with zero loops and the fourth-order Wilson matching of Lemma 12.6. The kernels and paths are chosen once for all periods. Finally $|\mathcal{F}_s| \le 2$ and $|\mathfrak{a}_i^{(k)}(U, x)| \le 2^k |\mathfrak{a}_i^{(k)}|$ give absolute convergence of the action. □

The scalar reset

Use Definitions 3.2, 3.3–3.4 and Lemmas 12.3–12.4. Write $P_{N,m}^{\mathrm{path}}(\beta, U) = \beta \mathsf{K}_m^{\mathrm{kin}}(U) + \mathcal{L}_m(U)$. For an input curve ρ'_u of Theorem 3.5, put $\mathcal{H}_u = \mathcal{B}_{\beta'_u}^{(m)} \operatorname{dens} \rho'_u$ and $b_u = \beta'_u + \alpha_{N,m}$. Lemma 13.8 supplies regular and forced sums $E_u^{\mathrm{sum}}, \mathcal{F}_u$, independent of d, with trial density

$$q_{u,d}(U) = e^{-(b_u + d)\mathsf{K}_m^{\mathrm{kin}}(U) - \mathcal{L}_m(U) - E_u^{\mathrm{sum}}(U)} \mathcal{F}_u(U) .$$

The flat-field normalizations give $q_{u,d}(1) = 1$. Fix $r_{\mathrm{res}} = 1$ before $\omega_j^{\mathrm{RG}} \ge d_j + 1$ as in Section 9.

Lemma 13.7 gives a preparatory sum E_u^{pre} with bound $c_{\mathrm{pre}} \epsilon_m^{\mathrm{reg}}/4$, for a stage constant $c_{\mathrm{pre}} \ge 4$. Lemmas 13.3, 13.8 give the final bound (3.12) and $q_{u,\Delta_u} = \mathcal{H}_u / \mathcal{H}_u(1)$ for some real Δ_u. The estimate below proves $|\Delta_u| \le 1$; that bound is no input to the identity.

The remaining scalar correction can be read from a single configuration with a definite kinetic energy.

Lemma 12.10 (Coupling reset)**.** *For each input curve of Theorem 3.5, the coupling in the representation of Lemma 13.8 is continuous and satisfies*

$$\beta_u = \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le d_j + 1 \le \omega_j^{\mathrm{RG}} .$$

The bound is independent of N, p, u and the input curve.

Proof. Use Definitions 2.1, 3.2, Lemmas 12.3, 12.9, and Lemmas 13.3, 13.7–13.9. Recall $t = t_m \le 1$ from row 7 of Section 9. Put a group element of chord $h_{\mathrm{pr}} = \tau t$ on direction-1 links whose second coordinate is even, and the identity elsewhere. Continuity along a nonconstant one-parameter subgroup gives $\varepsilon_G > 0$ such that every chord in $[0, \varepsilon_G]$ occurs. Choose

$$0 < \tau \le \min\left(\varepsilon_G, \frac{1}{4}, \frac{c_{\mathrm{dir},J_m}}{512(1 + A_{J_m}^{\mathrm{path}})}\right) .$$

Every $(1,2)$ plaquette has chord h_{pr} and all other plaquettes have chord zero. Thus this configuration V is everywhere healthy. All periods here are even because $4 \mid L_p$. Signed permutations preserve the sum of direct plaquette charges, so the direct contribution to $\mathsf{K}_m^{\mathrm{kin}}(V)$ is at least $|\Lambda| c_{\mathrm{dir},J_m} h_{\mathrm{pr}}^2/2$. Row squares are nonnegative. Using the bound $4h_{\mathrm{pr}} \le 1$ for the curvature factors, the cubic and quartic sums contribute at most

$$|\Lambda| A_{J_m}^{\mathrm{path}} ((4h_{\mathrm{pr}})^3 + (4h_{\mathrm{pr}})^4) \le |\Lambda| c_{\mathrm{dir},J_m} h_{\mathrm{pr}}^2/4 .$$

The same estimate holds after averaging. Consequently

$$K_{\mathrm{pr}} = \mathsf{K}_m^{\mathrm{kin}}(V) \ge |\Lambda| \chi_{\mathrm{pr}} t^2, \qquad \chi_{\mathrm{pr}} = c_{\mathrm{dir},J_m} \tau^2/4 > 0 .$$

Lemma 13.7 gives $\mathcal{H}_u(1), \mathcal{H}_u(V) > 0$. Put $v_u = |\Lambda|^{-1} \log \mathcal{H}_u(1)$ and $\varepsilon = \epsilon_m^{\mathrm{reg}}$. Its $\mathsf{K}_{2,m}^{\mathrm{kin}}$ is the nonlinear direct and row functional, omitting cubic and quartic atoms. Taking its logarithmic contrast gives

$$\log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} = \beta'_u \mathsf{K}_m^{\mathrm{kin}}(V) + \alpha_{N,m} \mathsf{K}_{2,m}^{\mathrm{kin}}(V) + \mathcal{L}_m(V) + E_u^{\mathrm{pre}}(V) - E_u^{\mathrm{pre}}(1) . \tag{12.5}$$

The pre-reset bound of Lemma 13.7, summed over anchors as in Definition 3.4, bounds each of the last two values by $|\Lambda| c_{\mathrm{pre}} \varepsilon/4$. No vanishing condition at 1 is needed for this preparatory sum.

On everywhere-healthy configurations every curvature slot has norm at most t. Lemma 12.3 and $t \le 1$ therefore give

$$|\mathsf{K}_m^{\mathrm{kin}} - \mathsf{K}_{2,m}^{\mathrm{kin}}| \le 2|\Lambda| A_{J_m}^{\mathrm{path}} t^3 .$$

After ℓ is fixed, Lemma 12.9 bounds $|\alpha_{N,m}|$ by a stage constant. Raise the coupling entry threshold until $2|\alpha_{N,m}| A_{J_m}^{\mathrm{path}} t^3 \le c_{\mathrm{pre}} \varepsilon/2$. This is possible since $t^3/\varepsilon = \widehat{\beta}_m^{-2/5} (M_m^{\mathrm{cut}})^3 \to 0$. Together with (12.5) this yields

$$\left| b_u K_{\mathrm{pr}} + \mathcal{L}_m(V) - \log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} \right| \le |\Lambda| c_{\mathrm{pre}} \varepsilon .$$

For the final regular sum, Lemma 13.3 and (3.12) give

$$|E_u^{\mathrm{sum}}(V)| \le \sum_x \sum_{o_i = x} e_i^{\mathrm{maj}} \le |\Lambda| \mathcal{E}_m \le |\Lambda| \varepsilon .$$

Since the activity families are fixed as d varies, the scalar discrepancy is affine:

$$\begin{aligned} f(u,d) &= (b_u + d)K_{\mathrm{pr}} + \mathcal{L}_m(V) + E_u^{\mathrm{sum}}(V) - \log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} \\ &= f(u,0) + dK_{\mathrm{pr}}\,. \end{aligned}$$

Its unique root satisfies

$$d_u = -f(u,0)/K_{\mathrm{pr}}, \qquad |d_u| \le \frac{(c_{\mathrm{pre}}+1)\epsilon_m^{\mathrm{reg}}}{\chi_{\mathrm{pr}} t_m^2} \le 1\,.$$

Increase the entry threshold to obtain the final inequality, using $\epsilon_m^{\mathrm{reg}}/t_m^2 = \widehat{\beta}_m^{-1/10}(M_m^{\mathrm{cut}})^{-2} \to 0$. The block integral and activity sums are continuous in u by Lemma 13.9, so d_u is continuous.

Since V has no flagged sites, $\mathcal{F}_u(V) = 1$. Evaluating the identity of Lemma 13.8 at V therefore gives $f(u, \Delta_u) = 0$. Uniqueness gives $\Delta_u = d_u$. Hence multiplication by $\mathcal{H}_u(1) = e^{v_u|\Lambda|}$ gives the density identity at d_u. Its coupling is $\beta_u = \beta'_u + \alpha_{N,m} + d_u$. Lemma 12.9 and $|d_u| \le 1$ prove the claimed drift. □

13 Activities and the inductive step

This section is under construction.

Definition 13.1 (Regular activities)**.** To be stated here.

Definition 13.2 (Forced activities)**.** To be stated here.

Lemma 13.3 (Activity bounds and covariance)**.** *For every representation in the chosen classes, both activity families are countable and equivariant under translations and H_4, including supports, anchors, inventories and tests. The nonnegative majorants of Definitions 3.3–3.4 have convergent weighted series and satisfy*

$$\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}}\,.$$

The proof of Lemma 13.3 is deferred to this section's activity estimates.

Lemma 13.4 (Specified recursion)**.** *To be stated here.*

Definition 13.5 (Continuity of representations)**.** A curve has fixed activity addresses and tests, and continuous coupling and vacuum scalar. At u_0, the regular derivatives of $E_i(u) - E_i(u_0)$ and forced values of $K_i(u) - K_i(u_0)$ have nonnegative majorants $e_i(u), k_i(u)$ in the sense of Definition 3.3, with

$$\sum_i e^{\sigma_{J_m} \operatorname{cov}_{B_m}(X_i)} e_i(u) + \sum_i e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Y_i)} k_i(u) \longrightarrow 0 \quad (u \to u_0)\,.$$

Both sums run over all addresses. This gives continuity for coupling tuning without differentiating flag boundaries.

The initial activities retain the difference between the Wilson energy and the clipped path action, including configurations with flagged plaquettes.

Lemma 13.6 (Wilson initialization)**.** *For every N, p, there is a continuous curve $\beta \mapsto \rho_\beta^{\mathrm{W}} \in \mathcal{I}_{N,p}$ on $\mathfrak{B}_N$ with coupling β, vacuum scalar zero, and*

$$\operatorname{dens} \rho_\beta^{\mathrm{W}} = e^{-\beta E^{\mathrm{W}}_{L_p D_N}}\,.$$

Together with the same classes and relation, this curve satisfies Theorem 3.5.

The proof of Lemma 13.6 is deferred to this section's Wilson initialization.

Fix the constructed path arrays and an input curve ρ'_u of Theorem 3.5. Use the regular and forced norms of Definitions 3.3–3.4. Set

$$\mathcal{H}_u = \mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens} \rho'_u, \qquad P^{\mathrm{path}}_{N,m}(\beta, U) = \beta \mathsf{K}^{\mathrm{kin}}_m(U) + \mathcal{L}_m(U)\,.$$

Let $\mathsf{K}^{\mathrm{kin}}_{2,m}$ be the direct and row part of $\mathsf{K}^{\mathrm{kin}}_m$, with the cubic and quartic atoms omitted; this is a nonlinear function of the links. The next expansion is needed only when $\mathrm{Flag}_m(U) = \varnothing$.

Lemma 13.7 (Pre-reset expansion)**.** *There is a preparatory regular family with sum E^{pre}_u and weighted bound $c_{\mathrm{pre}}\epsilon^{\mathrm{reg}}_m/4$, such that*

$$\mathcal{H}_u(U) = \exp\{\tau_u|\Lambda| - \beta'_u \mathsf{K}^{\mathrm{kin}}_m(U) - \alpha_{N,m}\mathsf{K}^{\mathrm{kin}}_{2,m}(U) - \mathcal{L}_m(U) - E^{\mathrm{pre}}_u(U)\}$$

whenever $\mathrm{Flag}_m(U) = \varnothing$. *Here* $c_{\mathrm{pre}} \geq 4$ *depends only on the fixed stage. In particular,* $\mathcal{H}_u(1), \mathcal{H}_u(U) > 0$ *on this set; no vanishing at* 1 *is imposed on* E^{pre}_u.

The proof of Lemma 13.7 is deferred to this section's preparatory connected expansion.

Write E^{sum}_u for the final regular sum and $\mathcal{F}_u$ for the final disjoint-support forced sum in Definition 3.2. They are held fixed as a trial correction d varies. Define

$$q_{u,d}(U) = e^{-(\beta'_u+\alpha_{N,m}+d)\mathsf{K}^{\mathrm{kin}}_m(U) - \mathcal{L}_m(U) - E^{\mathrm{sum}}_u(U)} \mathcal{F}_u(U)\,.$$

The next identity fixes the entire density before the scalar estimate of Lemma 12.10 is used.

Lemma 13.8 (Exact density representation)**.** *The final activities satisfy Lemma 13.3, vanish at the flat field in their regular part, and are independent of d. For every u there is $\Delta_u \in \mathbb{R}$ such that*

$$q_{u,\Delta_u}(U) = \frac{\mathcal{H}_u(U)}{\mathcal{H}_u(1)} \qquad \text{for every } U\,.$$

Moreover, $\mathcal{F}_u(U) = 1$ *when* $\mathrm{Flag}_m(U) = \varnothing$. *No bound on Δ_u is assumed here.*

The proof of Lemma 13.8 is deferred to this section's exact activity identities; it uses no conclusion of Lemma 12.10.

The scalar equation also uses continuity along the input curve, without differentiating a flag boundary.

Lemma 13.9 (Continuity of the activity sums)**.** *For the input curves of Theorem 3.5, the final activity families have fixed addresses and are continuous in the weighted sense of Definition 13.5. For every fixed configuration U,*

$$u \longmapsto \mathcal{H}_u(U), \qquad u \longmapsto E^{\mathrm{sum}}_u(U), \qquad u \longmapsto \mathcal{F}_u(U)$$

are continuous.

The proof of Lemma 13.9 is deferred to this section's continuity estimates.

14 Trajectories, marked fields and ultraviolet convergence

This section is under construction.

14.1 Trajectories and tuning

Lemma 14.1 (Scale and period choices)**.** *To be stated here.*

Lemma 14.2 (Exact trajectory identities)**.** *To be stated here.*

Lemma 14.3 (Analytic endpoint tuning)**.** *To be stated here.*

14.2 Marked fields and volume-uniform estimates

Definition 14.4 (Marked density)**.** To be stated here.

Lemma 14.5 (Reference normalization)**.** *To be stated here.*

Lemma 14.6 (Symmetry and the unit)**.** *To be stated here.*

Lemma 14.7 (Marked remainders and transport)**.** *To be stated here.*

Lemma 14.8 (Marked local format)**.** *To be stated here.*

Write $\langle F\rangle_{N,p} = \int F\, d\mu_{N,p}$. A bond tiling has cell boundaries through link midpoints. The reflected copies below place one bounded gauge-invariant observable in every cell.

Definition 14.9 (Chessboard root)**.** For a tiling $\mathcal{T}$ into congruent reflected cells and an observable F in its base cell, let $(F)_C$ be its reflected copy in C, conjugated on odd reflections. Set

$$\|F\|_{\mathcal{T}} = \left| \left\langle \prod_{C\in\mathcal{T}} (F)_C \right\rangle_{N,p} \right|^{1/|\mathcal{T}|} .$$

Interior means dependence only on links with both endpoints strictly inside the base cell. The centered tiling $\mathcal{T}_m(y)$ has physical cell side a_m and base cell centered at y. This quantity converts reflected-product estimates into bounds on one insertion.

Lemma 14.10 (Centered bound)**.** *To be stated here.*

Lemma 14.11 (Difference estimate)**.** *To be stated here.*

Corrections to marked insertions. The following estimates concern the insertions of Definition 3.8, with the terminal maps $\mathsf{Z}_{N,p,d}$ of Lemma 14.5. They will be used to identify the quotient fields in Section 15. Fix strictly increasing spacing and volume subsequences $N = \mathsf{n}(j)$ and $p = P(k)$. All the estimates below hold for every such choice and survive further refinements. Write

$$\operatorname{olim} X_{N,p} = x \quad\Longleftrightarrow\quad \forall\varepsilon>0\ \exists k_0\ \forall k\ge k_0\ \exists j_0(k)\ \forall j\ge j_0(k) : |X_{\mathsf{n}(j),P(k)} - x| < \varepsilon\,. \tag{14.1}$$

For vectors use their fixed coefficient norm. An assertion holds eventually in this order with the same quantifiers after ε is omitted. A family $c_{N,p}\in V_d$ is admissible when

$$\forall\varepsilon>0\ \exists k_0\ \forall k\ge k_0\ \exists j_0(k)\ \forall j\ge j_0(k) : \|\mathsf{Z}^{-1}_{\mathsf{n}(j),P(k),d} c_{\mathsf{n}(j),P(k)}\|_\infty \le a_{\mathsf{n}(j)}^{-\varepsilon}\,. \tag{14.2}$$

This permits the coefficient growth needed to correct a positional difference.

We use the chessboard root and interior condition of Definition 14.9. Translates of a bond tiling are by lattice vectors. An anchor is the lattice site at which an insertion is placed. In a base cell of side a_k, let z^a be its integer coordinate measured from the first site in direction a, and put

$$M_{q,k,N} = \left\lfloor \frac{a_k}{8(q+1)a_N} \right\rfloor , \qquad M_{q,k,N} \le z^a, \quad z^a + M_{q,k,N} + 1 \le a_k/a_N \, . \tag{14.3}$$

These inequalities specify a relative interior margin.

Definition 14.12 (Small errors)**.** An observable family $E_{N,p}(y)$ is small if some $\eta_{N,p} \ge 0$ with $\operatorname{olim} \eta_{N,p} = 0$ satisfies the following condition. For every arity q and lower level m, there is $k \ge m$ such that, eventually in the order (14.1), $k \le N$ and every translated bond tiling $\mathcal{T}$ of side a_k and every base-cell anchor satisfying (14.3) obey

$$E_{N,p}(y) \text{ is interior}, \qquad \|E_{N,p}(y)\|_{\mathcal{T}} \le \eta_{N,p} \, .$$

The tolerance family is chosen before q, m, k. This is stronger than smallness in just one centered cell.

The equation projections separate the coefficient directions controlled by exact cutoff identities. The free equation space consists of polynomial labels that vanish on curvature jets satisfying the Maxwell equations. Its transported image is the kernel of the projections chosen below; the free quotient has dimension $\dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. The projections and the equation observables in Lemmas 14.13–14.14 are chosen together, before the spacing and volume subsequences.

Lemma 14.13 (Equation projections)**.** *For every d, there are complex linear idempotents $\Pi_{d,N,p} : V_d \to V_d$ and $C_d < \infty$ such that*

$$\|\Pi_{d,N,p}\| \le C_d, \qquad \dim V_d \le \dim \ker \Pi_{d,N,p} + \dim_{\mathbb{C}} M_d^{\mathrm{Max}} \, .$$

These projections can be chosen so that Lemma 14.14 holds for every spacing and volume subsequence.

The proof is deferred to the marked-field estimates and the independent free equation count in this subsection.

The next estimate converts a coefficient equation into an observable with exact separated moment identities.

Lemma 14.14 (Equation observables)**.** *For these same projections, every admissible family $c_{N,p} \in \ker \Pi_{d,N,p}$ admits a measurable observable family $Z_{N,p}(y)$, bounded at each anchor and cutoff eventually in (14.1), for which $O^{(d)}_{N,p}(c_{N,p}, y) - Z_{N,p}(y)$ is small. For every fixed spectator list and $\delta > 0$, eventually in that order,*

$$\int Z_{N,p}(y_i) \prod_{h \ne i} O_{N,p}(v_h, y_h) \, d\mu_{N,p} = 0 \qquad (\operatorname{sep}_{\mathrm{tor}}(y) \ge \delta) \, .$$

The proof is deferred to the local equation identities and marked comparison in this subsection; boundedness here is not uniform in the cutoffs. Each polynomial in the free equation space is a sum of terms with a linear equation factor. The local representatives are formed from combinations of Wilson action derivatives, with the remaining factors chosen independent of the varied link. Haar integration then gives zero against exterior spectators; linear combinations and gauge averaging retain that identity. At fixed positive separation, all spectator stencils eventually lie outside its fixed lattice neighbourhood.

Changing the degree changes the insertion normalization. The next estimate corrects that change before taking limits.

Lemma 14.15 (Corrected inclusion)**.** *For every $d \le e$ and $c \in V_d$, there is an admissible family $b_{N,p} \in V_e$ such that*

$$O^{(e)}_{N,p}(b_{N,p}, y) - O^{(d)}_{N,p}(c, y) \quad \textit{is small}.$$

The proof is deferred to the marked transport estimates in this subsection, including $d = e$.

The physical difference quotient requires a correction in the next degree.

Lemma 14.16 (Corrected difference)**.** *For every degree d, $c \in V_d$ and axis a, there is an admissible family $b_{N,p} \in V_{d+1}$ such that*

$$O^{(d+1)}_{N,p}(b_{N,p}, y) - \frac{O^{(d)}_{N,p}(c, y + e_a) - O^{(d)}_{N,p}(c, y)}{a_N} \quad \textit{is small}.$$

The proof is deferred to the marked difference estimates in this subsection.

Short-distance rank tests. Fix d and a real basis $m_1 = 1, m_2, \dots, m_{q_d}$ of M_d^{Max}, where $q_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. Choose finitely many one- or two-point tests. In row i, the first label is the variable m_j; the optional second label is a fixed basis element. Let $y^i \in \mathrm{Gen}_{n_i}$, $n_i \in \{1, 2\}$, be the row's configuration. The free test matrix is

$$\mathcal{G}_{ij} = S^{\mathrm{Max}}_{n_i}(m_j,\ \text{row } i \text{ spectator}; y^i)\,. \tag{14.4}$$

The next lemma supplies a finite family that distinguishes every free field of degree at most d.

Lemma 14.17 (Free test rows)**.** *For every d and real basis $m_1 = 1, m_2, \dots, m_{q_d}$, there are finitely many one- or two-point rows at generic configurations, with every second label a basis element, for which the matrix* (14.4) *has rank q_d.*

The proof uses only the Maxwell correlations of Definition 4.9 and the free positivity in Lemma 16.1, independently of Section 15.

Proof. Let $c \neq 0$ in M_d^{Max}. Lemma 16.1 gives a strictly positive reflected two-point value for c and the reflection of $\bar{c}$. Expand that second label in the basis (m_j). At least one of the resulting two-point functionals is nonzero on c. Smoothness of the free correlations, also in Lemma 16.1, permits a perturbation of the pair to Gen_2 preserving this nonzero value. Thus the common kernel of all permitted row functionals is zero. Starting with M_d^{Max}, choose a row nonzero on a nonzero vector in the current common kernel. Each choice strictly lowers its dimension. After finitely many choices the common kernel is zero, so $\mathcal{G}$ has rank q_d. □

Fix such a family for the rank comparison below.

For a trial ν, a common coefficient list $b^{\nu}_{j,N,p} \in V_d$ and lattice-compatible configuration $x^{\nu,i} = r_\nu y^{\nu,i}$ define the row insertions $I_h = O^{(d)}_{N,p}(b^{\nu}_{j_h,N,p}, x^{\nu,i}_h/a_N)$, with $j_1 = j$ and the spectator index fixed by row i. Physical points are read modulo the lattice period. Let $\mathcal{G}^{\nu}_J$ be the Maxwell moment of the labels $(m_{j_h})_{h\in J}$ at $(y^{\nu,i}_h)_{h\in J}$, with $\mathcal{G}^{\nu}_{\varnothing} = 1$. Define the relative errors by

$$R_{\varnothing} = 1, \qquad R_I = \prod_{h\in I} I_h - \sum_{\varnothing \neq J \subseteq I} \mathcal{G}^{\nu}_J R_{I\setminus J}\,. \tag{14.5}$$

Only singleton and pair errors occur. The lifts b^{ν} include the scale normalization, so I_h carries no additional power of r_ν.

The following rank estimate is deferred to the short-distance analysis in this subsection. Each relative error is interior whenever its insertions are interior, since (14.5) is a polynomial in those insertions.

Lemma 14.18 (Relative rank comparison)**.** *For every such finite test family, there are $r_\nu \downarrow 0$, $y^{\nu,i} \to y^i$, levels $k_\nu \le t_\nu$ tending to infinity, $\gamma_\nu \ge 0$ tending to zero, and real lifts $b^\nu_{j,N,p}$, bounded in N, p for fixed ν and common to all rows. They satisfy $b^\nu_{1,N,p} = \mathbf{u}_d$. Each $x^{\nu,i}$ is generic on the spacing-a_{t_ν} grid. Eventually in N, uniformly in p, all row insertions are interior to $\mathcal{T}_{k_\nu}(0)$, and, for every nonempty I,*

$$\|R_I\|_{\mathcal{T}_{k_\nu}(0)} \le \gamma_\nu \,.$$

14.3 Full ultraviolet convergence

Lemma 14.19 (Lattice-point convergence)**.** *To be stated here.*

Lemma 14.20 (Interpolation and smearing)**.** *To be stated here.*

15 Infinite volume and field calculus

We pass from the fixed-volume kernels of Theorem 4.2 (UV) to correlations on $\mathbb{R}^4$. The order of limits matters: the lattice spacing tends to zero at every fixed admissible volume, and only then does the volume increase. Lemmas 15.1–15.5, using the ultraviolet estimates of Lemmas 14.13–14.18, prove Theorem 4.7 (IV). The same volume refinement supplies the coefficient maps needed for Proposition 4.10 (FC). Lemma 15.6 controls repeated insertions and will be used to construct the bounded slab operators in Lemma 17.1.

There are two compactness arguments. The bounds of Proposition 4.1 (UB), including one additional difference, make the ultraviolet kernels and their derivatives equicontinuous on compact separated sets. A countable diagonal extraction therefore chooses the volume limit for all polynomial fields. Finite-dimensional compactness also retains certain corrected coefficient maps. These maps are needed because a positional difference changes the degree and the insertion normalization; the polynomial embedding between two coefficient spaces need not preserve correlations.

The identification of null coefficients is the additional difficulty. An approximation to a derivative may have unbounded coefficients. We normalize its projected part and use correlation tests to exclude a nonzero null vector in the range of the limiting projection. This produces one fixed coefficient vector that represents the derivative against every spectator. Finally, chessboard estimates give bounds uniform in the positions, including configurations of arbitrarily large diameter. For repeated words, two cell sizes keep the cost of the exterior independent of the number of repetitions.

Throughout we use the Wilson family, tokens, differences and separated tests of Section 4. Write $f_{p,v} = M^{\mathrm{UV}}_{p,v}$ and $\langle F \rangle_{N,p} = \int F \, d\mu_{N,p}$. All constants below may depend on the fixed construction. Any further dependence is indicated.

15.1 Volume limits

Fix a strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$. The next extraction chooses the table and the volume sequence together. The periodic generic restrictions retain the fixed-volume kernels required in IV.

Lemma 15.1 (Generic volume kernels)**.** *There are a strictly increasing* q *and a table* S*. Put* $P = \mathsf{p} \circ \mathsf{q}$*. For every* p, n, v*, the restriction* $S_v^{(p)} = f_{p,v}$ *to*

$$\{x : x_i^a - x_j^a \notin L_p\mathbb{Z} \quad (i \neq j,\ a = 1, \dots, 4)\}$$

is continuous. Every compact $K \subset \mathrm{Gen}_n$ *eventually lies in these domains for* $p = P(k)$*, and* $S_v^{(P(k))} \to S_n(v;\cdot)$ *uniformly on* K*.*

The same choice also gives, for every $n, v, \boldsymbol{l}$,

$$S_n(v;\cdot) \in C^\infty(\mathrm{Conf}_n), \qquad \partial_{\boldsymbol{l}} f_{P(k),v} \longrightarrow \partial_{\boldsymbol{l}} S_n(v;\cdot) \quad \text{locally uniformly on } \mathrm{Conf}_n\,. \tag{15.1}$$

The proof establishes both assertions and ordered convergence against every separated test. After Lemma 15.4, P denotes its final refinement.

Proof. Fix a compact $K \subset \mathrm{Conf}_n$ and a compact neighbourhood K' contained in Conf_n. For all sufficiently large p, K' lies in $\mathrm{NoHit}_{L_p,n}$, and its torus separation has a positive lower bound $2r$ independent of p. UB bounds $\Delta_{\boldsymbol{l}} M_{N,p}^{\mathrm{lat}}$ by $B(v, \boldsymbol{l}, r)$ on lattice tuples near K', for all sufficiently large N, uniformly in p. By Lemma 14.20, the multilinear interpolants converge at each fixed p to $\partial_{\boldsymbol{l}} f_{p,v}$. Consequently

$$\sup_{p \geq p_0} \sup_{x \in K'} |\partial_{\boldsymbol{l}} f_{p,v}(x)| \leq B(v, \boldsymbol{l}, r). \tag{15.2}$$

The ultraviolet threshold for taking this limit may depend on p. One more derivative bounds variation along short segments in K', giving equicontinuity on K.

Apply Arzelà–Ascoli, in the form for a pointwise bounded equicontinuous sequence of continuous complex functions on a compact metric space [3, Theorem 7.25]. Diagonalize over compact exhaustions of Conf_n, arities, degree lists, basis coefficient lists and derivative lists. These indices are countable. Starting from any prescribed strictly increasing p, we obtain a strictly increasing q and $P = \mathsf{p} \circ \mathsf{q}$. Multilinearity extends the limits from basis lists to every token list. Integration by parts against a compactly supported test identifies the limit of each first derivative with the derivative of the limiting kernel. Repeating this for all derivative lists gives a smooth multilinear table S, with local convergence of every derivative.

Restrict $f_{p,v}$ to the displayed domain to define $S_v^{(p)}$. A compact generic set has bounded diameter and a positive minimum coordinate separation, so it eventually lies in that domain. The asserted uniform convergence follows from the extraction. For every separated test f, UV gives, at every admissible p, the full inner limit $C_p^{\mathrm{UV}}(v; f) = \int f f_{p,v}$. Its support is eventually admissible along P, and local uniform convergence gives

$$C_{P(k)}^{\mathrm{UV}}(v; f) \longrightarrow \int f(x) S_n(v; x)\, dx. \tag{15.3}$$

Thus Definition 4.3 holds. Equation (15.1) and ordered convergence persist on every further volume refinement. □

Symmetries of the limit. We first prove E1–E3 and E6, using Definition 3.8, Lemmas 14.6 and 14.20 and (15.1). Permutation symmetry, the empty correlation, deletion of the distinguished unit, conjugation and signed-permutation covariance hold on the lattice once N exceeds the finitely many reference levels. Approximate any fixed separated configuration by lattice points and use Lemma 14.20, then (15.1). For translations, approximate the translation vector on the same lattice. Translation invariance and (15.2) pass the identity to arbitrary real translations. This proves E1–E3 independently of the coefficient corrections. Equation (15.1) gives E6; multilinearity was retained in the extraction.

15.2 Interior roots and errors

We use the chessboard root of Definition 14.9. A signed image means the image under H_4; conjugation is also allowed. Bond tilings, anchors, margins and small errors are specified before Definition 14.12. We consider tilings at the construction levels. Two such tilings are aligned when their cells have the same coordinate axes and every coarse boundary is a fine boundary. Their side ratio is an odd integer, since the block ratios are odd.

For interior functions in the base cell of a bond tiling $\mathcal{T}$, the chessboard inequality reads

$$\left| \left\langle \prod_{C\in\mathcal{T}} (F_C)_C \right\rangle_{N,p} \right| \le \prod_{C\in\mathcal{T}} \|F_C\|_{\mathcal{T}} \,. \tag{15.4}$$

Apply the abstract chessboard theorem [2, Theorem 4.1] axis by axis, on real vector spaces of complex observables, with reflection followed by conjugation. The functional has $L_p D_k$ factors in each axis, an even number since $4 \mid L_p$. Lattice translation invariance gives cyclicity. Reflection positivity across every translated band cut gives the required Schwarz inequality. For the Wilson action this is Osterwalder–Schrader positivity on the periodic lattice [4, Theorem 2.1, pp. 14–15], applied to a compact matrix gauge group, positive coupling, and observables supported on one side of the bond planes. Their boundaries pass through link midpoints, as required there.

The centered root estimate must also control anchors near arbitrary, separated configurations. Write $v = (d, c)$ for the insertion token.

Lemma 15.2 (Interior root bound)**.** *Fix $m \le k \le N$, $N \ge m_{\mathrm{ref}}(d)$, and $A \ge 0$. Suppose, for every anchor y', every signed image or image under conjugation of $O_{N,p}(v, y')$ has root at most A in $\mathcal{T}_k(y')$ and stencil within a_k of y'. In every translated bond tiling $\mathcal{T}$ of side a_m, an anchor at distance at least $3a_k$ from its cell boundary satisfies*

$$O_{N,p}(v, y) \text{ is interior}, \qquad \|O_{N,p}(v, y)\|_{\mathcal{T}} \le A \,. \tag{15.5}$$

The proof transfers the centered bound first at aligned cell centers, then between them.

Proof. Use Definition 3.8, its symmetry in Lemma 14.6, the cell interior condition of Definition 14.9, and (15.4). At an aligned fine-cell center, reflect all coarse copies into distinct fine cells. Their signs and conjugations are among the assumed images. Apply (15.4) with 1 in every unoccupied fine cell. Taking the coarse root bounds it by A.

For intermediate anchors, let $H(z)$ be the coarse root after z lattice translations in one axis, with the other coordinates fixed. Translation invariance and reflection give

$$H(z)^2 \le H(z-1)H(z+1) \,. \tag{15.6}$$

Indeed, translate the reflected product by one lattice step. Alternate cells then contain shifts $z-1$ and $z+1$; apply (15.4). On a finite interval a nonnegative sequence satisfying this inequality cannot exceed its endpoint maximum: an interior positive maximum extends to both neighbours and hence to an endpoint. This includes zero endpoints. Apply this argument in each axis between the surrounding fine-cell centers. A margin $3a_k$ keeps these centers and every intermediate stencil inside the coarse cell. Interior follows from the stencil radius a_k. Thus (15.5) holds. □

Small errors are defined by a root tolerance chosen before the cell scale. This order makes them negligible against each fixed spectator test.

Lemma 15.3 (Vanishing errors)**.** *For every small error E of Definition 14.12, fixed token list v, distinguished slot i, and $\delta > 0$, there is $C(v,\delta) < \infty$ such that, eventually in (14.1),*

$$\left| \left\langle E_{N,p}(y_i) \prod_{h \neq i} O_{N,p}(v_h, y_h) \right\rangle_{N,p} \right| \leq C(v,\delta)\eta_{N,p} \qquad (\mathrm{sep}_{\mathrm{tor}}(y) \geq \delta)\,. \tag{15.7}$$

Consequently its smear against any fixed separated test tends to zero in that order.

The proof uses distinct cells for the error and each spectator.

Proof. Use Definition 3.8, the centered estimate of Lemma 14.10, Definition 14.12, Lemma 15.2 and (15.4). Let n be the arity. Take m with $a_m < \delta$; the small-error condition gives $k \geq m$. Choose a finer spectator level k_{sp} with $3a_{k_{\mathrm{sp}}} \leq a_k/[8(n+1)]$. Lemma 14.10 bounds all signed spectator roots and their images under conjugation there by a fixed constant.

In each axis choose the translation of the level-k tiling so that no boundary meets a strip of half-width $a_k/[4(n+1)]$ around any anchor. The excluded length is at most $na_k/[2(n+1)] < a_k$, so such a lattice translation exists for sufficiently large N. It supplies the integer margins (14.3), including their extra lattice unit, and margin $3a_{k_{\mathrm{sp}}}$ for spectators. Any pair of anchors differs in some coordinate by at least $\delta > a_k$, so they occupy distinct cells. Lemma 15.2 and (15.4) give (15.7). On a compact test support, the number of lattice tuples multiplied by a_N^{4n} is bounded independently of N, p. Multiplication by the test supremum therefore proves the smeared assertion. □

15.3 Corrected coefficients

We now use the equation projections and corrections of Lemmas 14.13–14.16, and the relative rank comparison of Lemma 14.18. The other inputs are Proposition 4.1, Definitions 3.8, 4.8 and 4.9, Lemmas 14.6–14.10, Lemma 14.20, Lemmas 15.1, 15.2 and 15.3, and E1–E3 established above. Lemma 14.17 supplies the free test rows independently of the interacting arguments here. Auxiliary coefficient limits may use a common spacing subsequence n at every p; the full ultraviolet kernel limits remain those of UV.

These estimates distinguish coefficients that vanish in every separated correlation. The inclusions and derivatives in the next statement are the maps asserted in FC; the following proof constructs them and proves their defining correlation identities.

Lemma 15.4 (Quotient compatibility)**.** *For every $d \leq e$, the inclusions of FC preserve correlations and units and compose compatibly. Inclusions and derivatives commute with conjugation, and*

$$h\partial_a = \sum_b h_{ba}\partial_b h \qquad (h \in H_4).$$

Each response map is injective; every response lies in one $\iota_d F_d$; every field has a token representative. The unit is $\mathbf{1}_d = [a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}]$.

The proof below constructs the inclusions and derivatives asserted in FC, proves its dimension equality, and then derives the compatibility properties. E1–E3 are already available.

Proof. We use the inputs listed above. Bounded coefficient families satisfy (14.2). In fact Lemma 14.7 bounds their inverse terminal coordinates by a fixed norm factor times $\prod_{m<N} C_{J_{m+1}}$, and row 5 of Section 9 gives $\log C_{J_{m+1}} \leq J_{m+1}^{-2} \log \ell_m$. Split the sum at a fixed level. Its finite initial part is $o(s_N)$, while its tail divided by $s_N = \sum_{m<N} \log \ell_m$ is bounded by the largest J_{m+1}^{-2} in that

tail. Sending the splitting level to infinity gives $\sum_{m<N} \log C_{J_{m+1}} = o(s_N)$. There are countably many projection matrices and comparison coefficient lists. The uniform bound in Lemma 14.13 gives convergent subsequences by finite-dimensional compactness. The same countable diagonal extraction retains every projection and each comparison lift, first in N at every p, then in volume, retaining (15.1). Thus, in operator norm,

$$\Pi_{d,\mathsf{n}(j),p} \longrightarrow \Pi_d^{(p)}, \qquad \Pi_d^{(P(k))} \longrightarrow \Pi_d, \qquad \Pi_d^2 = \Pi_d\,. \tag{15.8}$$

Continuity of composition gives idempotence at both stages. For later use, the convergence also holds in the order (14.1).

Let $L_{N,p} : V_d \to \mathbb{C}$ be any fixed one-slot smeared correlation map, with arbitrary fixed spectators and test. Its operator-norm limit is L, by finite dimensionality and (15.3). Apply Lemma 14.14 and (15.7) to $c - \Pi_{d,N,p}c$. This gives $L((1-\Pi_d)c) = 0$ for every L. Since S is smooth by (15.1),

$$\ker \Pi_d \subset \mathcal{N}_d(S). \tag{15.9}$$

For the reverse dimension inequality, use the rank-q_d free matrix $\mathcal{G}$ of Lemma 14.17 and the common lifts of Lemma 14.18. Rearrange (14.5) to isolate $\mathcal{G}_I^\nu$. Every remaining term contains a nonempty R_J; its expectation is bounded by γ_ν using (15.4). Thus

$$\left| \left\langle \prod_{h\in I} I_h \right\rangle - \mathcal{G}_I^\nu \right| \le \gamma_\nu \sum_{J\subsetneq I} |\mathcal{G}_J^\nu|. \tag{15.10}$$

The finite sum stays bounded as $y^{\nu,i} \to y^i$. Pass through both coefficient and kernel limits. Each matrix entry is a fixed spectator functional on V_d, applied to the limiting lift of m_j. That functional vanishes on $\mathcal{N}_d(S)$, so each matrix factors through F_d and has rank at most $\dim F_d$. The matrices tend to $\mathcal{G}$. A nonzero q_d-minor therefore gives $\dim F_d \ge q_d$. Ranks of converging idempotents are eventually constant: if $\|P-Q\| < 1$, P is injective on $\operatorname{ran} Q$, and conversely. Passing the dimension inequality in Lemma 14.13 to Π_d using (15.8) and using (15.9) now gives

$$\mathcal{N}_d(S) = \ker \Pi_d, \qquad \dim_{\mathbb{C}} F_d = q_d. \tag{15.11}$$

For inclusion, Lemma 14.15 gives a corrected family that has target smeared limit $\int fS(c,\dots)$. For the corrected difference of Lemma 14.16, torus summation by parts gives exactly

$$a_N^{4n} \sum_y f(\hat{y}) \Delta_{i,a} M_{N,p}^{\mathrm{lat}}(v;y) = a_N^{4n} \sum_y \frac{f(\hat{y} - a_N e_{i,a}) - f(\hat{y})}{a_N} M_{N,p}^{\mathrm{lat}}(v;y). \tag{15.12}$$

Here $e_{i,a}$ is the coordinate vector in slot i, axis a. The backward quotient tends uniformly to $-\partial_{i,a} f$; its residual is bounded by $a_N \|\partial_{i,a}^2 f\|_\infty / 2$. UB at order zero controls the residual sum on a fixed compact neighbourhood. Thus the derivative target is $-\int (\partial_{i,a} f) S(c,\dots)$.

For either corrected family $w_{N,p}$ in its representing degree e, write $\Pi = \Pi_{e,N,p}$. The normalized remainder $(w - \Pi w)/(1 + \|\Pi w\|)$ is admissible and belongs to $\ker \Pi$: its first term is a bounded scalar multiple of an admissible family, and its second term is bounded. Lemma 14.14 and Lemma 15.3 give

$$L_{N,p}\big((w - \Pi w)/(1 + \|\Pi w\|)\big) \longrightarrow 0. \tag{15.13}$$

Take a cluster point (λ, z) of $((1 + \|\Pi w\|)^{-1}, \Pi w/(1 + \|\Pi w\|))$ along one ultrafilter containing every ordered tail

$$\{(k,j) : k \ge k_0,\ j \ge j_0(k)\}\,.$$

These tails form a filter base; bounded finite-dimensional data have a cluster point along such an ultrafilter. All spectator limits hold on this same filter. Then $\Pi_e z = z$, $\lambda + \|z\| = 1$, and for every spectator test L, $L(z) = \lambda t_L$, where t_L is its target. If $\lambda = 0$, (15.11) puts z in $\ker \Pi_e$, contradicting $\|z\| = 1$. Hence z/λ represents the target for all spectators simultaneously. Choose these representatives on a basis and extend linearly, obtaining $\kappa_{d,e}$ and $\tau_{d,a}$ with

$$S(\kappa_{d,e}c, \dots) = S(c, \dots), \qquad S(\tau_{d,a}c, \dots) = \partial_{x_0^a} S(c, \dots). \tag{15.14}$$

Equation (15.1) converts the distributional identities to pointwise identities.

These maps preserve null coefficients, so they descend to $\jmath_{d,e}$ and ∂_a. A response vanishes precisely on $\mathcal{N}_d(S)$; hence ι_d is injective. Equality of responses gives injectivity, composition and uniqueness of inclusions. E1 and Lemma 14.6 give preservation of $[a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}]$. Any finite sum of responses lies in the largest degree involved; quotient surjectivity gives token representatives. Finally E2, conjugation and the chain rule applied to $S(hc; hx) = S(c; x)$ give the asserted symmetry identities by injectivity of the response maps. The same comparison without differentiation gives $h\jmath_{d,e} = \jmath_{d,e}h$. Permuting the distinguished slot proves FC's positional identity in every slot. This proves FC and the lemma using Lemmas 14.13–14.18. □

15.4 Euclidean properties

The lattice symmetries and positivity pass through the ordered limits. The centered root estimate gives the additional control as separated points approach one another.

Lemma 15.5 (Euclidean properties)**.** *The table supplied by IV is complex multilinear and satisfies E1–E7, including joint-test reflection positivity, absolute integrability of its pairings, and bounds for every derivative order:*

$$\|D_x^k S_n(c; x)\| \le C(n, d, c, k) r^{-A(n,d,c,k)} \quad (0 < r \le 1,\ \mathrm{sep}_\infty(x) \ge r).$$

The proof below passes reflection positivity to the limit and proves the power bound; its positive derivative orders use (15.14).

Proof. We use Definitions 3.8 and 4.8, Lemmas 14.6, 14.10 and 14.20, equations (15.1) and (15.4), Lemma 15.2, and the derivative maps (15.14) from Lemma 15.4. E1–E3 and E6 were proved after Lemma 15.1; only the positive derivative orders of E7 use the corrected maps.

For E5, fix finitely many joint smooth tests f_i with compact support on positive-side separated configurations in one axis. Their lattice smears F_i are supported strictly inside the positive half-torus for large p, N: the support has positive distance from the reflection plane, and the physical radius of each fixed stencil tends to zero. Use the bond reflection $x^a \mapsto a_N - x^a$, with planes $x^a = a_N/2$ and $x^a = (a_N + L_p)/2$. It is a lattice symmetry at every cutoff. The odd block ratios preserve the midpoint boundaries of the bond tilings. For large p, the supports also avoid a neighbourhood of the second plane $x^a = L_p/2$; for large N their stencils avoid both actual bond planes. The periodic Wilson reflection positivity cited before (15.4) gives

$$\sum_{i,j} \bar{z}_i z_j \langle \overline{F_i \circ \mathsf{r}_{a,N}} F_j \rangle_{N,p} \ge 0.$$

Here $\mathsf{r}_{a,N}$ maps a configuration to its reflected configuration. Insertion covariance reflects the positions and applies $\mathsf{s}_a^{\mathrm{refl}}$ to their labels; conjugation acts on the labels in the first word. The reflected joint test converges, with all derivatives, to $\bar{f}_i \otimes f_j$ with its first group reflected through $x^a = 0$.

Its support is compact and separated, including across the reflection plane. All these supports lie in one compact separated set. UB at order zero bounds the difference between the moving tests and their limiting test by their uniform difference times a bounded weighted site count. It tends to zero. Apply Lemma 14.20 and (15.3) to the fixed limiting test in each entry. Smoothness bounds the kernel on the support, so every pairing is absolutely integrable. The limiting matrix is positive semidefinite. Concentrating normalized smooth bumps at prescribed positive-side words gives E4 by continuity. This works in every axis with its corresponding reflection.

For the power bound first consider order zero and a nonempty token list. Put $d_{\mathrm{pow}} = 1 + \max_i d_i$. Lemma 14.10 with $\varepsilon = 1$, for all the finitely many signed images and images under conjugation, bounds their centered roots by $C_{\mathrm{ctr}} a_k^{-d_{\mathrm{pow}}}$ for $k \geq k_0$. The constants k_0, C_{ctr} precede the choice of separation. The scale condition $\log \ell_u / s_u \to 0$ in Lemma 14.1 gives

$$a_{u+1} \geq c a_u^2. \tag{15.15}$$

Indeed $\log \ell_u / \log a_u^{-1} \to 0$ implies $\ell_u \leq a_u^{-1}$ eventually; decrease $c > 0$ for the remaining levels. The first $u \geq k_0$ with $a_u < \delta \leq 1$ consequently satisfies $a_u \geq c_{k_0} \delta^2$, including $u = k_0$.

Choose the first $m \geq k_0$ with $a_m < r/2$, then the first $k \geq k_0$ with $a_k < a_m/[6(n+1)]$. Thus $k \geq m$ and $a_k \geq c' r^4$. In each coordinate the boundary strips of half-width $3a_k$ excluded by the n anchors have total length less than a_m. Choose a translated tiling outside them. All anchors occupy distinct cells. Lemma 15.2, followed by (15.4), gives the lattice bound $(C_{\mathrm{ctr}} a_k^{-d_{\mathrm{pow}}})^n$. Passing to the limits by Lemma 14.20 and (15.1), with slightly smaller separation to accommodate lattice approximation, gives

$$|S_n(v; x)| \leq C r^{-4 d_{\mathrm{pow}} n}. \tag{15.16}$$

All constants are independent of the positions and the volume.

For higher orders use (15.14): every coordinate derivative is a correlation with fixed, degree-raised tokens. At a fixed order there are finitely many such lists. Apply (15.16) to each, take the largest exponent, and bound the Fréchet operator norm by the sum of its coordinate coefficients in an orthonormal basis of $\mathbb{R}^{4n}$. This proves E7 using Lemma 15.4 and hence Lemmas 14.13–14.18. For arity zero, $S_0 = 1$ and all positive-order derivatives vanish. □

Proof of Theorem 4.7. Compose the volume refinements in Lemmas 15.1 and 15.4. Equation (15.3) proves ordered convergence, retaining UV's full inner limit at every admissible volume. Lemma 15.5 proves E1–E7. The coefficient corrections do not change any kernel limit already obtained. □

15.5 Repeated reflected words

Use the OS space of Definition 6.1. For any $s > 0$ and word w with times in $(0, s)$, set

$$A_w v = w\, \tau_{s e_4} v, \qquad w^\dagger = \tau_{s e_4} \mathsf{s}_4^{\mathrm{refl}} w^*.$$

Here A_w initially acts on words; boundedness on OS classes is not assumed. A buffer r means pairwise sup-distance at least r and times in $[r, h_{\mathrm{sl}} - r]$ for a slab of width h_{sl}.

The next estimate bounds the growth of repeated insertion before taking any operator extension.

Lemma 15.6 (Repeated reflected words)**.** *For every degree bound d and coefficient bound $R \geq 0$, there are $C_0(d, R) > 0$, $Q(d, R) \in \mathbb{N}$. For each nonempty length-n word w with these bounds, slab width $h_{\mathrm{sl}} > 0$ and buffer $0 < r \leq 1$, some $0 < s \leq h_{\mathrm{sl}}$, $k, n_{\mathrm{lat}} \in \mathbb{N}$ satisfy $s = n_{\mathrm{lat}} a_k$, and the slab of width s contains w. For every separated exterior word v with positive times, some $B_v < \infty$ gives*

$$\|[(A_{w^\dagger} A_w)^j v]\|_{\mathrm{OS}}^2 \leq B_v [C_0^n (n/r)^{Qn}]^{4j} \qquad (j \geq 0).$$

The operators in the statement use the chosen width s. Its choice and the constants C_0, Q precede v; B_v may also depend on w, n, r and s.

Proof. We use Definitions 3.8 and 6.1, Lemmas 14.10 and 14.20, Lemmas 15.2 and 15.5, and (15.4), (15.1) and (15.15). Enlarge the coefficient ball to contain every signed image and image under conjugation in degrees at most d. Lemma 14.10, with $\varepsilon = 1$, gives $k_* \in \mathbb{N}$ and $B_* > 0$ such that all these centered roots at levels $k_{\mathrm{w}} \geq k_*$ are bounded by $B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}}$, where $d_{\mathrm{pow}} = d + 1$. This choice precedes n, r, w.

Choose the first $m \geq k_*$ with $a_m < r/32$, and put $s = a_m \lfloor h_{\mathrm{sl}}/a_m \rfloor$. The word is nonempty with buffer r, so $s > 0$, $s \leq h_{\mathrm{sl}}$, and it still contains w with buffer at least $r/2$. Equation (15.15) gives $a_m \geq c_1 r^2$. Fix $0 < \alpha \leq 1$ with $20\alpha < c_1$, and choose the first $k_{\mathrm{w}} \geq k_*$ with $a_{k_{\mathrm{w}}} < \alpha(r/n)^2$. Then

$$20n a_{k_{\mathrm{w}}} < a_m, \qquad a_{k_{\mathrm{w}}} \geq c_2 (r/n)^4, \qquad (B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}})^n \leq (B_* c_2^{-d_{\mathrm{pow}}})^n (n/r)^{4 d_{\mathrm{pow}} n}. \tag{15.17}$$

These choices precede the exterior. In the statement take $k = m$ and $n_{\mathrm{lat}} = \lfloor h_{\mathrm{sl}}/a_m \rfloor$.

Expanding the OS norm gives a correlation of $4jn$ main insertions and two copies of v. The main insertions are mutually separated, and separated from the exterior, by at least $r/2$. Translations by multiples of s do not change residues modulo $a_m \mathbb{Z}^4$. Thus the main anchors have at most $2n$ residue patterns, those of w and its reflection with conjugation, independently of j.

For nonempty v, choose $m_{\mathrm{v}} \geq k_{\mathrm{w}}$ with $a_{m_{\mathrm{v}}}$ smaller than its internal separation and twice its minimum positive time. This separates the two exterior copies for every j. Next choose $k_{\mathrm{v}} \geq m_{\mathrm{v}}$, beyond the exterior reference levels, with $12|v| a_{k_{\mathrm{v}}} < a_{m_{\mathrm{v}}}$. Lemma 14.10 gives a common finite bound $X \geq 0$ for all exterior roots and their images at level k_{v}. Both $m_{\mathrm{v}}, k_{\mathrm{v}}$ are independent of j.

We apply the chessboard estimate at two cell sizes. In each axis, avoid strips of half-width $5a_{k_{\mathrm{w}}}$ around the $2n$ main residues modulo a_m. This is possible by (15.17). Then change the translation by less than $a_{m_{\mathrm{v}}}$ to place all $2|v|$ exterior anchors at distance at least $3a_{k_{\mathrm{v}}}$ from the finer boundaries; $12|v| a_{k_{\mathrm{v}}} < a_{m_{\mathrm{v}}}$ permits this. Since $a_{m_{\mathrm{v}}} \leq a_{k_{\mathrm{w}}}$, the main anchors retain margin at least $3a_{k_{\mathrm{w}}}$. Main insertions occupy distinct coarse cells, none containing an exterior anchor.

Several exterior factors may occupy one coarse cell. Reflect their product into every coarse cell, and apply (15.4) to the aligned tiling of side $a_{m_{\mathrm{v}}}$. The fine and coarse tilings share an origin and have an odd integer side ratio. Coarse reflections preserve the fine grid and its interior margins. Within each reflected coarse cell, exterior anchors are separated by more than $a_{m_{\mathrm{v}}}$ and so occupy distinct fine cells; across coarse boundaries their fine cells are disjoint. The uncentered estimate of Lemma 15.2 bounds every factor by X. If the coarse cell contains n_C exterior factors, its root is therefore at most X^{n_C}. Applying the coarse chessboard estimate now gives

$$\|[(A_{w^\dagger} A_w)^j v]\|_{\mathrm{OS}}^2 \leq X^{2|v|} (B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}})^{4jn}.$$

For each fixed j, take lattice approximations of all anchors, preserving these strict margins and separations, then apply Lemma 14.20 and (15.1). No cutoff threshold uniform in j is needed. The constants on the right are already independent of j. Choose $B_v = \max(1, X^{2|v|})$, $C_0 = B_* c_2^{-d_{\mathrm{pow}}}$, and $Q = 4d_{\mathrm{pow}}$. Equation (15.17) proves the assertion. For empty v omit its factors and take $B_v = 1$. □

16 Free fields, Ward identities and rotations

This section is under construction.

16.1 Free fields

Lemma 16.1 (Free positivity and stress)**.** *To be stated here.*

16.2 Free comparison and the conserved line

Lemma 16.2 (Stable weak comparison)**.** *To be stated here.*

Lemma 16.3 (Conserved line)**.** *To be stated here.*

Lemma 16.4 (Raw shell charges)**.** *To be stated here.*

16.3 Ward identities and rotations

Lemma 16.5 (Rotation and charge identification)**.** *To be stated here.*

Lemma 16.6 (Covariance and normalized tensor)**.** *To be stated here.*

Lemma 16.7 (Transported infinitesimal actions)**.** *To be stated here.*

17 Slabs, phases and reconstruction

This section is under construction.

17.1 Slabs and phases

Lemma 17.1 (Slab operators)**.** *To be stated here.*

Lemma 17.2 (Compression algebra)**.** *To be stated here.*

Lemma 17.3 (Phase fields and selection)**.** *To be stated here.*

17.2 Reconstruction

Lemma 17.4 (Fields and boundary distributions)**.** *To be stated here.*

Lemma 17.5 (OS comparison and irreducibility)**.** *To be stated here.*

Lemma 17.6 (Uniform low-energy bounds)**.** *To be stated here.*

Lemma 17.7 (Phase dependence and continued stress)**.** *To be stated here.*

18 The trace law and non-free comparisons

This section is under construction.

18.1 Trace and spectrum

Lemma 18.1 (Centering as an operator identity)**.** *To be stated here.*

Lemma 18.2 (Unique trace spectral measure)**.** *To be stated here.*

Lemma 18.3 (High-energy trace limit)**.** *To be stated here.*

18.2 Precise non-free comparisons

Lemma 18.4 (Finite Wick obstruction)**.** *To be stated here.*

Definition 18.5 (Polynomial mass-shell pairings)**.** To be stated here.

Lemma 18.6 (Polynomial countertest)**.** *To be stated here.*

Definition 18.7 (Maxwell Fock space)**.** To be stated here.

Definition 18.8 (Coherent comparison vectors)**.** To be stated here.

Lemma 18.9 (Fock countertest)**.** *To be stated here.*

Lemma 18.10 (Localized averaged countertest)**.** *To be stated here.*

19 Products and contact subtractions

This section is under construction.

19.1 Products and nested clusters

Definition 19.1 (Finite correlation projection)**.** To be stated here.

Definition 19.2 (Symmetric projection)**.** To be stated here.

Definition 19.3 (OPE coefficients)**.** To be stated here.

Lemma 19.4 (One channel prescription)**.** *To be stated here.*

Lemma 19.5 (Separated estimates)**.** *To be stated here.*

Lemma 19.6 (Integrated separated estimate)**.** *To be stated here.*

Definition 19.7 (Nested geometry and rates)**.** To be stated here.

Definition 19.8 (Iterated coefficient)**.** To be stated here.

Lemma 19.9 (Nested coefficient estimate)**.** *To be stated here.*

Lemma 19.10 (Nested tested estimate)**.** *To be stated here.*

Lemma 19.11 (Integrated nested estimates)**.** *To be stated here.*

19.2 Contact subtractions

Definition 19.12 (Contact detector projection)**.** To be stated here.

Definition 19.13 (Coherent contact projections)**.** To be stated here.

Definition 19.14 (Contact grades and clearance)**.** To be stated here.

Lemma 19.15 (Real bases and scales)**.** *To be stated here.*

Lemma 19.16 (Contact extension)**.** *To be stated here.*

Lemma 19.17 (Local distribution order)**.** *To be stated here.*

References

[1] Gerald B. Folland, *A course in abstract harmonic analysis*, 2 ed., Textbooks in Mathematics, CRC Press, Boca Raton, FL, 2015.

[2] Jürg Fröhlich, Robert Israel, E. H. Lieb, and Barry Simon, *Phase transitions and reflection positivity. I. General theory and long range lattice models*, Communications in Mathematical Physics **62** (1978), no. 1, 1–34.

[3] Walter Rudin, *Principles of mathematical analysis*, 3 ed., McGraw-Hill, New York, 1976.

[4] Erhard Seiler, *Lattice gauge theories*, Gauge Theories as a Problem of Constructive Quantum Field Theory and Statistical Mechanics, Lecture Notes in Physics, vol. 159, Springer, Berlin, Heidelberg, 1982, pp. 3–98.

[5] Gerald Teschl, *Ordinary differential equations and dynamical systems*, Graduate Studies in Mathematics, vol. 140, American Mathematical Society, Providence, RI, 2012.